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Design and Development of iDetect: An OCR-Based Discount ID Verification Application for Business Establishments under the Business Permit and Licensing Office in Valenzuela City

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CHAPTER I THE PROBLEM AND ITS BACKGROUND

Background of the Study

Senior citizens and persons with disabilities (PWDs) in the Philippines are legally entitled to significant government-mandated benefits under Republic Acts 9994 (Expanded Senior Citizens Act of 2010) and 10754 (An Act Expanding the Benefits and Privileges of Persons with Disability). These landmark legislations guarantee a 20% discount on goods and services and exemption from the 12% value-added tax (VAT), representing the government's commitment to protecting vulnerable populations and ensuring equitable access to essential commodities and services.

However, the implementation of these mandated discount programs faces a critical crisis that undermines their fundamental purpose. The proliferation of fraudulent identification cards has reached alarming proportions, with Senate investigations revealing that fake PWD IDs outnumber legitimate credentials by nearly 5 to 1 approximately 8.5 million fraudulent IDs circulating compared to only 1.89 million legitimately registered PWDs (Bordey, 2024). This systematic exploitation resulted in ₱88.2 billion in lost government tax revenues in 2023 alone, as calculated by the Bureau of Internal Revenue, which has classified fraudulent PWD ID use as criminal tax evasion (Arceo, 2024).

The business sector bears substantial operational and financial burdens from this widespread fraud. Restaurant operators documented dramatic increases in discount claims from approximately 5% of transactions in 2019 to over 30% by 2024, with industry associations

estimating losses approaching $1 billion in 2023, Empirical evidence from major restaurant chains reveals the severity of verification challenges; one establishment reported rejecting 726 of 4,212 presented PWD IDs (17.24%) over a three-month period due to suspected fraudulent use (Arceo, 2024). Industry-specific data demonstrates operational impact on businesses. More critically, the business sector bears substantial financial burden from fraudulent discount absorption. According to the Restaurant Owners of the Philippines, every fraudulent discount comes directly from restaurant margins, with data showing that of 4,212 PWD cards presented to member establishments over a two-month period in 2024, 17% were rejected as fake or expired, representing a substantial number potentially used successfully elsewhere (Arnaldo, 2024).

These verification failures create unintended consequences for legitimate beneficiaries. Businesses respond to pervasive fraud by implementing stricter verification procedures, developing skepticism toward all discount claimants, and subjecting authentic Senior Citizens and PWDs to uncomfortable interrogations, delayed transactions, and wrongful denials of entitled benefits. The Department of Justice issued clarifications in January 2024 noting that establishments cannot legally withhold privileges even when unable to verify authenticity immediately, as the Department of Health PWD Registry "does not yet contain complete listing" for various reasons (Bautista, 2025). This places businesses in untenable positions legally required to honor potentially fraudulent credentials or face ₱50,000-100,000 fines and 2-6 years imprisonment, yet lacking verification mechanisms distinguishing legitimate from fraudulent claims.

Recognizing these challenges, local government units have initiated various anti-fraud measures. Valenzuela City Government launched the "Bawal ang Fake ID" campaign and introduced the VALID (Valenzuela ID Validation) Portal in May 2024, featuring newly designed ID cards with holographic seals and web-based verification capabilities (City Government of Valenzuela, 2024). Similarly, other municipalities developed dedicated platforms including the OSCA Taguig Senior Citizen ID Verification System and PDAO Taguig PWD ID Checker. At the national level, the Department of Social Welfare and Development announced development of a unified PWD ID system targeted for implementation by 2025 (Serquiña, 2024), while Bacolod City implemented QR-coded credentials to strengthen authentication (Bacolod City Government, 2024).

Despite these initiatives, a critical gap persists in the verification ecosystem. Existing systems require manual data entry consuming 30-45 seconds per transaction, operate as one-way verification tools without mechanisms to automatically report suspicious IDs back to issuing authorities, are geographically restricted to single municipalities preventing cross-jurisdictional verification, and function as isolated solutions rather than integrated networks enabling coordinated fraud intelligence. Contemporary research on digital identity verification demonstrates that integrated systems achieve 70%+ fraud reduction through coordinated intelligence sharing and real-time cross-verification (Protecht Group, 2025), yet no current Philippine platform combines automated data capture, real-time dual-database verification, mobile accessibility for point-of-sale environments, and bidirectional fraud reporting within a unified architecture.

International examples provide proven architectural patterns for addressing these challenges. Estonia's X-Road infrastructure processes over 2.7 billion data queries annually through a decentralized data exchange layer connecting independent agency databases, demonstrating that distributed verification architectures can maintain institutional sovereignty while enabling comprehensive integration (Team, 2026). Research on optical character recognition establishes that contemporary systems achieve 95-99% accuracy on standardized documents, with mobile implementations capable of real-time processing at 62 frames per second on mid-range smartphones (Liu et al., 2022; Gayer et al., 2023). These technological capabilities, when adapted to Philippine implementation contexts, create opportunities for transformative improvements in discount program verification.

Contemporary data privacy legislation establishes beneficiary rights extending beyond passive credential verification. The Data Privacy Act of 2012 (Republic Act 10173, Section 16) grants data subjects the right to reasonable access to their personal information and the ability to dispute inaccuracies or errors in such data. This legal framework, combined with Department of Justice clarifications requiring provisional approval even for unverifiable credentials, creates regulatory expectations that verification systems must enable beneficiary participation in validation processes rather than positioning vulnerable populations as passive subjects of surveillance and control.

Project Context

This study is situated within the broader context of digital transformation efforts in Philippine local governance, specifically addressing the intersection of social welfare service delivery, fraud prevention, and technology-enabled business-government coordination. The project responds to the documented systemic failures in Senior Citizen and PWD discount verification detailed in the preceding section, where conventional manual approaches have proven inadequate in preventing the exploitation crisis.

The Philippine implementation context presents unique challenges absent in international digital identity systems. Unlike Estonia's unified infrastructure with 99% digital ID adoption or Singapore's centralized SingPass platform, Philippine beneficiary data resides in over 1,780 independent local government unit databases with varying digitization levels, technical sophistication, and data quality standards. Research by the University of the Philippines Center for Local and Regional Governance found that only 60% of Philippine LGUs have established operational Persons with Disability Affairs Offices as mandated by RA 10070, implying that 40% of municipalities and cities lack dedicated offices for PWD services, credential issuance, and verification support (Yang, 2018).

This institutional fragmentation creates fundamental verification challenges. Senior Citizen and PWD identification cards lack standardized formats across issuing authorities, with designs varying by municipality (Respicio Law, 2024). While the National Council on Disability Affairs announced a unified PWD ID design in January 2024, actual implementation across all

LGUs remains incomplete. Consequently, establishments cannot authenticate credentials through visual security inspection alone, and forged IDs remain difficult to detect without database verification yet database fragmentation prevents comprehensive cross-jurisdictional validation.

The technological landscape of existing verification solutions reveals substantial functional gaps. Local platforms (OSCA Taguig, PDAO Taguig, VALID Valenzuela) provide real-time database connectivity but remain fundamentally limited by the characteristics described in Section 1.1: manual data entry requirements, geographic restriction to single municipalities, lack of optical character recognition automation, and operation as one-way verification tools without bidirectional fraud reporting. National systems (eVerify) focus on PhilSys credentials rather than LGU-issued Senior Citizen and PWD IDs, creating scope misalignment with discount verification requirements. International commercial solutions (Jumio, VeriScan, BlinkID) demonstrate advanced technical capabilities but lack integration with local OSCA and PDAO databases and absence of fraud reporting mechanisms enabling coordination between businesses and government authorities.

Within this context, iDetect represents a response grounded in both proven technological approaches and careful adaptation to Philippine institutional realities. The system will address verification challenges through integrated multi-portal architecture synthesizing automated OCR-based data extraction, real-time cross-database verification with both OSCA and PDAO, mobile-first design optimized for point-of-sale transaction speed, and bidirectional fraud reporting creating intelligence-sharing networks between businesses and government enforcement authorities. This comprehensive approach distinguishes iDetect from existing

fragmented solutions, positioning it as a scalable model for strengthening social welfare program integrity while protecting legitimate beneficiaries' statutory rights.

These contextual realities, institutional fragmentation, technological gaps, and adaptation requirements frame the specific problems that iDetect addresses, as detailed in the following section.

Problem Statement

The continued misuse of fake Senior Citizen and PWD identification cards has created a multi-dimensional crisis affecting government revenues, business operations, and legitimate beneficiaries' access to statutory entitlements. As documented in Section 1.1, the proliferation of fraudulent credentials has reached unprecedented proportions, with 8.5 million fake PWD IDs circulating compared to only 1.89 million legitimate registrations a 4.5:1 ratio that fundamentally undermines the integrity of government-mandated discount programs under Republic Acts 9994 and 10754.

The economic consequences are severe and measurable. Government tax revenues lost ₱88.2 billion in 2023 due to fraudulent discount transactions, while businesses absorbed ₱166.1 billion in fraudulent discounts exceeding the ₱42.3 billion provided to legitimate beneficiaries (Bordey, 2024). One major restaurant chain's experience exemplifies operational impact, rejecting 17.24% of presented PWD IDs as suspected fraudulent over a three-month period, representing nearly one in five credentials failing basic authenticity checks (Arceo, 2024). These verification failures translate to substantial financial losses for establishments: the combined

20% discount and 12% VAT exemption means businesses lose approximately 32% of transaction value when fraudulent IDs succeed.

Current verification approaches have proven inadequate in addressing this crisis. Existing platforms including OSCA Taguig Senior Citizen ID Verification System, PDAO Taguig PWD ID Checker, and Valenzuela City's VALID Portal demonstrate four critical limitations that prevent comprehensive fraud prevention:

Manual data entry requirements consuming 30-45 seconds per transaction create operational bottlenecks in high-volume business environments, where verification delays accumulate to cause customer queue extensions and workflow disruptions.

Geographic restrictions limiting verification to single municipalities create problems for businesses serving broader populations, requiring multiple platform usage for comprehensive coverage.

Absence of optical character recognition automation forces staff to manually type ID details from physical cards, introducing transcription errors and consuming time that compounds across thousands of daily transactions.

One-way verification architecture represents the most critical limitation establishments can query databases to check credentials, but no automated mechanisms exist to report suspicious IDs back to issuing authorities, creating a fundamental disconnect in the fraud prevention ecosystem.

This fragmented approach produces cascading consequences across stakeholder groups. Businesses face untenable legal positions. Department of Justice clarifications prohibit denying discounts even for unverifiable credentials, yet establishments absorb fraudulent costs approaching $1 billion annually while risking ₱50,000-100,000 fines for wrongful denial. Local government units lack fraud intelligence necessary for targeted investigations and enforcement, as suspicious credentials identified by establishments during transactions are not systematically reported or tracked. Legitimate Senior Citizens and PWDs experience prolonged verification scrutiny, inconsistent treatment across establishments, and wrongful denials as businesses develop skepticism toward all discount claimants. Government revenues continue declining as systematic fraud exploitation persists unchecked.

The core problem is the absence of an integrated verification system combining six essential capabilities: (1) automated data capture through OCR technology eliminating manual entry bottlenecks, (2) real-time cross-database validation querying both OSCA and PDAO records simultaneously, (3) mobile accessibility optimized for point-of-sale use without requiring desktop access, (4) bidirectional fraud reporting enabling suspicious credentials to be automatically flagged to appropriate government authorities, (5) coordinated intelligence sharing creating fraud prevention networks rather than isolated verification points, and (6) beneficiary accountability mechanisms enabling Senior Citizens and PWDs to verify their own records, dispute erroneous denials, and report unauthorized use of their credentials. Without such comprehensive integration, Valenzuela City continues experiencing escalating fraud incidents,

financial losses for establishments, incomplete fraud intelligence for local government units, and operational barriers affecting all stakeholders in the verification ecosystem.

A critical accountability gap exists when verification systems generate false positives; legitimate beneficiaries wrongly flagged as fraudulent due to database errors, OCR misreads, or outdated records. Current platforms provide no mechanism for beneficiaries to view their verification status, understand denial reasons, or dispute incorrect flags. This creates a power imbalance where vulnerable populations (Senior Citizen and PWDs) face system-generated barriers to statutory benefits without recourse, while businesses and government agencies possess complete information asymmetry. The Data Privacy Act (Republic Act 10173, Section 16) grants data subjects the right to access and correct their personal information, yet existing verification systems do not provide beneficiary-facing interfaces enabling exercise of these rights. Without such accountability mechanisms, false denials disproportionately harm legitimate beneficiaries, erode trust in discount programs, and perpetuate the very stigmatization that fraud prevention aims to address. Without such comprehensive integration addressing both fraud prevention and beneficiary rights protection, Valenzuela City continues experiencing escalating fraud incidents, financial losses for establishment, incomplete fraud intelligence for local government units, wrongful denials affecting legitimate beneficiaries, and operational barriers affecting all stakeholders in the verification ecosystem.

General Objective

To design, develop, and implement iDetect, an integrated OCR-based identification verification system that will connect food service establishments in Valenzuela City with government databases (OSCA and PDAO) through a multi-portal architecture enabling efficient, accurate, and automated validation of Senior Citizen and PWD identification cards. The system will provide establishments with a reliable, secure, and mobile-accessible platform for verifying credentials, preventing fraud through coordinated intelligence sharing, and facilitating real-time bidirectional communication with local government units to ensure the integrity of government-mandated discount programs under Republic Acts 9994 and 10754.

The system will operate through **five integrated portals** serving distinct stakeholder groups: (1) a **Client Portal** (mobile application) enabling business establishments to perform automated OCR-based verification with real-time database cross-checking, (2) a **BPLO Portal** (web-based) enabling the Business Permit and Licensing Office to manage establishment accounts and monitor technical operations without accessing sensitive beneficiary data, (3) an **OSCA Portal** (web-based) enabling the Office of Senior Citizens Affairs to receive flagged Senior Citizen IDs, investigate suspicious records, and update official databases, and (4) a **PDAO Portal** (web-based) providing parallel functionality for Persons with Disability Affairs Office to manage PWD credential verification and fraud investigation, and (5) a **Beneficiary Mobile Application** enabling Senior Citizens and PWDs to verify their ID status in real-time, receive transaction notifications, dispute erroneous denials through structured review processes managed by OSCA/PDAO staff, and report unauthorized use of their credentials, Through this

comprehensive approach, iDetect aims to reduce fraudulent credential acceptance, accelerate verification processing time, strengthen business-government coordination, protect legitimate beneficiaries from wrongful denial through dispute mechanisms, ensure Data Privacy Act compliance by enabling beneficiary data access rights, and support equitable implementation of discount programs across Valenzuela City.

Specific Objectives

This study aims to achieve the following objectives across three development phases:

A. Design Objectives

The system will design:

- **Multi-portal architecture** integrating Client (mobile), BPLO (web), OSCA (web), and PDAO (web) portals with role-based access control and strict data compartmentalization between Senior Citizen and PWD databases.
- **OCR-based verification workflow** optimizing mobile camera capture with real-time image quality assessment, automated preprocessing (deskewing, contrast enhancement, noise reduction), and instant visual feedback guiding users toward optimal ID positioning and lighting conditions.
- **Bidirectional fraud intelligence system** enabling one-tap suspicious credential flagging by establishments with automatic routing to appropriate LGU portals (OSCA for Senior

Citizen IDs, PDAO for PWD IDs), complete transaction context transmission, and investigation workflows with fraud confirmation synchronization across all system users.

- **Beneficiary accountability mechanisms** providing Senior Citizens and PWDs with mobile application access to verification status monitoring, real-time transaction notifications, dispute submission with supporting documentation upload, and resolution tracking, implementing Republic Act 10173 data privacy compliance through access, correction, and transparency rights.
- **Friendly-accessible interfaces** employ large touch targets, high-contrast color schemes, bilingual support (English/Filipino), minimal text input requirements, and progressive disclosure of complex features while maintaining WCAG 2.1 AA accessibility standards.
- **Fraud analytics dashboards** visualizing temporal patterns, geographic distribution, establishment-specific trends, and flagged-to-confirmed conversion rates supporting evidence-based LGU enforcement strategies and resource allocation decisions.

B. Development Objectives

The system will develop:

- **Tesseract OCR integration** with field-level extraction of ID number, name, beneficiary type, validity period, and issuing office, implementing confidence scoring determining automatic processing versus manual review thresholds, and preprocessing pipeline maximizing extraction accuracy despite non-standardized Philippine LGU credential formats.

- **Real-time dual-database verification engine** querying OSCA and PDAO databases simultaneously through secure RESTful API integration, returning binary verification status (VALID, SUSPICIOUS, INVALID) without exposing comprehensive personal data to establishments, and completing verification cycles in under 30 seconds from capture to result display.
- **Automated fraud reporting pipeline** transmitting suspicious credential flags from Client Portal to appropriate LGU portals with captured ID image, extracted data, establishment identification, timestamp, and verification outcome, enabling OSCA/PDAO administrators to mark confirmed fraudulent credentials preventing future successful verification across the network.
- **Cross-establishment fraud prevention synchronization** propagating database status updates in real-time where fraudulent credential marking by one LGU administrator immediately reflects in all Client Portal verification attempts at other establishments, transforming isolated detection into coordinated network response.
- **Beneficiary dispute resolution workflow** enabling structured report submission with transaction reference numbers, pre-populated establishment details, denial reason selection, optional supporting documentation (ID photos, barangay certifications), automatic routing to investigation queues, status tracking with OSCA/PDAO staff responses, and resolution notification delivery via SMS and push notifications.
- **Mobile-first responsive architecture** supporting Android 8.0+ and iOS 12.0+ with offline capability caching recently verified credentials, transaction history viewing when

network unavailable, and automatic synchronization upon connectivity restoration, while SMS transaction notifications function independently without smartphone data access.

- **Role-based authentication and authorization system** implementing two-factor authentication for beneficiaries (ID credentials + mobile OTP), business permit validation for establishment accounts, and administrative credential verification for BPLO/OSCA/PDAO personnel with session management and password policies appropriate for each user category.
- **Comprehensive audit logging infrastructure** recording all verification attempts, fraud flags, database modifications, dispute submissions, administrative actions, and system access with immutable timestamps, user attribution, IP addresses, and complete transaction context supporting compliance investigations and security incident response.

C. Evaluation Objectives

The IDETECT system will be evaluated against the following ISO 25010 quality characteristics:

1. Functional Suitability

- *Functional Completeness:* All specified user functions for verification, management, and reporting are implemented.
- *Functional Correctness:* OCR accuracy, database cross-checking, and discount calculations produce correct results.
- *Functional Appropriateness:* Features align with stakeholder needs and regulatory compliance requirements.

2. Performance Efficiency

- *Time Behavior*: Verification transactions complete in under 30 seconds; system response time meets real-time requirements.
- *Resource Utilization:* Optimal use of mobile device resources, server capacity, and network bandwidth.
- *Capacity:* System handles concurrent users across multiple establishments without degradation.

3. Compatibility

- *Co-existence:* System operates alongside existing LGU databases and business systems.
- *Interoperability:* Seamless integration with OSCA and PDAO databases; API compatibility for future extensions.

4. Usability

- *Appropriateness Recognizability:* Users can readily understand system capabilities and functions.
- *Learnability*: New users achieve proficiency in verification processes with minimal training.
- *Operability:* Interface design supports efficient task completion for all user roles.
- *User Error Protection:* System prevents incorrect operations and provides clear guidance.

- *User Interface Aesthetics:* Visual design is professional, accessible, and culturally appropriate.
- *Accessibility:* Interface accommodates users with varying technical literacy and abilities.

5. Reliability

- *Maturity*: System operates without frequent failures during normal operation.
- *Availability*: System maintains 99% uptime during business hours.
- *Fault Tolerance:* Graceful handling of network interruptions, poor image quality, and database connection issues.
- *Recoverability*: Data integrity maintained and quick recovery from failures.

6. Security

- *Confidentiality*: Beneficiary personal information protected from unauthorized access.
- *Integrity*: Verification data and audit logs cannot be tampered with.
- *Non-repudiation:* All transactions are logged with complete traceability.
- *Accountability:* User actions are tracked and attributable to specific accounts.
- *Authenticity:* User identities and establishment credentials are verified.

7. Maintainability

- *Modularity*: System components can be modified independently.
- *Reusability*: Code components can be reused for future enhancements.
- *Analyzability*: System logs and diagnostics support troubleshooting.

- *Modifiability*: System can be updated to accommodate new requirements or regulations.
- *Testability*: Comprehensive testing can be performed on all system functions.

8. Portability

- *Adaptability*: System functions across different mobile devices and operating systems (Android, iOS).
- *Installability*: Easy deployment to business establishments and government offices.
- *Replaceability*: System can integrate with or replace existing manual verification processes.

Purpose and Description

The purpose of this study is to design, develop, and implement iDetect, an integrated OCR-based identification verification system that will enable food service establishments in Valenzuela City to efficiently and accurately authenticate Senior Citizen and PWD identification cards while creating coordinated fraud prevention networks between businesses and local government units. As established in the problem statement, the project addresses the documented crisis where fake IDs outnumber legitimate credentials 4.5 to 1.

iDetect responds to fundamental limitations in existing verification approaches by integrating five capabilities absent in current fragmented solutions: automated data extraction through Optical Character Recognition technology, real-time cross-database validation querying both OSCA and PDAO records simultaneously, mobile-first architecture optimized for point-of-sale transaction speed, bidirectional fraud reporting enabling automatic flagging to

appropriate government authorities, and coordinated intelligence sharing transforming isolated verification points into networked fraud prevention ecosystem.

The system comprises four interconnected portals serving distinct stakeholder groups with appropriate functional boundaries and data access controls:

Client Portal (Mobile Application) enables food service establishment staff to capture ID images through smartphone cameras, automatically extract credential data via Tesseract OCR with field-level validation, verify authenticity in real-time against authoritative government databases, receive instant verification results (VALID, SUSPICIOUS, or INVALID) with clear visual indicators, flag suspicious credentials for automatic transmission to appropriate LGU offices, access transaction history for audit purposes, and manually input ID details when OCR confidence is insufficient. The mobile-first design eliminates desktop access requirements, enabling verification directly at point of sale without workflow disruption.

BPLO Portal (Web-based) enables the Business Permit and Licensing Office to verify and approve legitimate establishment accounts, manage business access including approval, suspension, and revocation capabilities, monitor technical performance metrics, coordinate technical issue resolution, and access operational dashboards. Critical functional boundary: BPLO maintains administrative oversight without accessing flagged ID records or personal beneficiary information, ensuring clear separation between business account management and verification intelligence functions.

OSCA Portal (Web-based) enables the Office of Senior Citizens Affairs to manage the Senior Citizen ID database, receive real-time notifications of flagged credentials with complete transaction context, review and investigate suspicious credentials, mark confirmed fraudulent IDs preventing future successful use, and generate fraud analytics reports including temporal patterns and geographic distribution. Strict data compartmentalization ensures OSCA accesses only Senior Citizen data without visibility into PWD records.

PDAO Portal (Web-based) provides parallel functionality for the Persons with Disability Affairs Office with identical capabilities for PWD credential management: database administration, real-time flagged ID notifications, investigation workflows, fraudulent credential marking, and analytics generation. Data compartmentalization ensures PDAO accesses only PWD data without visibility into Senior Citizen records.

Beneficiary Mobile Application (Mobile Application for iOS and Android) enables Senior Citizens and PWDs to exercise Data Privacy Act rights and participate actively in verification processes rather than serving as passive subjects of system decisions. The application addresses the power imbalance inherent in one-way verification architectures by providing four core accountability capabilities.

Real-Time Transaction Transparency upon each verification attempt at participating establishments, beneficiaries receive immediate notifications via SMS and push notifications detailing business name, transaction timestamp, verification outcome (Approved or Denied), and specific denial reasons when applicable such as ID number mismatch, database record not found,

or ID expired. This transparency mechanism enables beneficiaries to detect unauthorized credential use within minutes rather than discovering fraud through unexpected service denials, and allows legitimate beneficiaries to immediately understand and contest system decisions affecting their statutory rights.

Self-Service Status Verification. Beneficiaries authenticate using their ID number paired with their birthdate — matching information already printed on physical credentials — verified through a one-time PIN sent to their registered mobile number. Upon authentication, the application displays current ID status synchronized with OSCA/PDAO master databases, last successful verification details, upcoming expiration dates, and issuing office contact information. This capability allows beneficiaries to proactively verify their standing before attempting discount transactions, reducing anxiety associated with potential denial and enabling advance correction of database discrepancies.

Structured Dispute Resolution. When beneficiaries receive denial notifications for transactions they believe should succeed, the application provides a standardized dispute mechanism: (1) select the flagged transaction from notification history with pre-populated business details and denial reason, (2) choose a dispute category — Wrong Denial (I Am Valid), ID Used Without My Authorization, Business Refused Provisional Approval, or Database Contains Incorrect Information, (3) provide a brief written explanation, (4) optionally upload supporting documentation such as a physical ID photo or barangay certification, and (5) submit directly to the appropriate OSCA or PDAO portal for staff review. The system enforces service-level agreements: automatic acknowledgment notification within 24 hours, status updates as

OSCA/PDAO staff review evidence, and final resolution notification with explanation upon case closure. All disputes are reviewed by OSCA or PDAO personnel — not automated systems — ensuring human judgment and institutional authority govern accountability processes. This workflow transforms opaque denial processes into transparent, accountable review mechanisms where beneficiaries can challenge system errors with documented evidence and guaranteed response timelines.

Bidirectional Fraud Reporting. The application inverts traditional fraud detection by enabling beneficiaries to report incidents that businesses and administrative systems cannot independently detect: (1) unauthorized use alerts when their ID is used at establishments they did not visit, prompting immediate flagging to OSCA/PDAO if confirmed unauthorized, (2) business refusal reports when establishments deny discounts despite system approval, documenting potential DOJ Opinion violations for BPLO investigation, and (3) database discrepancy alerts when beneficiaries notice incorrect personal information in their displayed records, triggering OSCA/PDAO verification and correction processes. These beneficiary-initiated reports create upward accountability flows contributing fraud intelligence that businesses and government agencies cannot independently generate.

The integrated architecture creates a bidirectional accountability ecosystem; transforming one-way surveillance into mutual verification where businesses verify beneficiaries through OCR, beneficiaries verify system accuracy through dispute mechanisms, and OSCA/PDAO mediates conflicts with transparent, documented processes — distinguishing iDetect from existing one-way verification tools. When establishments flag suspicious

credentials during transactions, the system automatically routes these flags to appropriate LGU portals (OSCA for Senior Citizen IDs, PDAO for PWD IDs) with complete transaction context. LGU administrators investigate flagged credentials, update official verification status, and mark confirmed fraudulent IDs in master databases. These database updates propagate in real-time to all system users when one establishment's flag results in official fraudulent marking, subsequent verification attempts at other establishments immediately reflect this updated status, preventing continued fraudulent use. This bidirectional synchronization transforms passive verification into coordinated intelligence sharing, enabling systematic response to fraud rather than isolated detection.

Technical implementation leverages proven approaches adapted to Philippine contexts. Tesseract OCR engine processes captured ID images with preprocessing including contrast enhancement, noise reduction, and deskewing to maximize extraction accuracy despite non-standardized credential formats. Field-level validation applies pattern recognition to verify extracted data conforms to expected formats (ID number structures, date ranges, issuing office codes), with confidence scoring determining whether automatic processing proceeds or manual review becomes necessary. Real-time database queries utilize secure API integration with OSCA and PDAO systems, implementing privacy-preserving designs returning binary verification status without exposing comprehensive personal data to businesses. Mobile application architecture follows three-action limit principles (launch, scan, view) documented in point-of-sale usability research, eliminating unnecessary authentication dialogs or navigation steps that increase adoption resistance.

iDetect is designed to reinforce the integrity of government-mandated discount programs under Republic Acts 9994 and 10754 while protecting legitimate beneficiaries from undue scrutiny. By reducing verification time from 45-60 seconds (manual entry) to under 30 seconds (automated OCR), the system eliminates customer queue delays that create pressure for rushed, inadequate verification. Automated flagging removes confrontational aspects of fraud detection, suspicious credentials are reported to authorities without requiring establishment staff to make accusatory determinations during customer interactions. Consistent verification treatment across participating establishments ensures that legitimate Senior Citizens and PWDs receive entitled benefits uniformly without arbitrary denial resulting from individual establishment skepticism.

The following section examines the significance of addressing these verification challenges for multiple stakeholder groups and broader Philippine digital governance objectives.

Significance of the Study

This study holds significant value for multiple stakeholder groups affected by Senior Citizen and PWD discount verification challenges, with implications extending beyond immediate operational improvements to broader questions of social welfare program integrity, digital governance, and technology-enabled public service delivery.

For Business Establishments

Food service operators in Valenzuela City will gain immediate operational benefits addressing documented verification burdens. The system will reduce average verification time

from 45-60 seconds (manual entry in existing platforms) to under 30 seconds (automated OCR), enabling establishments to maintain customer service quality during peak hours without extended queue delays.

By providing reliable fraud detection mechanisms, the system will help establishments reduce fraudulent discount absorption currently estimated at ₱166.1 billion industry-wide in 2023, representing substantial financial burden particularly for small and medium enterprises operating on narrow profit margins. The automated bidirectional reporting architecture will remove confrontational aspects of fraud detection, as suspicious credentials are flagged to authorities through system mechanisms rather than requiring staff to make accusatory determinations during customer interactions.

This protects establishments from legal liability while ensuring compliance with Department of Justice clarifications requiring discount provision even for unverifiable credentials.

For Local Government Units

Valenzuela City's Office of Senior Citizens Affairs and Persons with Disability Affairs Office will gain fraud intelligence capabilities absent in current systems. The bidirectional reporting mechanism will automatically transmit suspicious credentials detected by establishments to appropriate LGU portals with complete transaction context, enabling targeted investigations that would be impossible without coordinated intelligence sharing.

LGU administrators will gain comprehensive fraud analytics revealing patterns across establishments, temporal trends, and geographic distributions actionable intelligence supporting evidence-based enforcement strategies and resource allocation decisions. The system will strengthen database accuracy through continuous validation, as discrepancies detected during verification processes are systematically reported back to issuing authorities for investigation and correction.

This closed-loop feedback transforms static databases into dynamically maintained systems reflecting current credential status.

For the Business Permit and Licensing Office

The BPLO Portal will provide regulatory oversight capabilities ensuring system integrity while maintaining appropriate functional boundaries. BPLO administrators will be able to verify establishment legitimacy, manage system access based on business permit status, monitor technical performance metrics, and ensure compliance with usage policies all without accessing sensitive beneficiary data or verification intelligence.

This separation of concerns enables effective regulatory oversight while protecting privacy and maintaining institutional sovereignty of OSCA and PDAO over beneficiary information.

For Legitimate Senior Citizens and Persons with Disabilities

iDetect's significance extends beyond convenience to fundamental dignity and rights protection. Faster verification processes eliminate prolonged scrutiny that many beneficiaries find embarrassing or demeaning, while consistent treatment across participating establishments ensures uniform benefit access without arbitrary denial from individual establishment skepticism. The reduction in fraudulent credential circulation protects legitimate beneficiaries from discriminatory responses—businesses becoming "wary of honoring discounts because of fake IDs in circulation" (Luna, 2020)—helping restore trust in discount programs and benefiting those whom the legislation was designed to protect.

Beyond operational improvements, iDetect addresses rights protections mandated by Republic Act 10173 (Data Privacy Act of 2012, Section 16), which establishes data subjects' rights to reasonable access to personal information, dispute inaccuracies, and receive transparency regarding decisions affecting statutory entitlements. Existing verification systems systematically fail these provisions by denying beneficiaries access to verification records and providing no mechanisms to dispute database errors. iDetect establishes compliance infrastructure enabling these rights: beneficiaries can view verification status and transaction history (satisfying access rights), dispute denials through structured OSCA/PDAO review processes (satisfying correction rights), and receive documented explanations (satisfying transparency principles). When legitimate beneficiaries experience wrongful denials due to database typos, OCR misreads, or administrative delays, the ability to dispute errors with supporting evidence and receive timely responses transforms frustrating powerlessness into

procedural fairness, demonstrating that technology-enabled verification serves all stakeholders in the discount program ecosystem.

For Government Revenue Protection

The potential economic impact is substantial. As documented in Section 1.1, Senate investigations revealed ₱88.2 billion in lost tax revenues during 2023 due to fraudulent discount transactions. Even achieving a 50% reduction in fraudulent credential acceptance substantially below the 70%+ fraud reduction documented in integrated digital verification systems internationally would recover ₱44 billion annually in tax revenues while reducing business discount absorption by ₱83 billion.

These savings could be redirected toward enhancing services for legitimate beneficiaries, improving social welfare program sustainability, or funding other critical government priorities.

For Information Technology and Computer Science Research

This study will contribute empirical evidence addressing gaps in existing literature regarding OCR-based verification in non-standardized identification ecosystems, mobile point-of-sale system usability in Philippine contexts, fraud detection patterns specific to discount programs rather than financial transactions, and public-private partnership implementation at micro-enterprise scale.

The project demonstrates practical application of Routine Activity Theory to social welfare fraud prevention, showing how technology-enabled "capable guardianship" can address

systematic exploitation. Research on data quality frameworks finds practical testing ground in examining how accuracy, completeness, timeliness, consistency, and accessibility dimensions affect verification system trustworthiness and adoption.

For Philippine Digital Governance and Public Administration

iDetect provides a scalable model for LGU-business coordination addressing social service delivery challenges through integrated technology. The distributed architecture respects institutional sovereignty while enabling integration, a critical design principle for Philippine contexts where fragmentation across 1,780+ independent LGUs prevents centralized solutions.

The graduated deployment approach beginning with digitized urban LGUs while building capacity in resource-constrained municipalities demonstrates how digital transformation can proceed incrementally, delivering immediate value while working toward comprehensive coverage. Success in Valenzuela City could inform replication strategies for other municipalities nationwide, contributing to broader digitalization objectives articulated in national development plans.

For Academic Institutions

For Pamantasan ng Lungsod ng Valenzuela's College of Engineering and Information Technology, this capstone project demonstrates applied research addressing real-world problems affecting the local community. The integration of theoretical frameworks (Routine Activity Theory, Data Quality Framework) with practical system development illustrates how academic

training prepares students for technology leadership roles solving contemporary societal challenges.

The project's documentation contributes to institutional knowledge supporting future capstone endeavors in public sector technology applications.

For Future Researchers and Developers

This study provides comprehensive documentation of design decisions, implementation challenges, stakeholder coordination strategies, and evaluation methodologies applicable to similar verification systems. The project identifies research opportunities including investigation of behavioral biometrics for continuous authentication, machine learning approaches to pattern recognition in discount fraud, blockchain applications for credential verification with privacy preservation, and comparative analysis of verification system effectiveness across different Philippine LGU contexts with varying institutional capacities.

Scope and Limitation

This study focuses on the design, development, and implementation of iDetect, an OCR-based identification verification system for Senior Citizen and PWD credentials in food service establishments within Valenzuela City.

Geographic and Jurisdictional Scope: The system verifies Valenzuela City-issued Senior Citizen and PWD identification cards, targeting food service establishments within municipal boundaries with direct connection to OSCA and PDAO databases. While the

architecture supports future multi-municipal expansion, initial implementation maintains single-LGU focus for manageable deployment and thorough validation.

Business Establishment Scope: The system targets commercial food service establishments (restaurants, cafes, fast-food chains) selected based on high transaction volumes, substantial fraud exposure (17.24% PWD ID rejection rate documented in Section 1.1), and operational characteristics representative of high-volume retail environments.

Functional Scope - Client Portal: The mobile application enables establishment staff to capture ID images, automatically extract credential data through Tesseract OCR with image preprocessing, verify details in real-time against OSCA/PDAO databases, receive verification results (VALID, SUSPICIOUS, or INVALID), flag suspicious credentials for automatic LGU transmission, access transaction history, and generate compliance reports. Manual data entry serves as fallback when OCR confidence is insufficient.

Functional Scope - BPLO Portal: The Business Permit and Licensing Office portal enables administrators to verify establishment registrations, manage business access, monitor technical performance metrics, and coordinate system updates. BPLO administrators do not access flagged ID records or beneficiary information, maintaining separation between business account management and verification intelligence.

Functional Scope - OSCA Portal: The Office of Senior Citizens Affairs portal enables administrators to manage the Senior Citizen ID database, receive real-time notifications of flagged credentials with transaction context, review and investigate suspicious IDs, mark

confirmed fraudulent IDs to prevent future verification, and generate fraud analytics reports. Data compartmentalization ensures OSCA accesses only Senior Citizen records.

Functional Scope - PDAO Portal: The Persons with Disability Affairs Office portal provides parallel functionality for PWD credential management with identical capabilities, maintaining data compartmentalization to access only PWD records.

Functional Scope - Beneficiary Mobile Application: The Beneficiary Mobile Application enables Senior Citizens and PWDs to register accounts using ID number and birthdate authentication verified against OSCA/PDAO databases with mobile PIN confirmation, view current ID verification status (Active, Under Review, Suspended, Expired) synchronized with government databases, receive real-time transaction notifications via SMS and push notifications when establishments verify credentials, dispute erroneous denials by submitting structured reports to OSCA/PDAO with optional supporting documentation, track dispute resolution progress with staff responses and outcome notifications, access transaction history showing verification attempts with establishment details, and report suspected unauthorized credential use through fraud alert mechanisms. Authentication employs two-factor design: primary credentials (ID number and birthdate) verified against database records, with secondary verification (mobile number and OTP) linking accounts to beneficiaries' registered devices, balancing security with accessibility for elderly populations who may have difficulty managing complex password systems.

Technical Scope: The system implements Tesseract OCR engine with preprocessing algorithms, RESTful API architecture for real-time database integration, Android and iOS mobile application support, and responsive web portals. Security features include secure authentication, encrypted data transmission, and role-based access controls.

Data Scope: Verification processes access OSCA and PDAO databases containing beneficiary information (ID numbers, names, validity periods, issuing office details). The system returns binary verification status without exposing comprehensive personal data, implementing privacy-preserving design. Flagged ID reports include extracted credential data, establishment identification, timestamps, and verification outcomes.

Geographic and Credential Limitations: The system verifies only Valenzuela City-issued credentials. IDs from other municipalities are not supported due to independent LGU database systems, though architectural design enables future federation.

Business Type Limitations: The system is designed specifically for food service establishments. Other business types (retail stores, pharmacies, transportation, healthcare facilities) are excluded to enable focused optimization for food service workflows.

Database Dependency Limitations: Verification accuracy depends on the completeness, currency, and accessibility of OSCA and PDAO databases. Outdated records, incomplete registrations, or system downtime directly impact verification reliability.

Connectivity Requirements: iDetect requires stable internet connectivity for real-time verification. Extended connectivity loss prevents database queries and automated flag transmission, reflecting architectural prioritization of real-time verification over offline capability.

Verification Authority Limitations: The system verifies existing identification cards but does not determine beneficiary eligibility, issue new IDs, or modify ID status. Final validation decisions rest with authorized LGU personnel through OSCA and PDAO portals.

OCR Accuracy Limitations: OCR performance may be affected by poor image quality, damaged cards, unclear text, non-standard formats, poor lighting, and image blur. Manual data entry serves as fallback when OCR confidence falls below acceptable thresholds.

Beneficiary Application Limitations and Dependencies: The application displays verification status and transaction outcomes without exposing comprehensive personal data beyond physical credential information (ID number, name, type, dates), implementing minimal data exposure while enabling dispute transparency. Final dispute adjudication rests with OSCA/PDAO personnel who investigate evidence and update official records; the application enables submission and tracking but not direct database modification. Service-level agreements establish 24-hour acknowledgment and 48-72 hour resolution for database errors, with complex cases extending timelines with explanatory notifications. Internet connectivity is required for real-time functions, though Capstone 2 implements offline caching for viewing previously loaded information; SMS notifications function independently without data access. The

application supports Android 8.0+ and iOS 12.0+, while OSCA/PDAO offices provide assisted dispute submission ensuring universal access for beneficiaries without compatible devices or digital literacy. The application is optional rather than mandatory—businesses provide discounts regardless of beneficiary registration, ensuring no new technological barriers for vulnerable populations.

Biometric Verification Exclusion: The system does not incorporate biometric verification (facial recognition, fingerprint matching, iris scanning). Verification relies on database record matching and pattern anomaly detection. This reflects current infrastructure limitations, privacy concerns, and system complexity considerations.

Training and User Competency Dependencies: System effectiveness depends on proper training of establishment staff in ID capture procedures, image quality assessment, and verification result interpretation. The system provides user interface guidance but cannot fully compensate for insufficient training or deliberate misuse.

Legal and Enforcement Limitations: iDetect does not alter existing legal frameworks under Republic Acts 9994 and 10754. Department of Justice clarifications prohibiting discount denial for unverifiable credentials remain in effect (Bautista, 2025). The system operates as an intelligence-gathering mechanism rather than real-time gatekeeping, with establishments continuing to provide discounts during verification processing.

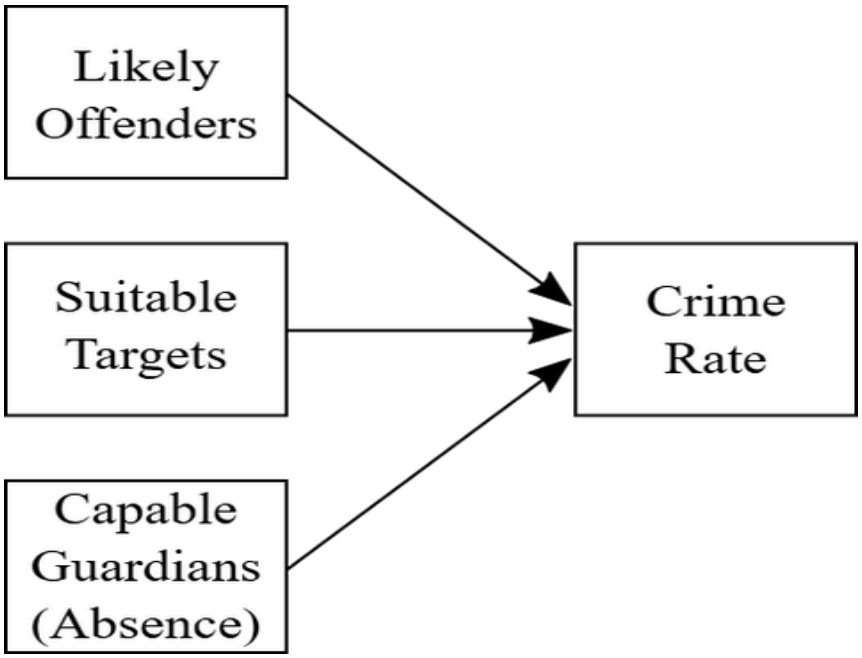
These scope parameters and delimitations establish realistic boundaries for system development, deployment, and evaluation within capstone project constraints, while identifying areas for future enhancement as technical capabilities and institutional infrastructure evolve.

Theoretical Framework

The design and development of iDetect is anchored in two complementary theoretical frameworks that provide conceptual foundation for understanding fraud prevention mechanisms and ensuring system reliability in the context of Senior Citizen and PWD identification verification. These frameworks Routine Activity Theory and Fraud Management Lifecycle Theory offer distinct but interconnected perspectives on how technology-enabled verification systems can effectively address systematic fraud exploitation while maintaining trustworthiness and user acceptance.

Figure 1.

Routine Activity Theory (Cohen & Felson, 1979)



Routine Activity Theory (RAT), originally developed by Cohen and Felson (1979) in criminology, posits that crime occurs when three elements converge in time and space: (1) a motivated offender willing to commit fraud, (2) a suitable target that can be exploited, and (3) the absence of a capable guardian providing effective oversight or protection. According to RAT, crime prevention is most effective when interventions focus on reducing target suitability and increasing capable guardianship rather than solely addressing offender motivation, as altering criminal inclinations is substantially more difficult than modifying situational opportunities for crime commission.

Application to Senior Citizen and PWD ID Fraud Context:

In the context of discount program fraud in Valenzuela City, all three RAT elements are demonstrably present. Motivated offenders exist in substantial numbers as documented in Section 1.1, Senate investigations revealed 8.5 million individuals using fraudulent PWD IDs compared to 1.89 million legitimate registrations, indicating large populations seeking financial gain through credential fraud. Reports indicate that the combined 20 % discount and 12 % VAT exemption can represent a 32 % reduction in transaction value, and many establishments particularly in the restaurant sector have reported significant increases in sales attributed to PWD customers, rising from about 5 % to around 30 %, much of which is suspected to involve fraudulent use of PWD IDs (Barcelonia, 2024).

Suitable targets manifest as business establishments with limited capacity to verify credential authenticity. The current manual verification process consuming 45-60 seconds per transaction creates vulnerabilities that offenders exploit during peak operational hours when thoroughness conflicts with customer service demands. The lack of standardized ID formats across municipalities and absence of real-time verification tools further increase target suitability, making establishments easy prey for fraudsters who understand that overwhelmed staff facing long customer queues are unlikely to conduct rigorous authentication.

Most critically, the absence of capable guardianship represents the fundamental weakness enabling systematic fraud. Existing verification systems function as weak guardians requiring manual data entry, lacking OCR automation, operating as one-way verification tools, and providing no mechanisms for reporting suspicious IDs back to authorities. This fragmented guardianship allows fraudulent credentials to circulate undetected across multiple establishments,

as coordinated intelligence sharing between businesses and government agencies does not exist systematically.

iDetect as Capable Guardian:

iDetect directly addresses the "absence of capable guardianship" by introducing technologically-enabled, multi-layered guardian mechanisms operating through three distinct channels:

Automated Surveillance: OCR technology combined with real-time database verification creates constant, automated surveillance of credential authenticity. Unlike human-dependent manual verification subject to fatigue, distraction, and varying diligence, this guardian operates consistently and tirelessly, examining every presented ID with 95% accuracy in under 30 seconds. The automation eliminates human judgment variability the system applies identical verification standards regardless of time pressure, customer behavior, or staff experience level.

Real-Time Intelligence: By cross-referencing credentials against authoritative OSCA and PDAO databases in real-time, iDetect transforms passive verification into active fraud detection. The system immediately identifies mismatches between presented credentials and official records, inconsistencies in extracted data, or credentials flagged previously by other establishments. This intelligence capability effectively blocks fraudulent attempts at the point of transaction rather than detecting fraud retrospectively through audit processes.

Network Guardianship: The bidirectional reporting mechanism creates what RAT theorists call "collective guardianship" a network effect where multiple establishments and government offices function as interconnected guardians rather than isolated verification points. When one establishment flags a suspicious credential, that intelligence is immediately shared with LGUs and propagated across the entire network through database synchronization. This network approach prevents fraudulent IDs from succeeding elsewhere even after initial detection, transforming isolated verification failures into systemic fraud prevention.

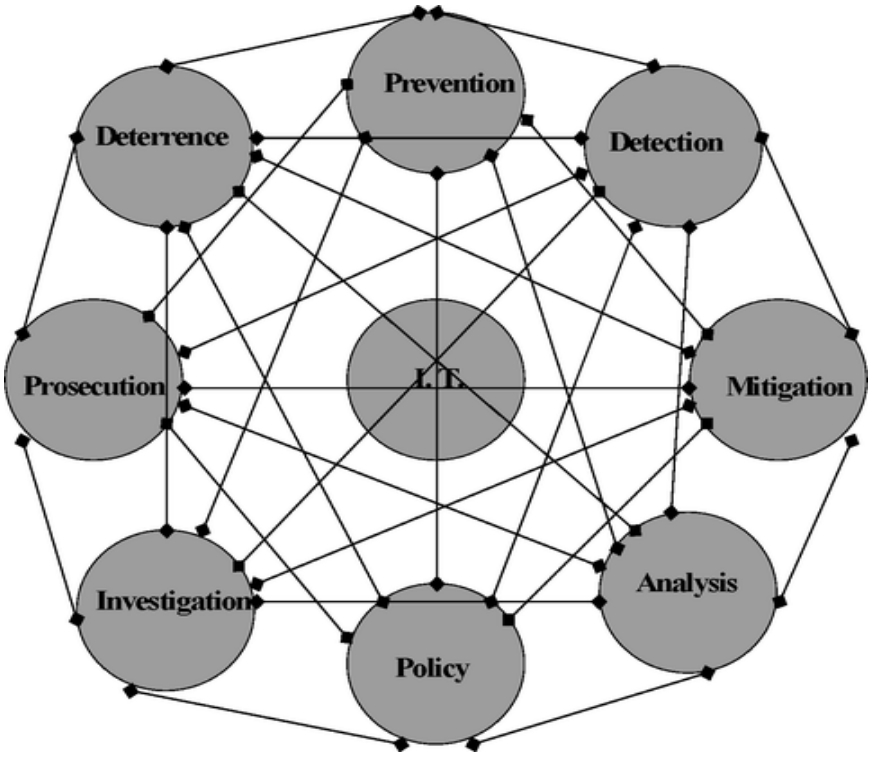
According to RAT principles, increasing capable guardianship reduces crime opportunities by raising the perceived risk of detection and apprehension for offenders (Felson, 2002). iDetect increases this perceived risk substantially: fraudsters can no longer assume that credentials rejected at one establishment will succeed at another, nor can they exploit intelligence gaps between businesses and government authorities. The systematic increase in guardianship capability addresses what Clarke and Eck (2003) call "situational crime prevention" modifying the environment to make fraud attempts more difficult, risky, and less rewarding through architectural changes rather than attempting to reform offender motivation.

RAT emphasizes that effective guardianship must be present (physically available at crime locations), capable (possessing necessary tools and authority), and willing to intervene (actively monitoring and responding rather than passively observing). iDetect satisfies all three criteria: presence through mobile accessibility enabling verification at every point of transaction, capability through advanced technology combining OCR, database integration, and automated flagging, and willingness through automated enforcement protocols requiring no discretionary

human judgment that might be swayed by social pressure or compassion appeals commonly used by fraudsters to circumvent verification.

Figure 2.

Fraud Management Lifecycle (Wilhelm, 2004)



The Fraud Management Lifecycle (FML) Theory, developed by Wilhelm (2004), provides a holistic and systematic approach to fraud management through eight interconnected stages that form a dynamic, evolving, and adaptive cycle. This framework recognizes that effective fraud management extends beyond simple detection to encompass comprehensive

lifecycle management from deterrence through prosecution, with each stage playing a critical role in overall fraud prevention and mitigation.

Wilhelm (2004) emphasizes that "effective fraud management requires a balance in the competing and complementary actions within the Fraud Management Lifecycle, highlighting that success depends not on focusing exclusively on any single stage but rather on coordinated activity across all stages. The framework has been validated across five industries with significant economic crime, demonstrating functional applicability across various organizational contexts including banking, telecommunications, insurance, and retail sectors (Wilhelm, 2004).

The Eight Stages of the Fraud Management Lifecycle:

1. Deterrence: The first stage is characterized by actions and activities intended to stop or prevent fraud before it is attempted; that is, to turn aside or discourage even the attempt at fraud (Wilhelm, 2004). Deterrence mechanisms create psychological and practical barriers that dissuade potential fraudsters from attempting fraudulent activities. In the context of identity verification systems, the visible presence of technology-enabled authentication serves as a deterrent signal that fraudulent credentials will be detected.

2. Prevention: Prevention involves proactive actions and activities designed to prevent fraud from occurring by eliminating opportunities for fraudulent exploitation (Wilhelm, 2004). Unlike deterrence which focuses on discouraging attempts, prevention focuses on blocking execution through system controls, validation mechanisms, and access restrictions that make fraud technically difficult or impossible to complete.

3. Detection: Detection stage involves actions and activities, such as statistical monitoring programs, used to identify and locate fraud prior to, during, and subsequent to the completion of the fraudulent activity (Wilhelm, 2004). The intent of detection is to uncover or reveal the presence of fraud or fraud attempts through automated systems, pattern recognition, anomaly detection, and manual review processes that identify discrepancies between presented credentials and authoritative records.

4. Mitigation: The goal of mitigation is to stop losses from occurring or continuing to occur and/or to hinder a fraudster from continuing or completing the fraudulent activity (Wilhelm, 2004). Wilhelm emphasizes that successful fraud management revolves around the appropriate allocation of sufficient, efficient, and early mitigation efforts, as timely intervention directly reduces financial impact and prevents fraud propagation.

5. Analysis: Analysis involves systematic study of fraud incidents using root cause analysis, pattern recognition, and data analytics to understand fraud factors, identify trends, and extract actionable intelligence from fraud events (Wilhelm, 2004). This stage transforms individual fraud incidents into organizational learning, enabling continuous improvement of fraud management capabilities through evidence-based insights.

6. Policy: The policy stage encompasses creation, evaluation, and deployment of fraud reduction policies based on analysis findings and organizational requirements (Wilhelm, 2004). Effective policies translate fraud intelligence into actionable guidelines, controls, and procedures that

standardize fraud management practices across the organization and establish accountability mechanisms.

7. Investigation: Investigation involves detailed examination of fraud incidents to gather evidence, establish facts, determine responsibility, and support enforcement actions (Wilhelm, 2004). Thorough investigations provide the evidentiary foundation necessary for both internal corrective actions and external legal proceedings while identifying systemic vulnerabilities requiring remediation.

8. Prosecution: Prosecution encompasses legal actions against fraudsters, whether civil or criminal, to impose consequences for fraudulent behavior (Wilhelm, 2004). Wilhelm emphasizes the interconnection between prosecution and deterrence: "a lack of attention to prosecution stage activities, be it civil or criminal, can result in a reduction of the various types of deterrence," demonstrating how backend accountability reinforces frontend prevention through visible consequences.

Application to Senior Citizen and PWD Discount Verification:

In the context of iDetect, the Fraud Management Lifecycle provides systematic framework for understanding how technology-enabled verification systems address fraudulent credential use through coordinated lifecycle management:

- **Deterrence**: The presence of OCR-based verification and database cross-checking deters fraudulent ID use by increasing perceived detection risk.

- **Prevention**: Real-time OSCA and PDAO database verification prevents fraudulent credentials from successfully obtaining discounts.
- **Detection**: Automated OCR extraction combined with pattern recognition identifies suspicious credentials during transactions.
- **Mitigation**: Instant flagging mechanisms stop fraud losses by alerting authorities while preventing continued fraudulent use.
- **Analysis**: Fraud analytics dashboards enable pattern recognition across establishments, revealing temporal trends and geographic distributions.
- **Policy**: BPLO portal facilitates policy implementation and system governance aligned with fraud intelligence.
- **Investigation**: OSCA and PDAO investigation workflows provide structured processes for examining flagged credentials.
- **Prosecution**: Evidence gathering capabilities support legal action against systematic fraud exploitation.

Wilhelm's emphasis on balanced activity levels across lifecycle stages directly informs iDetect's integrated multi-portal architecture. Rather than focusing exclusively on detection (the most visible stage), the system implements comprehensive lifecycle management through coordinated stakeholder coordination: business establishments perform detection, local government offices conduct investigation, and the integrated system enables analysis and policy refinement through fraud intelligence sharing.

Integration of Theoretical Frameworks

Routine Activity Theory and the Fraud Management Lifecycle Theory complement each other by addressing different aspects of the verification challenge. RAT provides a crime prevention lens, explaining how iDetect reduces fraud opportunities by introducing capable guardianship into a previously vulnerable ecosystem. The Fraud Management Lifecycle Theory provides a systematic fraud management lens, explaining how comprehensive lifecycle activities across deterrence, prevention, detection, mitigation, analysis, policy, investigation, and prosecution ensure that guardianship is not only present but also effective, adaptive, and sustainable.

Together, these frameworks guide iDetect's design principles:

From RAT: The system must function as a capable, present, and willing guardian through automated surveillance, real-time verification, and network intelligence sharing.

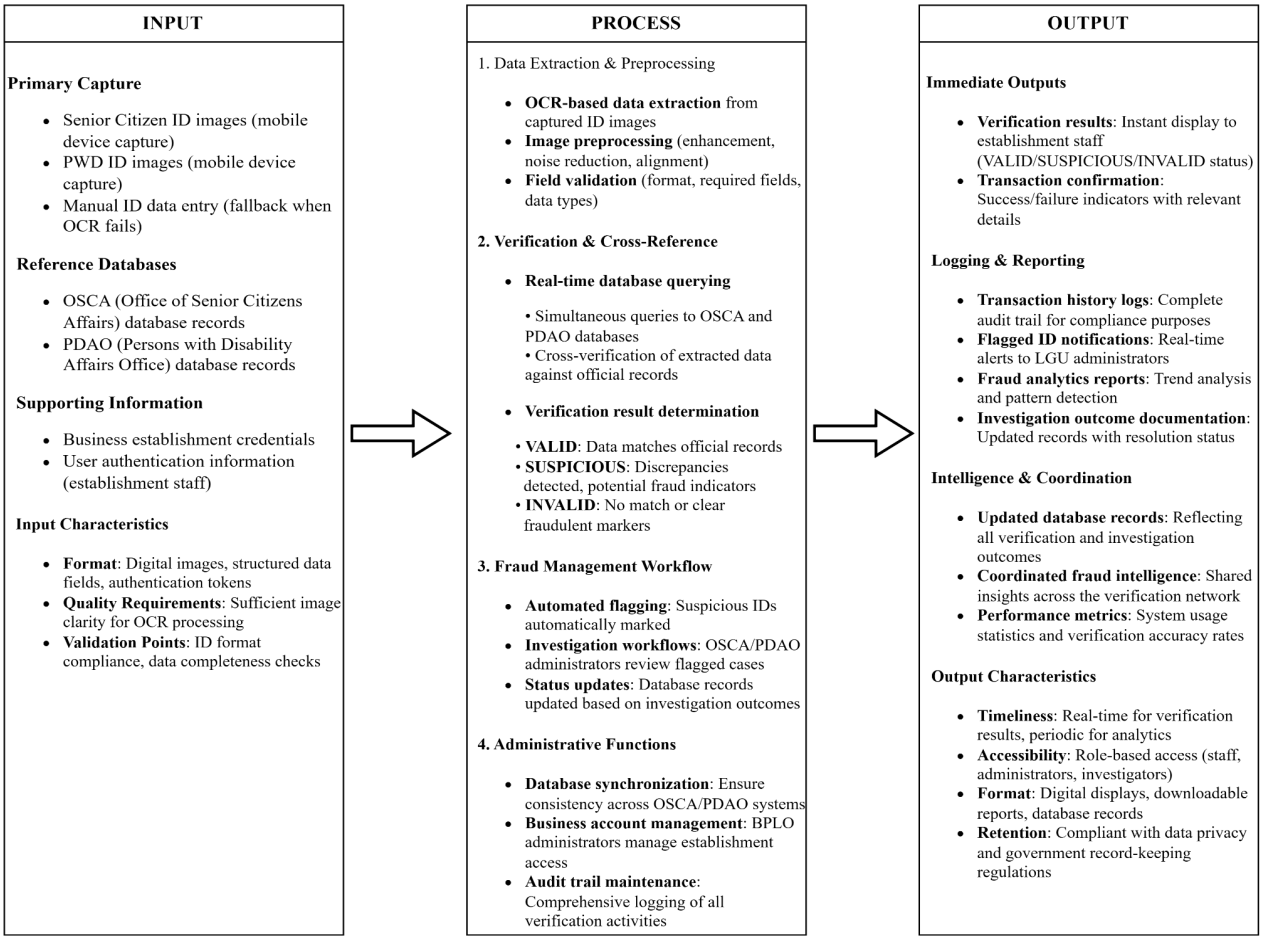
From Fraud Management Lifecycle: Guardianship effectiveness depends on systematic lifecycle management; therefore, the system must implement coordinated activities across all eight stages deterrence through visible verification presence, prevention through database cross-checking, detection through OCR and pattern recognition, mitigation through instant flagging, analysis through fraud analytics dashboards, policy implementation through BPLO governance, investigation through OSCA/PDAO workflows, and prosecution support through evidence gathering.

Synthesis: Effective fraud prevention requires both the presence of a guardian (RAT) and the systematic management of fraud across its complete lifecycle (FML). iDetect achieves this synthesis by combining technological guardianship with comprehensive lifecycle management mechanisms, creating a coordinated fraud prevention ecosystem where detection capabilities are reinforced by deterrence, prevention, mitigation, analysis, policy, investigation, and prosecution activities working in concert.

This integrated theoretical foundation informs not only system design but also evaluation criteria. Success will be measured not just by technical performance metrics (OCR accuracy, processing speed) but by crime prevention outcomes (reduced fraud incidents), lifecycle management indicators (deterrence effectiveness, detection rates, mitigation speed, investigation completion, prosecution support), and stakeholder coordination (intelligence sharing quality, policy implementation consistency, continued adoption) metrics reflecting both theoretical perspectives working in combination to produce meaningful impact.

Conceptual Framework

Figure 3.



The conceptual framework illustrates the integrated multi-portal architecture of iDetect, showing the relationships between system components, stakeholder interactions, data flows, and verification workflows. The framework demonstrates how the four portals (Client, BPLO, OSCA, and PDAO) function cohesively to create a comprehensive fraud prevention ecosystem.

The framework is based on the Input-Process-Output (IPO) model adapted for multi-stakeholder verification systems, showing:

Senior Citizen and PWD ID images captured via mobile devices, manual ID data entry when OCR is insufficient, OSCA and PDAO database records, business establishment credentials, and user authentication information.

OCR-based data extraction with image preprocessing and field validation, real-time database cross-verification querying both OSCA and PDAO simultaneously, verification result determination (VALID/SUSPICIOUS/INVALID), automated suspicious ID flagging and transmission to appropriate LGU portals, fraud investigation workflows conducted by OSCA/PDAO administrators, database status updates and synchronization, and business account management by BPLO administrators.

Instant verification results displayed to establishment staff, transaction history logs for audit purposes, flagged ID notifications to LGU administrators, fraud analytics reports and trend analysis, updated database records reflecting investigation outcomes, and coordinated fraud intelligence shared across the verification network.

The framework emphasizes bidirectional information flows distinguishing iDetect from existing one-way verification systems establishments not only query databases but also contribute fraud intelligence back to authorities, creating closed-loop fraud prevention rather than isolated verification points. The clear separation between portals with appropriate role-based access control is visually represented, showing how BPLO maintains business

oversight without accessing verification intelligence, while OSCA and PDAO maintain data compartmentalization preventing cross-access to different beneficiary populations.

Definition of Terms

This section provides operational definitions of key terms used throughout this study to ensure clarity and consistent understanding. Terms are organized alphabetically for easy reference.

Bidirectional Fraud Reporting: A system architecture enabling two-way information exchange where business establishments can both query government databases for verification and automatically transmit suspicious credential reports back to appropriate authorities, creating coordinated intelligence sharing rather than one-way verification.

BPLO (Business Permit and Licensing Office): The local government unit office responsible for issuing business permits, regulating commercial establishments, and ensuring compliance with municipal business operation requirements. In iDetect, BPLO manages establishment account verification and system access without handling verification intelligence or beneficiary data.

Capable Guardian: A theoretical concept from Routine Activity Theory referring to persons or systems present, capable, and willing to prevent crime through surveillance and intervention. In iDetect, technology-enabled verification serves as a capable guardian preventing credential fraud.

Client Portal: The mobile application component of iDetect used by food service establishment staff to capture ID images, perform OCR-based verification, receive real-time results, and flag suspicious credentials. Designed for point-of-sale use optimizing transaction speed.

Data Compartmentalization: Security principle ensuring that different user groups or system components access only the specific data necessary for their functions, preventing unauthorized access to sensitive information. In iDetect, OSCA accesses only Senior Citizen data, PDAO accesses only PWD data, and BPLO accesses only business account information.

Field-Level Validation: OCR verification process examining whether extracted data fields conform to expected patterns and formats (e.g., ID number structure, date ranges, issuing office codes) rather than accepting character recognition output without contextual verification.

Food Service Establishment (FSE): Commercial-type restaurants, cafes, fast-food chains, and similar food retail businesses serving prepared food and beverages to customers. The target establishment category for iDetect initial deployment.

Fraudulent Credential: A counterfeit, altered, or improperly obtained Senior Citizen or PWD identification card used to illegitimately claim statutory discounts and VAT exemptions. Includes both fake IDs manufactured externally and legitimately-issued IDs used by non-entitled individuals.

OCR (Optical Character Recognition): Technology enabling automated extraction of text from images, converting printed or handwritten characters into machine-readable digital format. In iDetect, Tesseract OCR processes ID images to extract credential details without manual data entry.

OSCA (Office of Senior Citizens Affairs): The local government unit office responsible for implementing programs and services for senior citizens, maintaining Senior Citizen ID database, and enforcing Republic Act 9994 provisions. In iDetect, OSCA administrators manage Senior Citizen verification records and investigate flagged credentials.

PDAO (Persons with Disability Affairs Office): The local government unit office responsible for implementing programs and services for persons with disabilities, maintaining PWD ID database, and enforcing Republic Act 10754 provisions. In iDetect, PDAO administrators manage PWD verification records and investigate flagged credentials.

Point-of-Sale (POS) Verification: Authentication of credentials conducted during customer transaction processing at retail checkout or service counter, requiring rapid processing to avoid workflow disruption and customer queue delays.

Real-Time Verification: Database query and response occurring immediately during transaction processing with minimal latency (typically under 5 seconds), enabling instant authentication decisions without requiring delayed processing or callback confirmation.

Republic Act 9994 (Expanded Senior Citizens Act of 2010): Philippine legislation mandating 20% discount and 12% VAT exemption for senior citizens (persons 60 years and above) on purchases of goods and services, with implementing rules specifying covered establishments and enforcement mechanisms.

Republic Act 10754: Philippine legislation expanding benefits and privileges for persons with disability, mandating 20% discount and 12% VAT exemption on purchases of goods and services, parallel to senior citizen provisions.

Role-Based Access Control (RBAC): Security approach granting system access and data visibility based on user roles (establishment staff, BPLO administrator, OSCA administrator, PDAO administrator), with each role having specific permissions aligned to functional responsibilities.

Routine Activity Theory (RAT): Criminological framework positing that crime occurs when motivated offenders, suitable targets, and absence of capable guardians converge in time and space. Forms theoretical basis for iDetect's fraud prevention approach.

Suspicious Credential: An identification card flagged during verification due to database mismatch, data inconsistencies, pattern anomalies, or establishment staff assessment, triggering investigation workflow without automatic determination of fraudulent status pending LGU administrator review.

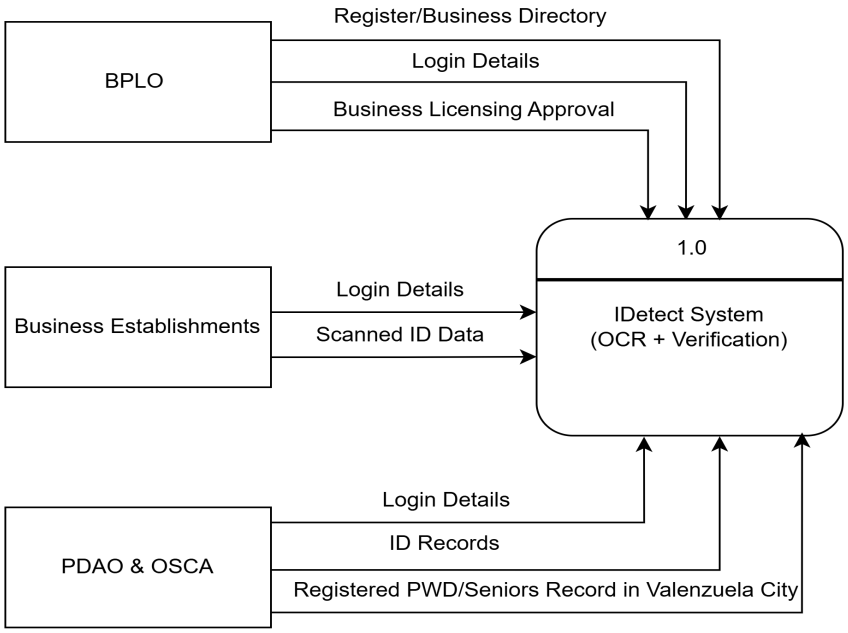
VALID (Valenzuela ID Validation Portal): Valenzuela City's existing web-based platform enabling verification of locally-issued Senior Citizen and PWD identification cards through manual data entry and database lookup. Launched May 2024 with new holographic ID designs.

Verification Result Status: Classification assigned to credential verification attempts: VALID (credential matches official records with no anomalies), SUSPICIOUS (credential shows discrepancies requiring investigation), or INVALID (credential does not match any official record or is confirmed fraudulent).

Context Diagram Level 0 and Level 1

Figure 4.

Context Diagram Level 0

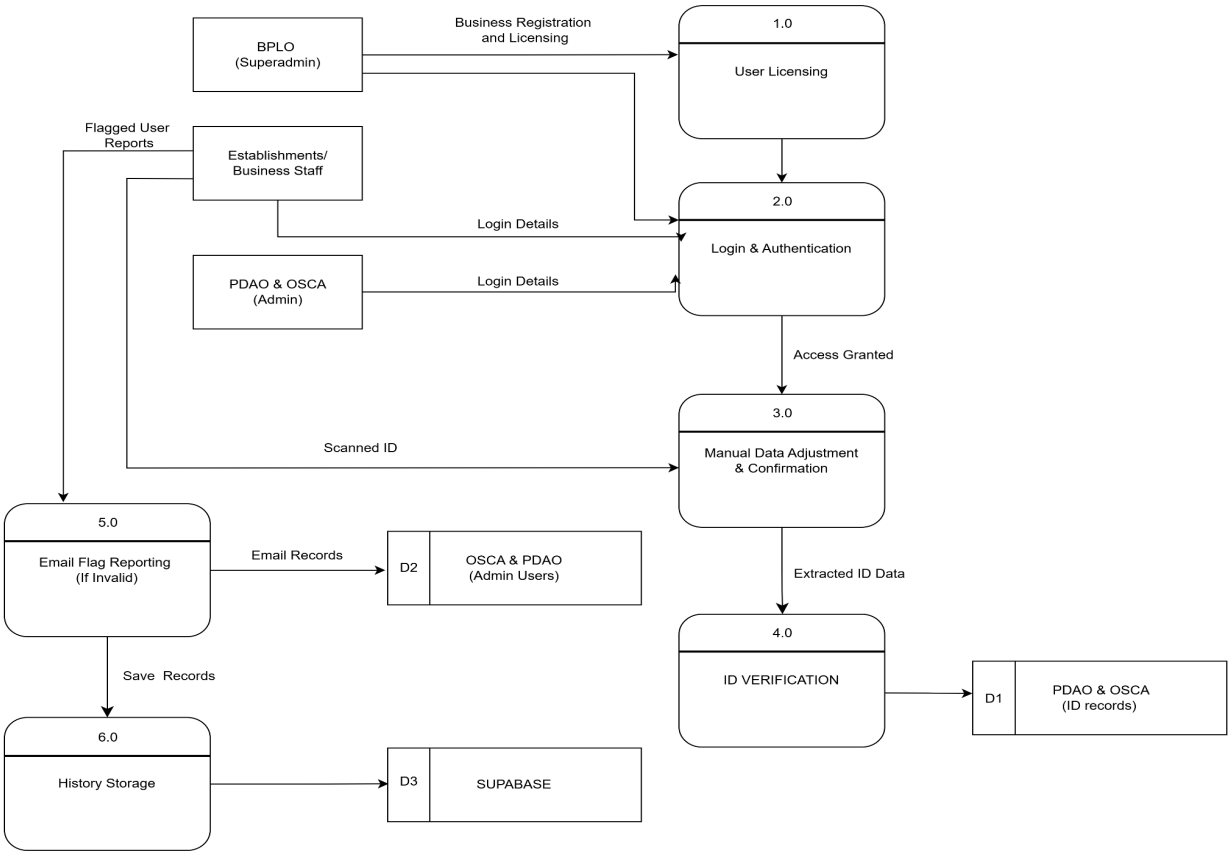


The Level 0 context diagram presents a high-level overview of the IDetect System, showing how it interacts with external entities without revealing internal processes. At the center of this diagram is the IDetect System, which performs OCR (Optical Character Recognition) and verification functions. Three external entities interact with this central system: BPLO (Business Permits and Licensing Office), Business Establishments, and PDAO & OSCA (Persons with Disabilities Affairs Office and Office of Senior Citizens Affairs). The BPLO serves as the regulatory body that provides the system with business registration data through a register/business directory function, along with login credentials and business licensing approvals. Business Establishments, on the other hand, are the users who need verification services, and they submit their login details and scanned ID data to the system for processing. The PDAO & OSCA agencies provide critical reference data including login credentials, ID records, and the registered database of PWDs and senior citizens specifically from Valenzuela

City. This context diagram effectively illustrates the system's boundaries and its relationship with external stakeholders, showing that the IDetect System acts as a central hub that receives information from government agencies and business establishments to perform identity verification and validation services.

Figure 5.

Context Diagram Level 1



The Level 1 diagram expands the IDetect System into six detailed processes that show exactly how data flows through the system from start to finish. The process begins with User Licensing (0.1), where the BPLO superadmin initiates business registration and licensing, which then flows into Login & Authentication (0.2). At this authentication stage, three different user types, establishment/business staff, BPLO personnel, and PDAO & OSCA administrators, all provide their login credentials to gain access to the system. Once access is granted, the flow moves Manual Data Adjustment & Confirmation (0.3), where human operators can review, correct, and confirm the information extracted by the OCR system to ensure accuracy. The extracted ID data then proceeds to ID Verification (0.4), where establishments and business staff

submit scanned identification documents. This process connects to data store D1, which houses PDAO and OSCA ID records for cross-referencing and validation.. If any issues or invalid entries are detected during this process, the system triggers Email Flag Reporting (0.5), which generates flagged user reports and sends email notifications to administrators whose information is stored in data store D2 (OSCA & PDAO Admin Users). Finally, all successfully processed and saved records flow to History Storage (0.6), where they are permanently archived in data store D3, the SUPABASE database. This comprehensive workflow ensures data accuracy through both automated and manual verification steps, provides accountability through email notifications to administrators, and maintains complete historical records for audit and compliance purposes.

CHAPTER 2 REVIEW OF RELATED LITERATURE AND SYSTEMS

Technical Background

The development of iDetect requires foundational understanding of several interconnected technologies and methodologies that enable automated identity verification in point-of-sale environments. This technical background section establishes the core technological concepts, implementation frameworks, and system architectures that inform the design and development of the proposed integrated verification system.

Optical Character Recognition (OCR) Technology

Optical Character Recognition is a foundational technology enabling automated extraction of text from images, converting printed or handwritten characters into machine-readable digital format. In identity verification contexts, OCR serves as the critical first step by extracting data from identification documents such as IDs, and government-issued credentials. Modern OCR systems have evolved from template-matching approaches to deep learning-based implementations capable of processing complex document layouts with high accuracy rates.

Contemporary OCR research demonstrates high accuracy rates when processing standardized identity documents. Liu, James, Gupta, and Raviv (2022) conducted comprehensive studies on Machine-Readable Zone (MRZ) extraction from passports and visa documents using convolutional neural networks, achieving 100% MRZ detection rate and 99.25% character recognition accuracy. This high accuracy makes OCR viable for automated verification systems where manual data entry would be time-prohibitive. However, accuracy varies significantly with image quality. Research comparing OCR engines on typewritten documents found that

commercial solutions like Abbyy FineReader achieved 95-98% character accuracy at optimal scanning resolutions between 200-500 dots per inch (Stančić & Trbušić, 2020).

Mobile OCR implementation introduces unique requirements beyond traditional document processing. Unlike office environments where documents can be rescanned if results are unsatisfactory, point-of-sale verification demands real-time performance with minimal user interaction. Gayer, Ershova, and Arlazarov (2023) demonstrated that mobile devices can perform real-time OCR extraction within the Machine-Readable Zone using YOLO-based models, achieving 62 frames per second on mid-range smartphones while maintaining high accuracy. Their optimized 1-megabyte model proved 83 times faster than previous implementations, demonstrating that computational constraints of mobile devices need not compromise performance. However, mobile implementations must accommodate variations in document quality, lighting conditions, and operator skill levels while maintaining accuracy sufficient for fraud detection.

The operational advantages of OCR over manual data entry manifest in both time savings and error reduction. Industry research demonstrates that OCR-based systems achieve accuracy rates of 99.95-99.99% compared to manual entry accuracy of 96-99% (Martinez, 2025). A study of laboratory point-of-care testing found that manual transcription had an error rate of 3.7%, with 14.2% of those errors containing significant and potentially dangerous discrepancies (Mays et al., 2019). A multi-center study across ICUs in three countries found that OCR implementation reduced data entry time by an average of 43.9%, with processing time decreasing from 6.0 minutes to 3.4 minutes per document (Nitayavardhana et al., 2025). For high-volume retail

environments processing thousands of daily transactions, these improvements compound to create substantial efficiency gains while simultaneously reducing fraud opportunities enabled by data entry errors.

OCR error rates reaching 50% on degraded documents with blurring and skewing as critical degradation factors (Hamdi et al., 2019) necessitates iDetect's comprehensive preprocessing pipeline including deskewing, denoising, contrast enhancement, and manual fallback mechanisms to maintain verification reliability despite non-standardized Philippine credential formats and variable capture quality. This sensitivity necessitates robust preprocessing including image quality assessment, deskewing algorithms, contrast enhancement, and user guidance during capture. Additionally, field-level accuracy proves more critical than character-level accuracy for identity verification research shows that 2.52% character error rates produced only 48.6% field-level accuracy for ID numbers and dates (Lerouge et al., 2024), requiring validation against known patterns such as ID number formats and date ranges.

While OCR technology enables efficient extraction of data from physical credentials, verifying the authenticity of that extracted data requires integration with authoritative databases. The following section examines the architectural patterns and implementation requirements for real-time verification systems that connect document scanning capabilities with government record databases.

Database Integration and Real-Time Verification Architecture

Database integration constitutes the foundational fraud detection mechanism for credential verification, confirming that presented credentials match records in authoritative

government databases. Real-time verification systems require instant connectivity to source databases maintained by respective agencies, with query responses provided with minimal latency to avoid transaction delays at point of sale. The architectural challenge lies in enabling distributed database access while maintaining data sovereignty, privacy protection, and query performance.

International examples demonstrate proven architectural patterns. Estonia's X-Road system processes over 2.7 billion data queries annually through a decentralized data exchange layer connecting independent agency databases (Frost & Sullivan Institute, 2024). Rather than centralizing identity data, X-Road functions as a secure query layer where agencies maintain database autonomy while enabling cross-system verification. This distributed architecture provides multiple advantages: reduced single-point-of-failure risk, maintained agency sovereignty over data, real-time currency since queries access source systems directly, and inherent auditability through mandatory query logging. Research on digital identity interoperability emphasizes that such systems must ensure data integrity, confidentiality, and privacy during exchange through proper authentication, encryption, and access control mechanisms (Ibor et al., 2024)

Real-time verification capabilities distinguish modern integrated systems from batch-processing legacy approaches. Big data analytics enable real-time fraud detection mechanisms that monitor transactions as they occur and flag suspicious activities for further investigation (Udeh et al., 2024). Financial fraud detection systems must process thousands of transactions per second with sub-100 millisecond latency constraints (Curry, 2025). While

identity verification operates at lower transaction volumes than financial systems, the architectural principles transfer: instant access to verification data requires high-memory configurations, NVMe storage for sustained operations, and database indexing strategies optimized for verification query patterns.

The Philippine context introduces specific challenges absent in standardized national ID systems. Unlike Estonia's unified infrastructure or Singapore's centralized SingPass, Philippine Senior Citizen and PWD beneficiary data reside in hundreds of independent local government unit databases with varying digitization levels, data quality standards, and technical sophistication. This fragmentation necessitates verification architecture tolerant to incomplete database coverage; the system must function optimally when authoritative verification is available while implementing alternative fraud detection mechanisms when database confirmation proves impossible. Privacy considerations further constrain implementation: businesses require binary valid/invalid confirmation without gaining access to comprehensive beneficiary information, requiring privacy-preserving query designs that return verification status without exposing detailed personal data.

Mobile Application Development for Point-of-Sale Environments

Mobile-first application architecture has transformed point-of-sale operations by enabling verification systems deployable on existing staff smartphones without requiring specialized hardware investment. However, mobile user experience design faces substantial challenges evidenced by research finding that zero percent of 138 major mobile e-commerce sites achieved 'good' overall user experience ratings, with 62% rated mediocre or worse (Baymard Institute,

2024). This pervasive usability challenge demonstrates that mobile interface excellence requires deliberate design effort rather than emerging naturally from functional implementation.

Point-of-sale systems possess specific usability requirements distinguishing them from general mobile applications. Research by Nielsen and Molich (2016) synthesizing usability evaluation models identifies twelve critical factors: learnability, efficiency, memorability, error frequency and severity, satisfaction, accessibility, visibility, feedback, consistency, error prevention, help availability, and aesthetic minimalism. These factors reflect POS operational realities: high staff turnover necessitates rapid learnability without extensive training, high transaction volumes demand efficiency minimizing time per verification, varying technical proficiency requires consistent interfaces minimizing memorization, and customer-facing operation mandates error prevention rather than error correction requiring visible recovery processes.

Research on technology adoption establishes the three-action limit as a critical threshold: systems requiring more than three discrete user actions face significant adoption resistance (Thompson et al., 2023). For identity verification, this constraint directly impacts workflow design; the optimal sequence of opening an application, scanning credential, and receiving verification consumes all three actions, leaving no room for login screens, menu navigation, or confirmation dialogs without exceeding the adoption resistance threshold. This necessitates architectural decisions enabling automatic background authentication, immediate camera activation on launch, and automatic processing without explicit confirmation steps.

Mobile interface design must accommodate device-specific constraints including limited screen real estate requiring careful information prioritization, touch interfaces demanding finger-sized interaction targets, and variable connectivity necessitating offline-capable functionality. Research by Budiu and Nielsen (2021) on mobile user experience emphasizes that mobile UX encompasses more than visual design; it extends to how users interact with and feel about the application, incorporating emotional responses, efficiency perceptions, and overall satisfaction. User preference studies prevent information overload by establishing what users actually need versus what designers assume, enabling selective display of essential verification information while relegating detailed logs and statistics to administrative interfaces accessed separately rather than cluttering transaction screens.

Review of Related Literature

This section presents a thematic review of scholarly and industry literature organized into four interconnected themes that inform iDetect's conceptual foundation, technical design, and implementation strategy. The review synthesizes international research on identity verification technologies with Philippine-specific studies on discount program implementation, creating contextually grounded understanding of both proven technological approaches and local implementation challenges.

Digital Identity Verification Systems and E-Government Integration

Digital identity verification systems represent fundamental shifts from paper-based identification to electronic credentials enabling secure, efficient service access. International

implementations provide evidence-based insights into database integration strategies, public-private partnerships, and fraud prevention mechanisms directly applicable to localized verification solutions. Estonia's comprehensive digital governance ecosystem provides the most extensively documented case of successful national-scale implementation, achieving 100% digitalization of government services by 2024 (Frost & Sullivan Institute, 2024). The system's foundation rests on X-Road data exchange infrastructure and mandatory digital ID cards issued to 99% of the population, processing over 2.7 billion data queries demonstrating both scalability and operational efficiency.

The Estonian model operates on the 'once-only' principle whereby citizens provide information to the government only once, after which agencies share data internally rather than requiring repeated submissions. This architectural decision produces measurable efficiency gains citizens file taxes in under 3 minutes and register businesses in under 15 minutes without physical office visits. Academic analysis documents how X-Road's combination with digital signatures enables comprehensive service integration including contract signing, medical record access, prescription ordering, and online voting (Margetts & Naumann, 2017). The measurable outcomes validate integrated verification approaches: Estonia saves an estimated 2% of GDP annually through digitalization, equivalent to one week of working time per citizen. For service verification specifically, unified architecture makes document forgery ineffective as credentials must pass database verification regardless of physical authenticity.

Singapore's National Digital Identity framework demonstrates complementary approaches emphasizing public-private sector integration. Built on the SingPass platform,

Singapore's system extends beyond government services to enable authentication for banking, e-commerce, and private transactions. Comparative research by Goede (2019) examining e-government development in Estonia, Singapore, and Curaçao reveals distinct pathways in digital transformation. While Estonia's success emerged from government-led innovation working closely with the banking sector and telecommunications companies with Estonia notably leading innovation from the public sector side, Singapore emphasized coordinated development with the private sector playing a more substantial role from inception, particularly through corporate ICT sector involvement. Both models demonstrate that comprehensive digital identity requires robust legal frameworks establishing digital signatures' legal equivalence to handwritten signatures, providing the trust foundations that enable subsequent service development and citizen adoption.

Critical to iDetect's design: integrated verification infrastructures connecting government databases with private sector users through standardized protocols achieve documented fraud reduction exceeding 70% through coordinated intelligence sharing and real-time cross-verification (Protecht Group, 2025). However, success depends on architecture, governance, and trust mechanisms rather than simply implementing technology. The distributed architecture principle where verification systems function as secure query layers connecting decentralized databases rather than centralizing all identity data proves essential for maintaining agency sovereignty while enabling integration. For Philippine implementation, this implies that iDetect should enable real-time lookups against authoritative LGU databases rather than attempting centralization that would face institutional resistance.

Fraud Detection and Prevention in Discount Programs

The proliferation of fraudulent credentials in Philippine discount programs has reached crisis proportions documented through comprehensive government investigations. Senate Committee hearings revealed 8.5 million fraudulent PWD IDs circulating compared to only 1.89 million legitimate credentials registered with the National Council on Disability Affairs, a 4.5:1 ratio translating to severe economic consequences (Bordey, 2024). Of ₱208.4 billion in total discount transactions during 2023, an estimated ₱166.1 billion involved fraudulent credentials, with Bureau of Internal Revenue calculating direct tax revenue losses at ₱88.2 billion for that year alone, classifying fraudulent PWD ID use as criminal tax evasion (Gonzales, 2024).

Industry-specific data demonstrates operational impact on businesses. Restaurant operators documented a dramatic 22.21% increase in the ratio of PWD to senior citizen discount claims within just two years, from 2022 to 2024, with establishments rejecting approximately 17% of PWD cards presented due to suspected fraud (Arnaldo, 2024). One major restaurant chain reported rejecting 726 of 4,212 presented PWD IDs during three months, a 17.24% rejection rate indicating nearly one in five credentials failed even basic authenticity checks (Arceo, 2024). This likely understates true fraud prevalence given legal constraints: Department of Justice opinion issued January 2024 clarified that establishments cannot legally deny discounts even when unable to verify authenticity, placing businesses in untenable positions of absorbing fraudulent costs or risking legal liability for denying potentially legitimate beneficiaries (Cervantes, 2025).

Multi-factor authentication technologies provide proven frameworks for credential verification by requiring multiple independent validation factors. Systematic review by Bonneau and colleagues (2024) analyzing 70 academic papers plus 13 industry tools documented widespread adoption with 77% achieving NIST standards compliance. Contemporary implementations increasingly incorporate behavioral biometrics continuous authentication through keystroke dynamics, mouse movements, and touchscreen gestures enabling fraud detection throughout transactions rather than only at initial authentication. Research on secure financial transactions implementing two-layer security combining MFA with machine learning achieved classification accuracies of 97.938% using logistic regression, 97.881% with decision trees, and 96.717% with random forest algorithms (Kumar & Singh, 2024).

Pattern recognition and anomaly detection constitute complementary fraud detection approaches enabling identification of behavioral patterns distinguishing legitimate from fraudulent use. Machine learning algorithms learn normal behavior baselines from historical data, creating behavioral profiles against which subsequent transactions are evaluated. Research documents that ensemble methods achieve 94.3% fraud detection accuracy while neural networks achieve 93.7% (Wang et al., 2024). These methods prove particularly effective for credential-based fraud where behavioral patterns, transaction frequency, geographic distribution, temporal patterns, establishment preferences provide fraud signals independent of credential physical authenticity. Geographic impossibility detection flags credentials used at multiple distant locations within impossible timeframes; frequency analysis identifies abnormal

transaction volumes; network analysis using graph databases identifies fraud rings where multiple credentials share suspicious commonalities.

Critical for iDetect implementation: The documented ₱166.1 billion in fraudulent transactions and 4.5:1 fraudulent-to-legitimate credential ratio establishes compelling economic justification for comprehensive technological intervention. Even achieving 50% fraud reduction substantially below the 70%+ documented in financial systems would save ₱44 billion annually in tax revenue while reducing business discount absorption by ₱83 billion. However, adaptation from financial fraud detection requires careful consideration: discount beneficiaries need simpler processes than sophisticated financial users, false positive rates tolerable in financial contexts become problematic when denying statutory entitlements, and fragmented Philippine database infrastructure precludes exclusive reliance on database verification, necessitating multi-layer detection combining database cross-referencing, pattern recognition, and network intelligence.

Mobile Point-of-Sale Verification Solutions and Usability Requirements

The transformation of point-of-sale operations through smartphone ubiquity enables verification systems deployable without specialized hardware, but success requires understanding mobile-specific usability principles. Systematic literature review by Harrison and colleagues (2024) analyzing 205 documents identified STFM framework Scenarios, Themes, Features, and Methodologies as organizing principle for mobile UX research, with usability, satisfaction, and engagement emerging as most frequently studied factors. The PACMAD quality model extends traditional evaluation by incorporating performance, accessibility, compatibility,

and mobility alongside conventional metrics, recognizing that mobile applications function in substantially different usage contexts than desktop software.

Industry benchmarking reveals pervasive usability deficiencies even among major implementations. Analysis of 138 leading mobile commerce sites found zero achieving 'good' overall user experience, with 62% rated mediocre or worse and only 38% achieving 'decent' ratings (Baymard Institute, 2024). This benchmark representing over 200,000 hours of mobile UX research demonstrates that mobile usability remains difficult even for well-resourced organizations, suggesting verification systems require deliberate design effort. Research on POS technology adoption identifies workflow integration and action simplicity as critical factors, documenting that systems requiring more than three discrete actions face significant resistance and additional steps create cognitive load extending transaction time (Thompson et al., 2023).

Point-of-sale contexts introduce unique requirements beyond general mobile applications: verification must complete within seconds to avoid customer queue delays, staff possess varying technical proficiency requiring intuitive designs minimizing training, and business environments often lack IT support necessitating systems maintainable by non-technical operators. Research by Seffah and colleagues (2016) synthesizing usability evaluation models identifies twelve critical POS factors including learnability, efficiency, memorability, errors, satisfaction, accessibility, visibility, feedback, consistency, error prevention, help, and aesthetic minimalism. These reflect operational realities where staff turnover necessitates rapid learning, high transaction volumes demand efficiency, varying proficiency requires consistent interfaces, and customer-facing operation mandates error prevention rather than correction.

Mobile interface design principles establish that quality of use must constitute prime design objective rather than ancillary consideration. Research by Constantine and Lockwood (2021) emphasizes that usability relates directly to business objectives applications failing to deliver quality experiences fail to achieve adoption regardless of functional capability. The distinction between user interface (graphical layout) and user experience (overall interaction quality) proves important: well-implemented UI requires well-implemented UX, where visually appealing interfaces failing to support efficient task completion provide poor experiences regardless of aesthetics. User preference research prevents information overload by establishing what users actually need versus designer assumptions, enabling selective display of essential verification information while relegating detailed data to separate administrative interfaces.

Critical for iDetect design: The three-action limit directly constrains verification workflow to launch-scan-view sequence, necessitating automatic background authentication, immediate camera activation, and automatic processing without confirmation dialogs. The finding that different evaluation methods identify non-overlapping problem sets implies comprehensive assessment requires expert heuristic evaluation during development, controlled testing with representative users, satisfaction surveys post-deployment, and continuous monitoring of operational metrics including completion rates and error frequencies. The 62% mediocre-or-worse mobile UX baseline establishes that simply avoiding major usability failures positions systems favorably, while the pervasive mediocrity demonstrates substantial design investment proves necessary to exceed industry standards.

Philippine Senior Citizen and PWD Discount Implementation Challenges

Philippine discount program implementation faces systemic challenges rooted in institutional capacity limitations, infrastructure fragmentation, and verification system inadequacies. Republic Act 9994 (Expanded Senior Citizens Act of 2010) and Republic Act 10754 (Expanding Benefits for Persons with Disability) mandate 20% discounts plus VAT exemptions, with implementing rules directing provisions be 'construed liberally and in favor' of beneficiaries when interpretation conflicts arise. However, the Department of Justice opinion issued January 2024 clarified that establishments cannot legally withhold privileges even when unable to verify authenticity immediately, noting the Department of Health PWD Registry 'does not yet contain complete listing' for various reasons (Cervantes, 2024). This places establishments in untenable positions legally required to honor potentially fraudulent credentials or face ₱50,000-100,000 fines and 2-6 years imprisonment, yet lacking verification mechanisms distinguishing legitimate from fraudulent claims.

Research by the University of the Philippines Center for Local and Regional Governance found that only 60% of Philippine LGUs established operational Persons with Disability Affairs Offices as mandated by RA 10070, with 33 provinces, 25 cities, and 282 municipalities lacking dedicated PDAO offices (Alampay et al., 2018). This implies 40% of municipalities and cities hundreds of LGUs across the archipelago lack dedicated offices for PWD services, credential issuance, and verification support. The advocacy report noted PDAO absence 'impaired implementation of Republic Act 10754,' demonstrating direct linkage between institutional capacity and program effectiveness. Funding structure compounds disparities as RA 9994 specifies that 'necessary appropriations for operation and maintenance shall be appropriated and

approved by LGUs concerned,' creating dependency on local budgets rather than guaranteed national funding. Wealthy urban LGUs invest in digital systems and trained staff while resource-constrained rural municipalities maintain paper records with minimal technical expertise, producing verification capacity directly correlated with LGU wealth.

Identification standardization challenges compound capacity gaps. Unlike national ID systems with standardized formats, Philippine senior citizen and PWD credentials vary across 1,780+ independent issuing authorities. Legal analysis by Respicio Law (2024) notes senior citizen IDs have 'no standard design/layout unlike passport or driver's license,' with formats varying by municipality. While NCDA announced unified PWD ID design in January 2024, actual implementation across all LGUs remains incomplete. This non-standardization creates multiple verification challenges; establishments cannot authenticate credentials through visual security inspection, forged IDs prove difficult to detect without database verification, and only 'some LGUs' implement holograms or digital verification capabilities. Database fragmentation compounds standardization challenges as each LGU maintains separate OSCA and PDAO databases with no centralized national registry enabling cross-jurisdictional verification.

Cultural factors influence implementation dynamics distinctly from technical challenges. Research by Inabangan, Garcia, and Abocejo (2019) evaluating RA 9994 implementation identified Filipino values affecting compliance: 'jackpot mentality, helpfulness and shyness' among senior citizens impacted law implementation. Jackpot mentality encourages maximum benefit exploitation; helpfulness norms facilitate credential sharing among family; shyness prevents asserting rights when establishments resist discounts. These dynamics create patterns

where some systematically maximize benefits while others underutilize entitlements, and confrontation-averse seniors accept non-compliance rather than filing complaints. The Senate Committee on Ways and Means held a December 5, 2024 hearing on fake PWD ID proliferation and its impact on government revenues. Restaurant owners reported PWD customer sales increasing from 5 percent to 30 percent, with most suspected fraudulent (Bacelonia, 2024). The crisis has led to 'legitimate PWDs whose disability is not apparent' being 'subjected to scrutiny whenever they use their IDs' (Lauengco, 2024), creating discriminatory treatment that harms the intended beneficiaries of discount programs.

Critical for iDetect implementation: The 60% PDAO coverage creates geographic verification inequality requiring graduated deployment strategies beginning with digitized urban LGUs while building capacity in under-resourced municipalities. The DOJ opinion preventing denial of unverifiable credentials creates fundamental design constraints verification systems cannot function as gatekeepers preventing transactions but must operate as intelligence-gathering mechanisms building evidence for subsequent enforcement while ensuring beneficiary access through provisional approval architectures. Cultural factors require non-confrontational verification design where results are presented transparently without accusation, with flags explained as data quality issues rather than fraud accusations, maintaining dignity while enabling intelligence gathering. The system's success depends on adapting proven verification principles to Philippine institutional realities, building capacity incrementally while delivering immediate fraud reduction value.

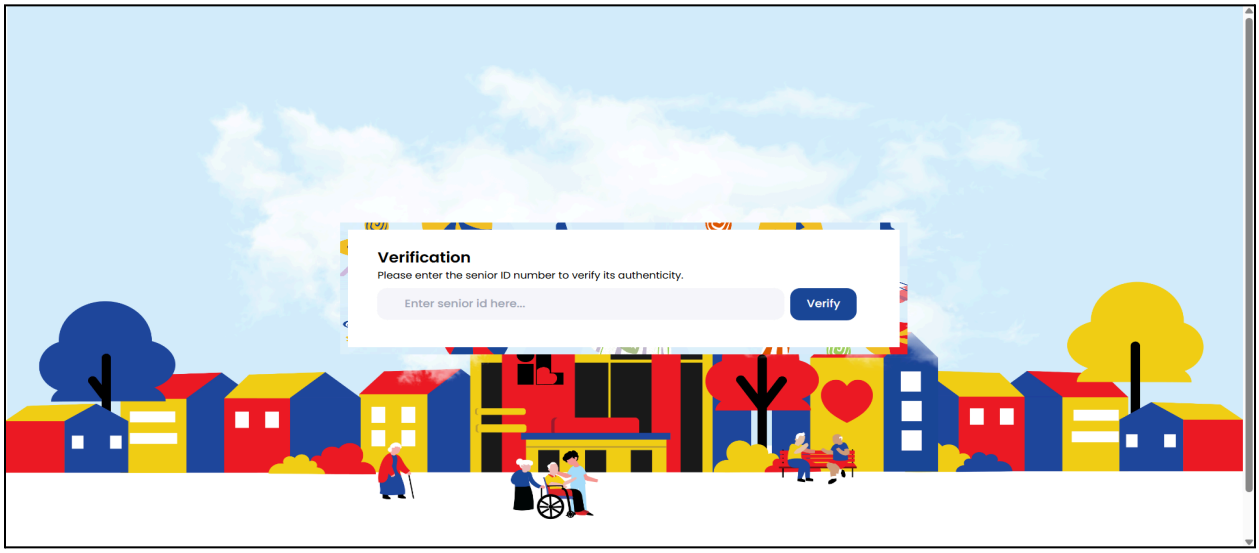
Review of Related Systems

This section examines existing identity verification systems relevant to iDetect's functional requirements, analyzing both local Philippine implementations and international commercial solutions. The review identifies technological capabilities, operational strengths, and critical limitations of current systems, establishing the functional and technical gaps iDetect addresses through its integrated multi-portal architecture.

Philippine Local Government Unit Verification Platforms

Figure 6.

OSCA Taguig

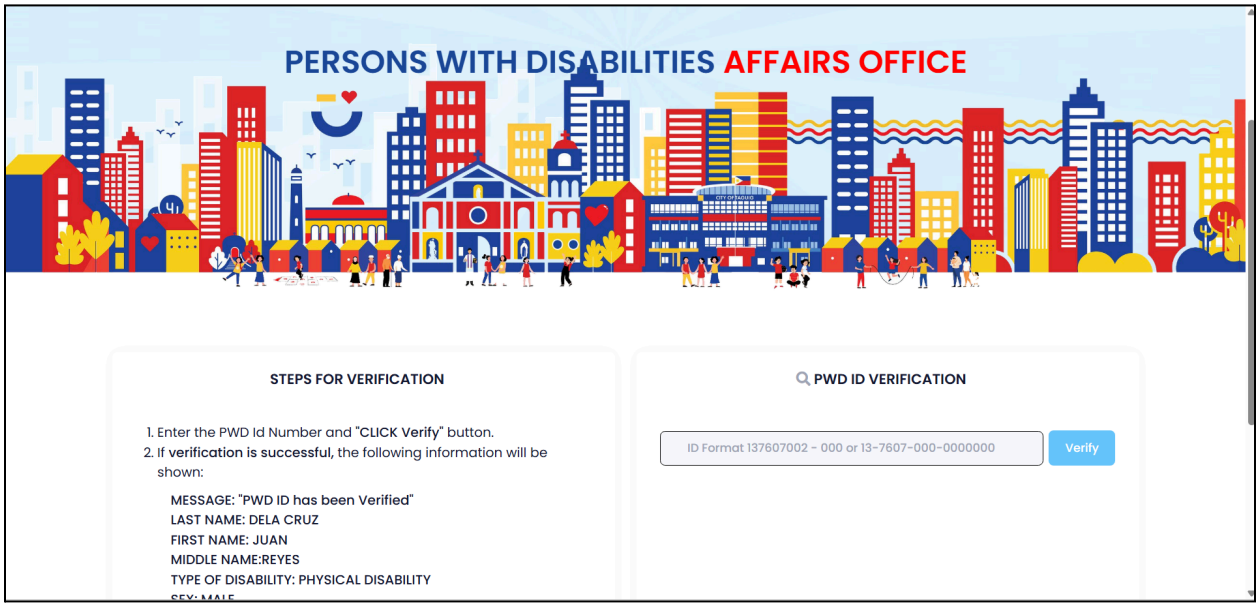


OSCA Taguig Senior Citizen Verification System. The Office of the Senior Citizens Affairs in Taguig City operates an online platform enabling establishments to verify Senior Citizen IDs by entering ID numbers into a web portal (Office of the Senior Citizens Affairs Taguig, n.d.). The system queries the official OSCA database and returns basic cardholder

information for validation. Primary strengths include real-time database connectivity, simple web interface accessible through standard browsers, and immediate verification results enabling establishments to confirm authenticity quickly without requiring specialized software or hardware. However, the system presents notable limitations: verification requires manual data entry of ID information consuming 30-45 seconds per transaction, functionality is restricted exclusively to Taguig City-issued IDs precluding verification of credentials from other municipalities, no OCR capability exists for automated data extraction from ID images, and the platform lacks automated fraud reporting mechanisms to notify OSCA authorities of suspicious IDs detected during verification processes.

Figure 7.

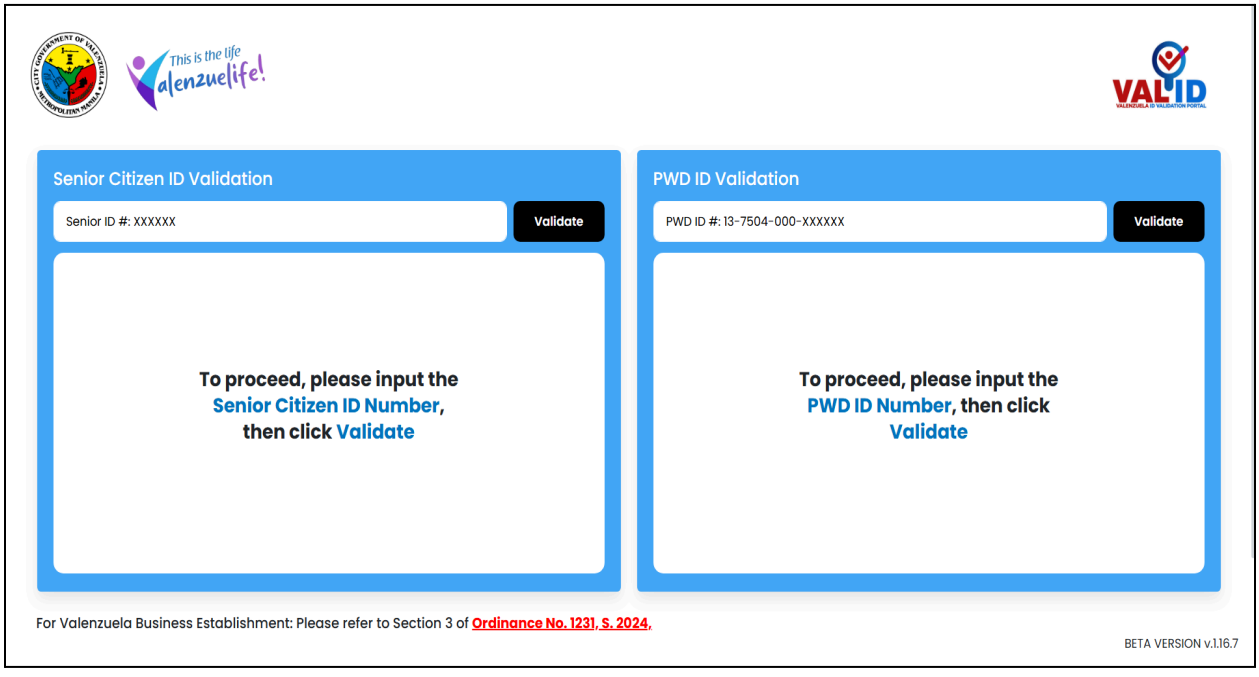
PDAO Taguig



PDAO Taguig PWD ID Verification System. The Persons with Disabilities Affairs Office in Taguig City provides parallel functionality for PWD credential verification through a web-based portal connected to the PDAO database (Persons with Disabilities Office Taguig, 2024). Authorized users search ID numbers to confirm legitimacy, benefiting from direct database access, controlled user authorization, and real-time verification. Despite these strengths, significant limitations exist: manual input of ID details remains required consuming verification time, restriction to Taguig PWDs exclusively limits utility for establishments serving broader populations, the system operates separately from senior citizen verification requiring establishments to use multiple platforms for comprehensive discount verification, and absence of automatic data extraction features makes the system time-consuming for high-volume business environments where transaction speed directly impacts customer experience and operational efficiency.

Figure 8.

VALID (Valenzuela ID Validation Portal)

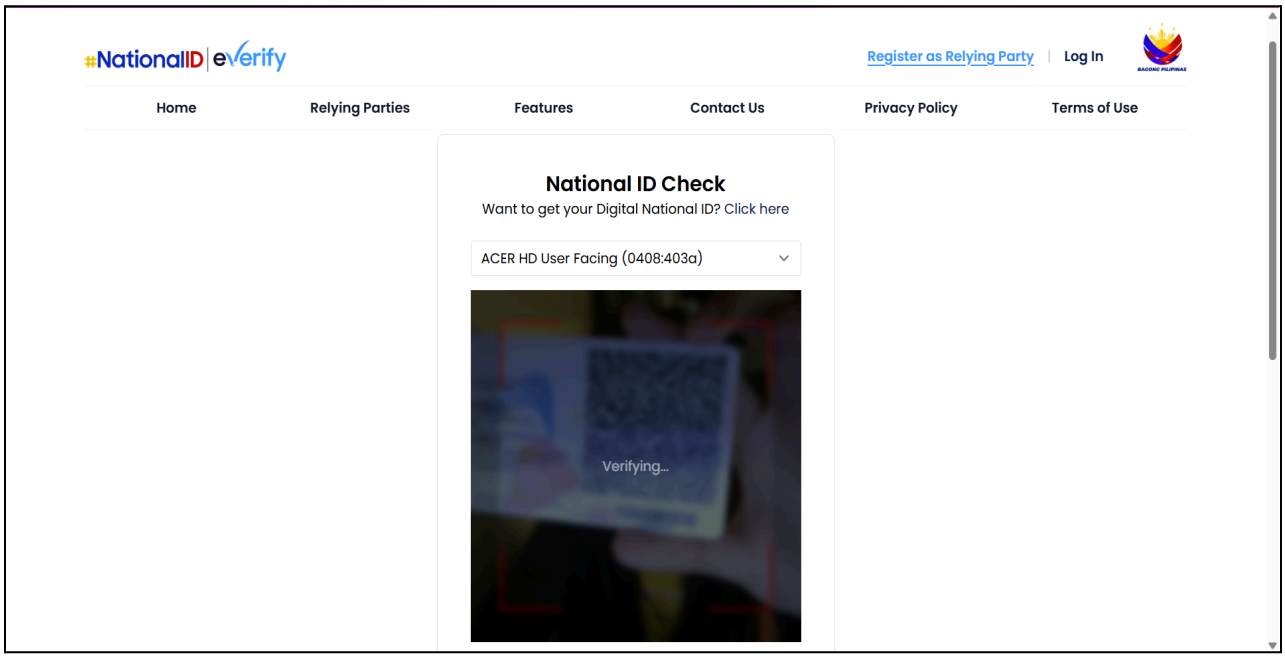


VALID (Valenzuela ID Validation Portal). VALID represents Valenzuela City's integrated platform for verifying both Senior Citizen and PWD IDs through a web-based system connecting to city databases maintained by OSCA and PDAO (City Government of Valenzuela, 2024). Launched officially May 20, 2024, the portal features newly designed ID cards with holographic seals and can be accessed via desktop or mobile devices at valid.valenzuela.gov.ph. The platform's key strengths include unified approach to multiple ID types reducing platform switching for businesses, real-time connectivity to local government databases enabling current verification status checks, and specific service to Valenzuela businesses addressing local verification needs. However, VALID presents operational challenges: verification requires manual entry of ID details consuming processing time, the system is not optimized for high-transaction retail environments where verification speed is critical to maintaining customer flow, no OCR integration exists for automated data capture from ID images, and the platform lacks automated fraud reporting mechanisms to notify government offices of suspicious IDs detected by establishments, functioning as one-way verification tool without bidirectional intelligence sharing between businesses and authorities.

National-Level Identity Verification Platforms

Figure 9.

eVerify



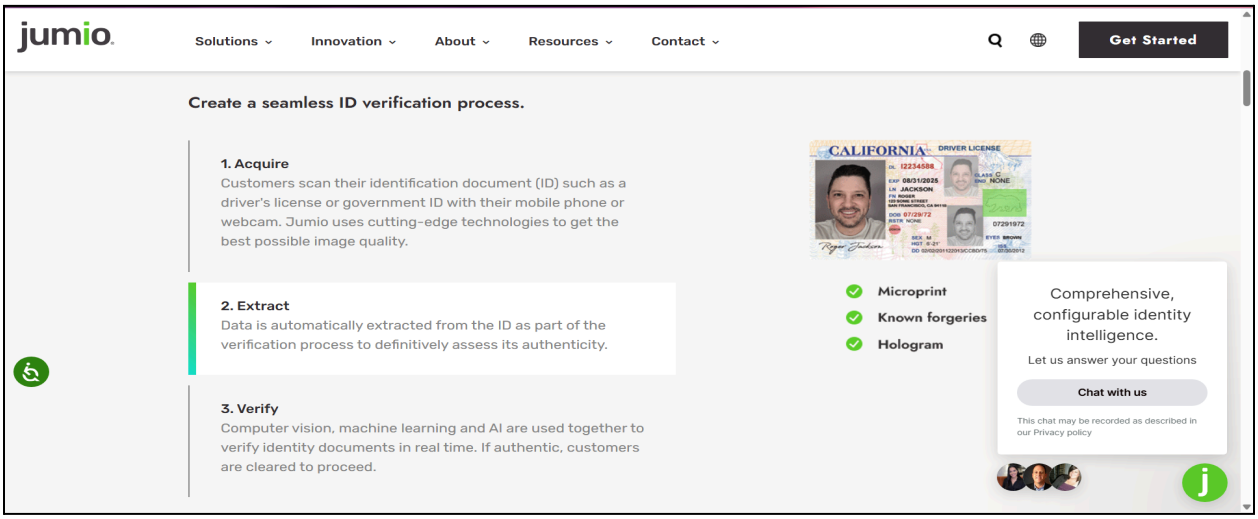
eVerify (Philippine Statistics Authority). The eVerify system, developed and managed by Philippine Statistics Authority, is a national-level digital verification platform designed to authenticate Philippine Identification System (PhilSys) credentials through secure API access for authorized institutional users. Officially launched June 2024, eVerify aims to strengthen identity verification by providing standardized, centralized, and secure mechanism for validating PhilSys numbers across participating organizations (Philippine Statistics Authority, 2024). Despite its robust framework, eVerify presents several limitations for Senior Citizen and PWD discount

verification. The system is primarily focused on PhilSys-based authentication and does not support validation of local government unit-issued senior citizen or PWD identification cards commonly used in availing statutory discounts. Furthermore, eVerify does not incorporate optical character recognition or image-based ID scanning features, limiting usability in point-of-sale environments. The verification workflow is not optimized for real-time discount processing, and formal institutional authorization requirements may pose challenges for small-scale businesses and retail establishments lacking institutional relationships with government verification systems.

International Commercial Identity Verification Solutions

Figure 10.

JUMIO

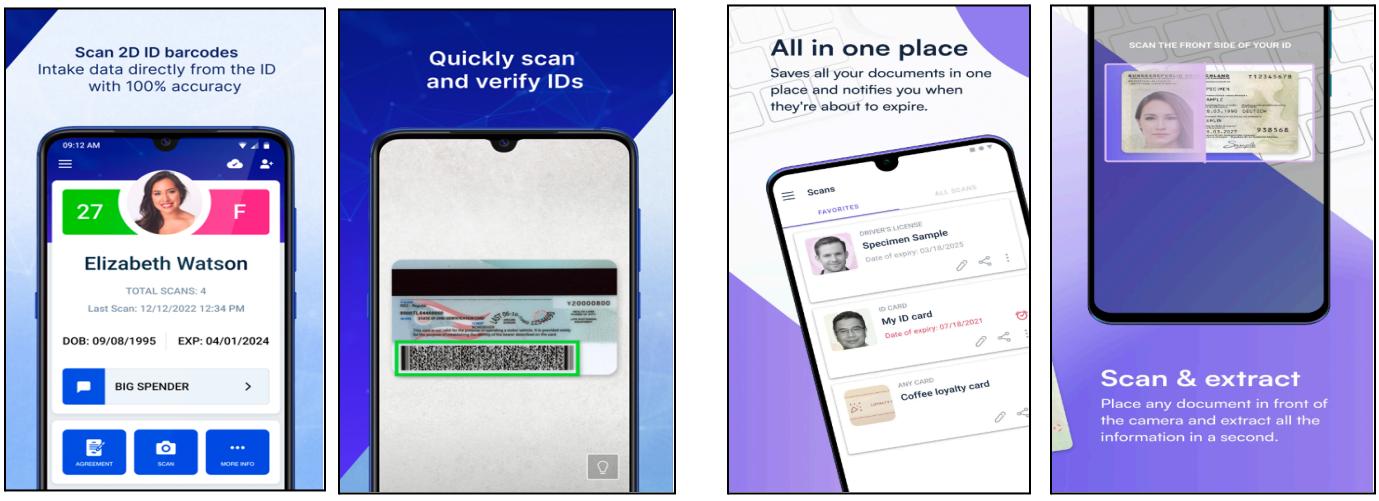


Jumio Identity Verification System. Jumio enables verification of government-issued IDs (passports, driver's licenses, national IDs) through user-facing workflow where individuals scan ID front and back and take selfie, with system automatically extracting data via OCR,

performing liveness and face-match checks, and returning verification result (Jumio Corporation, 2023). Strengths include real-time automated verification, support for diverse ID types globally, OCR-based data extraction, liveness detection and biometric face matching deterring fraud, and fast user-friendly interface suitable for high-volume or online onboarding. However, Jumio has critical limitations for Philippine discount verification context: designed mainly for eKYC/online identity verification rather than point-of-sale discount validation, not tailored for local Philippine Senior Citizen or PWD discount workflows, may not integrate with local government databases, depends on internet connectivity and quality of images, and is not specific to Senior/PWD IDs used by Philippine LGUs.

Figure 11.

Veri Scan & BlinkID

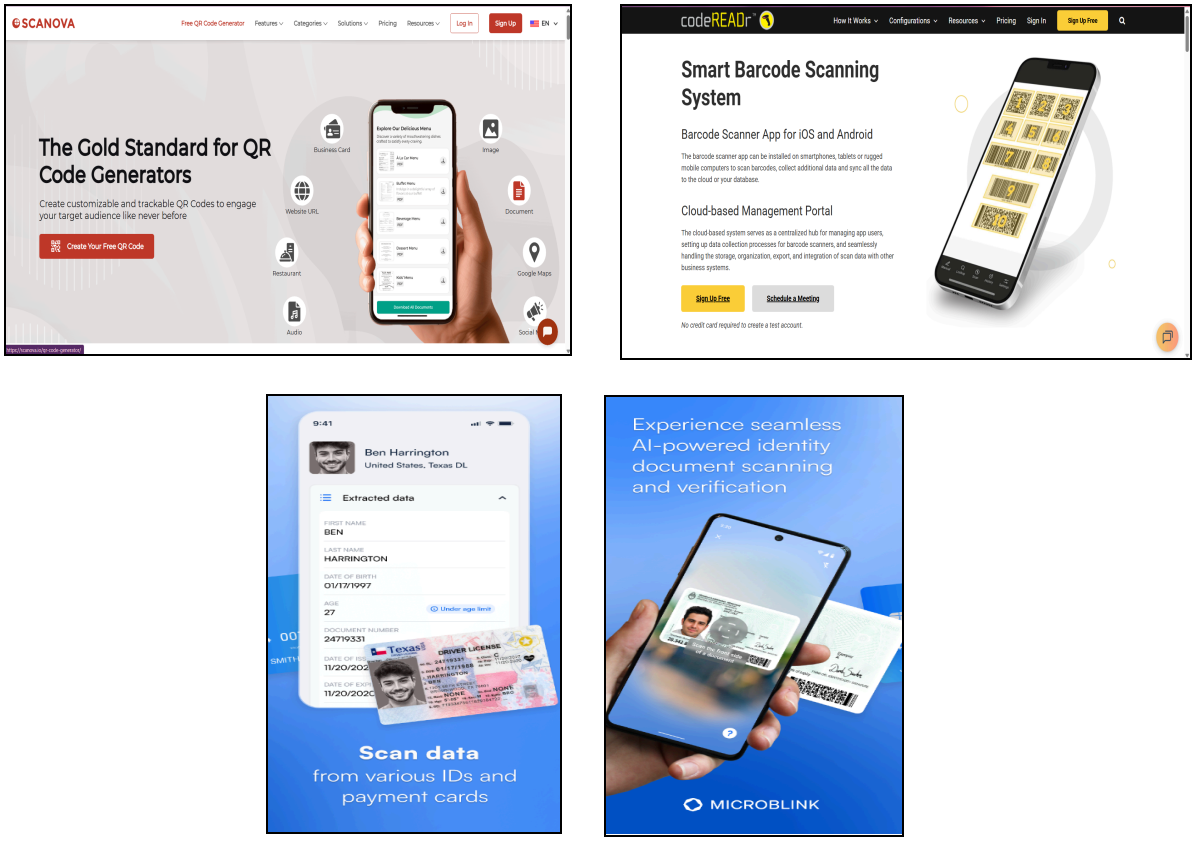


VeriScan and BlinkID/IDSafe. VeriScan is mobile ID scanning tool developed by IDScan.net that decrypts barcode data from government IDs and provides real-time age verification and visitor management features (IDScan.net, 2024). BlinkID (now IDSafe) uses

AI-powered OCR and NFC technology to extract and verify identity document data from over 190 countries, offering real-time authenticity checks (Microblink, 2024). Both systems demonstrate strengths in barcode/chip reading capabilities, global document coverage, offline functionality, and established commercial deployment. However, shared limitations include: limited support for Philippine non-barcode identification documents, lack of integration with Philippine government OSCA and PDAO databases, subscription requirements for full functionality, and absence of automated reporting mechanisms for flagging suspicious credentials to government authorities. These systems excel at data extraction but lack the government-database integration and fraud reporting architecture required for coordinated discount program verification.

Figure 12.

Scanova, CodeREADr, and Microblink Identity Verification



Scanova, CodeREADr, and Microblink Identity Verification. Scanova provides lightweight QR and barcode scanning with fast decoding and scan history logging (Scanova, 2024). CodeREADR offers enterprise data-capture across barcodes, QR codes, NFC, and OCR with configurable workflows and audit trails (CodeREADR Inc., 2024). Microblink Identity Verification uses AI-powered computer vision to scan and verify over 2,500 document types from 200+ countries (Microblink, 2024). These systems demonstrate advanced technical capabilities including multi-modal data capture, offline scanning, and robust audit logging. However, common limitations restrict Philippine discount verification applicability: designed as general-purpose solutions rather than being specifically tailored for Philippine LGU-issued credentials, lack native integration with local OSCA and PDAO databases requiring custom development, and absence of automated fraud reporting mechanisms enabling coordinated intelligence sharing between businesses and government enforcement authorities. While technically sophisticated, these international systems require substantial adaptation for Philippine discount program context.

Table of Comparison

Table 1.

Comparison of iDetect with Existing ID Verification Systems

Systems	OCR extraction	Real-time OSCA/PDAO verification	Auto fraud reporting	Mobile POS design	Dual database integration
OSCA Taguig Senior Citizen ID Verification System	✗	✓	✗	✗	✗
PDAO Taguig PWD ID Verification System	✗	✓	✗	✗	✗
VALID (Valenzuela ID Validation Portal)	✗	✓	✗	✗	✓
eVerify Philippines - National ID Verification System	✗	✓	✗	✗	✗
Jumio Identity Verification System	✓	✗	✗	✓	✗
VeriScan	✗	✗	✗	✓	✗
BlinkID	✓	✗	✗	✓	✗

Scanova	✗	✗	✗	✓	✗
CodeREADr	✓	✗	✗	✓	✗
Microblink Identity	✓	✗	✗	✓	✗
Proposed System: IDetect	✓	✓	✓	✓	✓

Note: OCR extraction = optical character recognition-based automated data extraction; Real-time OSCA/PDAO verification = real-time cross-database verification with Office of Senior Citizens Affairs and Persons with Disability Affairs Office; Auto fraud reporting = automated bidirectional fraud reporting to local government units; Mobile POS design = mobile-first design optimized for point-of-sale environments; Dual database integration = integration with both OSCA and PDAO databases. iDetect is the only system integrating all five capabilities. This comparison focuses on systems with similar scope (ID verification for discount programs). International digital identity platforms (X-Road, SingPass) are discussed in Section 2.2 as architectural models rather than direct competitors due to different scope and purpose.

Synthesis Matrix

Table 2.

System Matrix Table

Author /Study	OCR Technology & Accuracy	Digital Identity Systems	Fraud Detection Methods	Mobile POS Usability	Philippine Context Challenges	Implications for iDetect
Liu et al. (2022)	99.25% character recognition accuracy on MRZ extraction using CNNs; 100% MRZ detection rate on passports	-	-	-	-	Establishes OCR viability for ID verification; sets accuracy benchmark for iDetect
Gayer et al. (2023)	Real-time MRZ detection at 62 FPS on	-	-	Mobile OCR feasible for real-time point-of-sale	-	Demonstrates mobile OCR is technically feasible for

	mid-range smartphones; 1MB optimized model 83× faster than previous implementations			use		iDetect's client portal
Stančić & Trbušić (2020)	Abbyy FineReader: 95-98% accuracy at 200-500 DPI on typewritten documents	-	-	-	-	Shows image quality directly impacts OCR accuracy; informs preprocessing requirements
Hamdi et al. (2019)	OCR error rates reaching 50% on degraded documents (blurred/skewed images)	-	-	-	-	Necessitates robust preprocessing (deskewing, contrast enhancement) and manual fallback
Lerouge et al. (2024)	2.52% character error rate = only 48.6% field-level accuracy for ID numbers and dates	-	-	-	-	Field-level validation more critical than character-level; requires pattern matching for ID formats
Martinez (2025)	OCR reduces processing time 90-95% (15 min → 1 min per document); 99.95-99.99% accuracy vs 96-99% manual	-	-	-	-	Quantifies time savings and error reduction benefits for iDetect implementation

	entry					
Mays et al. (2019)	Manual transcription error rate: 3.7% in point-of-care testing	-	-	-	-	Manual entry introduces verification inaccuracies; OCR automation eliminates transcription errors that could enable fraudulent credentials to pass verification
Nitayavardhana et al.,2025)	OCR reduced data entry time by 43.9% (6.0 min → 3.4 min)	-	-	-	-	Validates iDetect's efficiency claims; projects similar reductions from 45-60 seconds to under 30 seconds in discount verification
Frost & Sullivan Institute (2024)	-	Estonia's X-Road: 2.7B queries/year; 99% digital ID adoption; distributed architecture maintains agency sovereignty	-	-	-	Provides architectural model: distributed database integration maintaining LGU sovereignty
Margetts & Naumann (2017)	-	Estonia's 'once-only' principle; 2% GDP annual savings; citizens file taxes	-	-	-	Demonstrates measurable efficiency gains from integrated verification systems

		in <3 min				
Goede (2019)	-	Compared Estonia (government-led) vs Singapore (public-private collaboration); both require robust legal frameworks	-	-	-	Shows multiple pathways to digital transformation; informs iDetect's public-private coordination
Protecht Group (2025)	-	Integrated systems achieve 70%+ fraud reduction through coordinated intelligence sharing	Real-time cross-verification and intelligence sharing critical for fraud reduction	-	-	Sets target: 50%+ fraud reduction (conservative vs 70%+ international benchmark)
Ibor et al. (2024)	-	Digital identity interoperability requires data integrity, confidentiality, privacy during exchange	-	-	-	Informs privacy-preserving design: binary valid/invalid without exposing full beneficiary data
Udeh et al. (2024)	-	-	Discusses big data analytics and ML for real-time fraud detection through continuous	-	-	Validates the shift from batch processing to real-time analysis to proactively identify and flag suspicious

			transaction monitoring.			activities.
Curry (2025)	-	-	Identifies performance benchmarks for fraud detection: processing thousands of transactions /sec at sub-100ms latency.	-	-	Defines the technical success criteria for the system's speed and throughput to ensure high-velocity transaction safety.
Bordey (2024)	-	-	8.5M fake PWD IDs vs 1.89M legitimate (4.5:1 ratio); ₱88.2B tax losses in 2023	-	Documents fraud crisis magnitude in Philippines	Establishes problem urgency and economic justification (₱44B recovery potential at 50% reduction)
Gonzales (2024)	-	-	₱166.1B fraudulent discounts in 2023; BIR classifies fake ID use as criminal tax evasion	-	Legal classification strengthens enforcement potential	Supports fraud reporting mechanism to authorities with legal backing
Arceo (2024)	-	-	Major restaurant chain: 17.24% PWD ID rejection rate (726 of 4,212 over 3 months)	-	Industry data quantifies fraud prevalence	Validates need for automated verification; establishes baseline metrics

Arnaldo (2024)	-	-	PWD discount claims increased from 5% (2019) to 30% (2024); 17% rejection rate at establishments	-	22.21% increase in PWD-to-Senior ratio (2022-2024)	Documents fraud trend escalation requiring urgent intervention
Bonneau et al. (2024)	-	-	Multi-factor authentication: 77% achieve NIST compliance; behavioral biometrics enable continuous authentication	-	-	MFA principles inform field-level validation and pattern recognition
Kumar & Singh (2024)	-	-	MFA + ML: 97.938% accuracy (logistic regression); 97.881% (decision trees); 96.717% (random forest)	-	-	ML fraud detection achievable; informs future pattern recognition enhancements
Wang et al. (2024)	-	-	Ensemble methods: 94.3% fraud detection accuracy; neural networks: 93.7% accuracy	-	-	Establishes multi-layer detection feasibility for behavioral pattern analysis
Baymard Institute	-	-	-	0% of 138 mobile commerce	-	Sets usability challenge: deliberate

(2024)				sites achieved 'good' UX; 62% rated mediocre/worse		design needed to exceed poor industry baseline
Thompson et al. (2023)	-	-	-	Three-action limit: systems requiring >3 actions face significant adoption resistance	-	Constrains workflow to launch-scan-view; necessitates auto-authentication
Harrison et al. (2024)	-	-	-	PACMAD model: performance, accessibility, compatibility, mobility + traditional metrics	-	Informs comprehensive usability evaluation framework for iDetect
Nielsen & Molich (2016)	-	-	-	12 POS usability factors: learnability, efficiency, memorability, errors, satisfaction, etc.	-	Defines evaluation criteria for mobile interface design
Cervantes (2025)	-	-	DOJ Jan 2024: establishments cannot deny discounts for unverifiable credentials	-	Legal constraint in Philippines	Requires provisional approval architecture vs gatekeeping; intelligence gathering approach
Alampay et al. (2018)	-	-	-	-	Only 60% of Philippine LGUs have operational PDAOs (33 provinces, 25 cities, 282 municipalit	Necessitates graduated deployment starting with digitized urban LGUs; explains incomplete database coverage

					ies lacking)	
Respicio Law (2024)	-	-	-	-	Senior citizen IDs have no standard design; formats vary by municipality across 1,780+ LGUs	Requires OCR optimization for non-standardized credentials vs international systems designed for standardized docs
Inabangan et al. (2019)	-	-	-	-	Filipino cultural factors: 'jackpot mentality, helpfulness, shyness' affect discount program implementation	Informs non-confrontational verification design; flags presented as data quality issues not fraud accusations

Project Synthesis and Research Gap Identification

The synthesis of technical background, thematic literature review, and existing system analysis establishes both the necessity and feasibility of iDetect while revealing critical gaps in current verification infrastructure. This section integrates findings across research themes to articulate iDetect's unique contribution to Philippine discount program verification, positioning the system as response to documented technological capabilities, implementation challenges, and systemic fraud reaching ₱88.2 billion annually in lost tax revenues.

Key Findings from Literature Review

The literature review reveals five fundamental findings informing iDetect's design:

OCR Technology and Accuracy: The foundational viability of the iDetect system is supported by the high-performance standards of modern Optical Character Recognition (OCR). Research by Liu et al. (2022) establishes a benchmark of 99.25% character recognition accuracy, suggesting that automated extraction is a reliable alternative to manual entry. Industry research further demonstrates that OCR-based systems achieve accuracy rates of 99.95-99.99% compared to manual entry accuracy of 96-99%, with manual transcription error rates reaching 3.7% in laboratory settings, of which 14.2% contain potentially dangerous discrepancies (Martinez, 2025; Mays et al., 2019). However, the transition from laboratory settings to point-of-sale environments introduces significant variables. Hamdi et al. (2019) and Stančić & Trbušić (2020) observe that environmental factors such as blurred or skewed images can cause accuracy to degrade as low as 50% error rate. Furthermore, Lerouge et al. (2024) posits that field-level accuracy, specifically for critical data like ID numbers, is more difficult to maintain than individual character recognition. These findings necessitate iDetect's focus on robust preprocessing and pattern-matching algorithms. A multi-center study across ICUs demonstrated that well-implemented OCR systems can reduce data entry time by an average of 43.9%, with processing time decreasing from 6.0 minutes to 3.4 minutes per document (Nitayavardhana et al., 2025), supporting the feasibility of significant efficiency gains in automated receipt verification systems.

Digital Identity Verification Systems and E-Government Integration The architectural philosophy of iDetect mirrors international best practices in e-government. Estonia's X-Road serves as a primary model, demonstrating how a distributed architecture can maintain agency sovereignty while facilitating billions of annual queries (Frost & Sullivan Institute, 2024). Whether through government-led models or public-private collaborations, successful integration requires robust legal frameworks and the "once-only" principle to maximize efficiency (Goede, 2019; Margetts & Naumann, 2017). Real-time financial fraud detection systems similarly employ distributed architectures to achieve sub-100 millisecond latency through in-memory caching and optimized query patterns (Curry, 2025; Udeh et al., 2024). iDetect adopts these architectural principles for instant verification."

Fraud Detection and Prevention in Discount Programs The urgency for an integrated verification system is driven by the severe economic impact of identity fraud. Bordey (2024) and Gonzales (2024) document a crisis in the Philippines where fraudulent PWD IDs outnumber legitimate ones by a 4.5:1 ratio, resulting in approximately ₱88.2 billion in tax losses in 2023. While integrated international systems have achieved 70% fraud reduction through coordinated intelligence sharing (Protecht Group, 2025), iDetect adopts a targeted approach to the local retail sector. By utilizing ensemble detection methods and behavioral pattern analysis approaches that Wang et al. (2024) and Kumar & Singh (2024) prove can reach detection accuracies of over 94% the system provides a proactive defense against the unauthorized use of discount privileges.

Mobile Point-of-Sale Verification Solutions and Usability Requirements The practical utility of iDetect depends on its seamless integration into the fast-paced retail workflow.

Baymard Institute (2024) notes that the majority of mobile commerce interfaces fail to meet basic usability standards, which leads to significant adoption resistance. As Thompson et al. (2023) suggests, systems requiring more than three user actions are likely to be abandoned by staff. iDetect addresses these constraints by leveraging optimized mobile OCR models that achieve 62 FPS on mid-range devices (Gayer et al., 2023) and following the PACMAD usability model (Harrison et al., 2024) to ensure that the verification process remains efficient under the pressures of a point-of-sale environment.

Philippine Senior Citizen and PWD Discount Implementation Challenges Finally, the system must navigate a highly fragmented domestic landscape. Respicio Law (2024) highlights that Senior Citizen IDs lack a standard design across the 1,780+ LGUs in the Philippines, while Alampay et al. (2018) notes that many municipalities still lack operational PDAO offices. This lack of standardization is complicated by legal mandates; Cervantes (2025) reports that establishments are legally prohibited from denying discounts based solely on unverifiable credentials. Moreover, Inabangan et al. (2019) emphasizes cultural factors such as "helpfulness" and "shyness" that can impede strict enforcement. iDetect therefore utilizes a non-confrontational flagging architecture, prioritizing data quality and intelligence gathering over direct rejection to remain compliant with local regulations and cultural sensitivities.

The synthesis of current literature reveals a critical research gap: while high-accuracy OCR and distributed identity models exist globally, they have not been integrated into a mobile-first solution tailored to the fragmented and legally complex Philippine discount program ecosystem. Current manual verification methods are insufficient to stem the billions in tax

leakages caused by a 4.5:1 fraud-to-legitimacy ratio. By harmonizing technical feasibility with the specific legal constraints and cultural nuances of the Philippine retail sector, iDetect provides a unique, real-time verification tool that balances efficient business operations with robust fraud detection. This research fills the void between high-level e-government theory and the practical, day-to-day requirements of local business establishments.

Critical Limitations of Existing Verification Systems

Existing systems, both local and international, present substantial functional gaps preventing comprehensive discount program fraud prevention:

Local government platforms (OSCA Taguig, PDAO Taguig, VALID Valenzuela) provide real-time database verification but remain fundamentally limited by manual data entry requirements consuming 30-45 seconds per transaction, geographic restriction to single municipalities preventing cross-LGU verification needed by regional businesses, lack of OCR automation forcing staff to manually type ID details from physical cards, and operation as one-way verification tools without automated fraud reporting mechanisms enabling suspicious credentials to circulate undetected across multiple establishments even after initial detection. Most critically, these systems function as isolated municipal solutions rather than integrated networks, preventing intelligence sharing and coordinated fraud response.

National systems (eVerify) focus on PhilSys credentials rather than LGU-issued Senior Citizen and PWD IDs, creating fundamental scope misalignment with discount verification requirements. The system's institutional authorization model proves impractical for small retail

establishments, while absence of mobile-optimized workflows and OCR capabilities limits point-of-sale applicability.

International commercial solutions (Jumio, VeriScan, BlinkID, Microblink) demonstrate advanced technical capabilities including multi-modal scanning, global document coverage, and robust data extraction, but present critical limitations for Philippine context: designed for international standardized documents rather than non-standardized Philippine LGU credentials, lack integration with local OSCA and PDAO databases requiring custom development work, absence of fraud reporting mechanisms enabling coordination between businesses and government authorities, and commercial subscription models potentially prohibitive for small establishments. These systems excel at technical data extraction but lack the government-ecosystem integration essential for coordinated fraud prevention.

The fundamental gap: no existing system integrates automated OCR-based data capture, real-time cross-database verification with both OSCA and PDAO, mobile-first point-of-sale design optimized for transaction speed, and bidirectional fraud reporting creating intelligence-sharing networks between businesses and government enforcement authorities. Current solutions address verification in isolation without creating the coordinated fraud prevention ecosystem necessary to combat systematic exploitation reaching ₱88.2 billion annually.

iDetect's Unique Contribution and Innovation

iDetect addresses identified gaps through integrated multi-portal architecture synthesizing proven technological approaches with Philippine-specific implementation requirements:

Automated OCR-based verification workflow. The Client Portal (mobile application) enables business establishments to capture ID images through smartphone cameras, automatically extract data using Tesseract OCR with field-level validation, and receive verification results in under 30 seconds reducing transaction time by 40-60% compared to manual entry systems. This automation directly addresses the operational burden documented in local platforms while implementing mobile-first design absent in national systems. Critical distinction from commercial solutions: optimization for Philippine LGU-issued credentials rather than international standardized documents, with preprocessing specifically tuned for non-standardized formats and manual fallback enabling operation when OCR confidence is insufficient.

Real-time dual-database integration. Unlike existing platforms verifying single credential types or operating in isolated municipalities, iDetect integrates simultaneously with both OSCA and PDAO databases enabling comprehensive discount verification through single application. The distributed database architecture following Estonia's X-Road principles maintains LGU sovereignty while enabling cross-verification Valenzuela OSCA and PDAO maintain independent databases accessed through a secure API layer rather than attempting centralization that would face institutional resistance. Privacy-preserving design returns binary valid/invalid confirmation without exposing comprehensive beneficiary data to businesses, balancing verification requirements against data protection obligations.

Bidirectional fraud reporting and intelligence sharing. iDetect's most significant innovation: automated mechanisms enabling suspicious credentials flagged by establishments to

be immediately transmitted to appropriate LGU portal (OSCA for Senior Citizen IDs, PDAO for PWD IDs) for investigation, with confirmed fraudulent records synchronized back into official databases and propagated across all system users. This creates an active fraud prevention ecosystem rather than passive verification, transforming disconnected verification points into coordinated intelligence networks. When establishment flags suspicious credentials, the system automatically notifies LGU administrators, provides establishment contact for investigation follow-up, updates database verification status preventing future use, and generates fraud analytics enabling pattern recognition across establishments capabilities absent in all reviewed systems.

Multi-portal role-based architecture. iDetect operates through four interconnected portals serving distinct stakeholder needs: Client Portal (mobile) for business verification, BPLO Portal (web) for business account management and technical oversight without accessing beneficiary data, OSCA Portal (web) for Senior Citizen database management and fraud investigation, and PDAO Portal (web) for PWD database management and fraud investigation. This separation maintains clear functional boundaries: business regulators oversee system operations without accessing verification intelligence, fraud investigators access relevant beneficiary data without business operational details, and businesses receive verification services without comprehensive database access. The architecture enables coordination while maintaining appropriate information compartmentalization protecting privacy and institutional sovereignty.

Philippine context adaptation. Critical design decisions directly address local implementation challenges: graduated deployment strategy beginning with Metro Manila

digitized LGUs accommodates 60% PDAO coverage reality, provisional approval architecture complying with DOJ prohibition on denying unverifiable credentials balances legal requirements with fraud detection through intelligence gathering, multi-layer detection combining database verification with behavioral pattern analysis compensates for incomplete database coverage, and non-confrontational interface design presenting flags as data quality issues rather than fraud accusations addresses cultural sensitivity requirements. The system builds incrementally toward comprehensive verification while delivering immediate fraud reduction value to participating LGUs and establishments.

Research Justification and Expected Contributions

The convergence of the documented fraud crisis (₱88.2 billion annual tax losses), proven technological capabilities (95-99% OCR accuracy, 70%+ fraud reduction through integrated systems), and identified verification gaps establishes compelling justification for iDetect development. The system addresses documented needs across multiple stakeholder groups:

Business establishments gain efficient verification tools reducing transaction time 40-60%, decreasing fraudulent discount absorption, and enabling compliance with legal verification obligations without requiring multiple platform usage or extensive staff training.

Local government units receive fraud intelligence unavailable in current systems, enabling targeted investigations, database accuracy improvements, and evidence supporting enforcement actions against systematic fraud transforming passive verification into active fraud prevention.

Legitimate beneficiaries experience faster verification without prolonged scrutiny, consistent treatment across participating establishments, and protection against wrongful denial as verification becomes more reliable and less dependent on individual establishment judgment.

Government revenue protection through substantial fraud reduction even 50% reduction in fraudulent transactions would recover ₱44 billion annually in tax revenues while reducing business discount absorption by ₱83 billion, creating measurable economic impact justifying implementation investment.

Beyond immediate operational benefits, iDetect contributes to broader digital transformation of Philippine local governance. The system demonstrates a scalable model for LGU-business coordination addressing social service delivery challenges through integrated technology rather than isolated solutions. Implementation experience will generate empirical data addressing research gaps identified in literature review: verification effectiveness in non-standardized ID ecosystems, public-private partnership implementation at micro-enterprise scale, fraud detection patterns specific to Philippine discount programs, and trust-building mechanisms in contexts characterized by institutional fragmentation and established fraudulent credential markets.

Success Metrics and Evaluation Framework

The system's success will be measured across multiple dimensions reflecting its comprehensive approach. Technical performance metrics include OCR accuracy rates under operational conditions, average verification time from credential capture to result display, and system uptime and reliability. Operational effectiveness metrics encompass fraud reduction

percentages comparing flagged suspicious credentials before and after implementation, transaction processing time improvements measured at participating establishments, and database verification adoption rates tracking the proportion of transactions receiving authoritative confirmation. Stakeholder adoption metrics monitor business establishment enrollment and active usage rates, staff training completion and proficiency levels, and user satisfaction scores from both establishments and legitimate beneficiaries. Fraud intelligence metrics track suspicious credential reports submitted, investigation outcomes and confirmation rates, and fraud pattern identification enabling systemic interventions. These multidimensional success criteria demonstrate that technology alone proves insufficient without corresponding attention to usability, governance, and contextual adaptation. iDetect's integrated approach synthesizing OCR automation, database integration, mobile usability, fraud intelligence sharing, and Philippine-specific design adaptations represents the comprehensive solution architecture required to address systematic verification failures enabling the documented ₱88.2 billion annual fraud crisis.

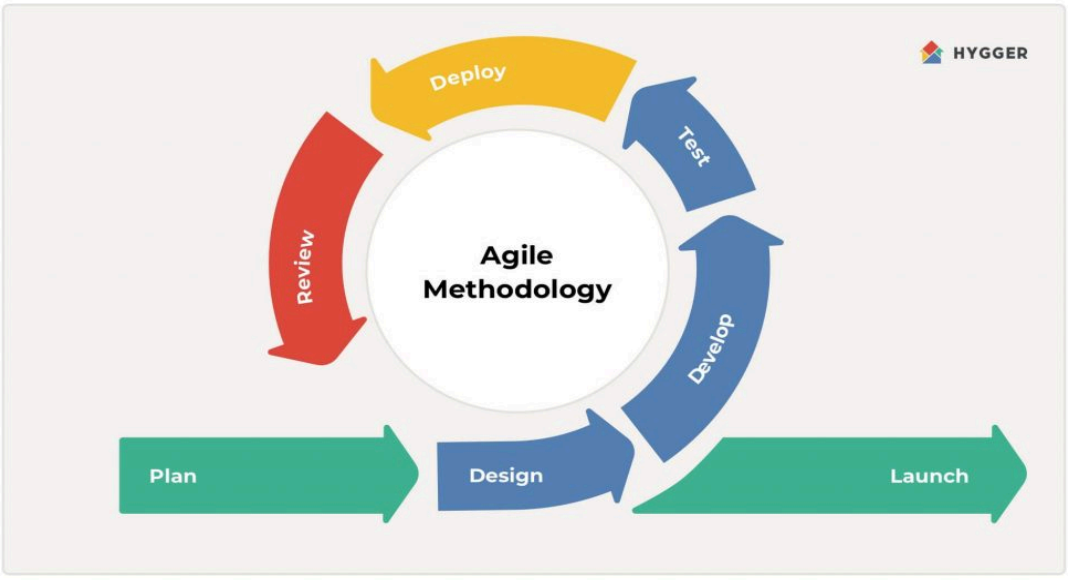
This comprehensive literature review and system analysis establishes the foundation for iDetect's design and methodology presented in Chapter 3. The integration of proven OCR technologies, established database architectures, documented fraud detection methodologies, and Philippine-specific implementation considerations creates a research-informed approach addressing the ₱88.2 billion fraud crisis while respecting the unique constraints of Philippine local government capacity and legal frameworks.

CHAPTER 3 DESIGN AND METHODOLOGY

Development Model (NA)

Figure 6.

Agile Methodology



The development of IDetect will follow the Agile methodology, which emphasizes iterative development, continuous feedback, and adaptive planning. The system will be developed through multiple sprint cycles, with each sprint focusing on specific features and components. The development process includes requirements gathering from business establishments and LGU offices (OSCA and PDAO), system design, implementation of OCR scanning functionality, backend and database integration, automated flagging features, and mobile application development.

Each sprint will conclude with testing iterations to ensure accuracy, usability, and reliability of implemented features, with feedback incorporated into subsequent sprints. Stakeholder reviews and demonstrations will occur at the end of each sprint cycle, allowing for continuous alignment with user needs and requirements. This iterative approach enables flexibility in addressing challenges, such as database access arrangements, and allows the team to adapt to technical constraints or changing requirements while maintaining systematic progress toward a fully functional iDetect mobile application.

Requirement Analysis/Documentation

This presents the Requirements Analysis for iDetect, an OCR-based Discount ID Verification Application developed as a capstone project at the Pamantasan ng Lungsod ng Valenzuela. The system aims to address the widespread misuse of fake Senior Citizen and PWD identification cards used to fraudulently claim government-mandated discounts under Republic Acts 9994 and 10754.

iDetect works by using Optical Character Recognition (OCR) to automatically scan and extract data from physical IDs, verify them against government databases in real time, and enable coordinated fraud reporting between businesses and local government units. The system is composed of five portals: the Client Portal (mobile app for business establishments to verify IDs at point of sale), the BPLO Portal (web app for managing registered business accounts), the OSCA Portal (web app for the Office of Senior Citizens Affairs to manage Senior Citizen ID records and fraud flags), the PDAO Portal (web app for the Persons with Disability Affairs

Office to manage PWD records), and the Beneficiary Mobile App (for Senior Citizens and PWDs to check their verification status and dispute wrongful denials).

Functional Requirements

1. Logging In and Authenticating Users

1.1 Description and Priority

This feature allows all system users: establishment staff, BPLO officers, OSCA administrators, PDAO administrators, and beneficiaries to securely log in to their respective portals using role-specific credentials. Two-factor authentication is enforced for beneficiary accounts to balance security with accessibility.

Priority: High

1.2 Stimulus / Response

1.2.1 Positive Flow

1.2.1.1 An establishment staff enters their registered username and password in the Client Portal login screen.

Response: The system shall authenticate the credentials, verify the establishment's active business permit status, and grant access to the verification dashboard.

1.2.1.2 A beneficiary enters their ID number and birthdate, then submits the OTP sent to their registered mobile number.

Response: The system shall validate both authentication factors against the OSCA/PDAO database records and grant access to the Beneficiary Mobile Application.

1.2.1.3 A BPLO, OSCA, or PDAO administrator enters their institutionally issued credentials in their respective portal login screen.

Response: The system shall authenticate the credentials against the administrator account database, verify the account is active, and grant access to the appropriate administrative portal dashboard.

1.2.2 Negative Flow

1.2.2.1 A user enters incorrect login credentials three consecutive times.

Response: The system shall temporarily lock the account for a defined cooling-off period and display a lockout notification with instructions to contact their administrator.

1.2.2.2 A beneficiary enters a correct ID number and birthdate but submits an expired or incorrect OTP.

Response: The system shall deny access and prompt the beneficiary to request a new OTP, with a maximum resend limit per session.

1.3 Functional Requirements

REQ-1: The system shall authenticate establishment staff credentials by verifying the username and password against the account database, and confirm that the associated establishment account has an approved and active status within the Client Portal.

REQ-2: The system shall implement two-factor authentication for beneficiaries by requiring their ID number paired with their birthdate as the primary factor, and a one-time PIN delivered to their registered mobile number as the secondary factor within the login workflow.

REQ-3: The system shall pre-configure administrator accounts for BPLO, OSCA, and PDAO officers prior to system deployment, with each office responsible for managing its own personnel credentials through its respective portal, and no cross-portal administrative access permitted.

REQ-4: The system shall enforce role-based access control so that each authenticated user type: establishment staff, BPLO administrator, OSCA administrator, PDAO administrator, and beneficiary can only access portal features and data corresponding to their assigned role.

REQ-5: The system shall temporarily lock a user account after three consecutive failed login attempts and notify the account holder through their registered contact, requiring administrator intervention or a cool-down period before access is restored.

2. Provisioning and Managing Establishment Accounts

2.1 Description and Priority

This feature enables BPLO officers to directly create, manage, and control iDetect Client Portal accounts for food service establishments in Valenzuela City. Rather than self-registering establishments, BPLO initiates account creation on behalf of eligible establishments, both newly registered businesses and existing ones — upon verification of a valid Valenzuela City business permit. BPLO officers retain full authority to activate, suspend, or revoke establishment accounts at any time, without accessing beneficiary data or verification intelligence.

Priority: High

2.2 Stimulus / Response

2.2.1 Positive Flow

2.2.1.1 A food service establishment (new or existing) approaches the BPLO office and requests iDetect access. The BPLO officer retrieves or verifies the establishment's business permit from the BPLO registry and initiates account creation on behalf of the establishment.

Response: The system shall allow the BPLO officer to create an establishment account by entering the valid business permit number and designated staff credentials directly within the BPLO Portal, and shall immediately activate the account upon successful submission.

2.2.1.2 The BPLO officer completes account creation and provides the establishment's staff with their login credentials.

Response: The system shall generate and display the newly created account credentials to the BPLO officer, send an automated notification to the establishment (via email or SMS) containing their login details and instructions for accessing the Client Portal verification dashboard.

2.2.1.3 An establishment staff initiates a password reset by entering their registered email or mobile number on the Client Portal login screen.

Response: The system shall send a one-time reset link or PIN to the registered contact, allow the staff to set a new password upon verification, and log the reset event in the system audit trail.

2.2.2 Negative Flow

2.2.2.1 A BPLO officer attempts to create an account using a business permit number that is not found in the BPLO registry, is expired, or does not correspond to a food service establishment in Valenzuela City.

Response: The system shall block account creation, display a specific error message identifying the invalid or ineligible permit,

and prevent the record from being saved until a valid permit is entered.

2.2.2.2 A BPLO officer suspends or revokes an active establishment account due to a policy violation, business permit expiry, or closure.

Response: The system shall immediately revoke the establishment's Client Portal access, notify the establishment staff of the suspension or revocation with a stated reason, and log the action including the BPLO officer's identity, timestamp, and reason in the audit trail.

2.2.2.3 An establishment staff cannot complete self-service password reset because their registered contact details are outdated or inaccessible.

Response: The system shall direct the staff to contact their BPLO officer, who may update the registered contact details and trigger a new reset through the BPLO Portal.

2.3 Functional Requirements

REQ-6: The system shall allow a BPLO officer to directly create a Client Portal account for a food service establishment by entering a valid Valenzuela City business permit number and designated staff credentials within the BPLO Portal, covering both newly registered and existing business establishments.

REQ-7: The system shall validate the submitted business permit number against the BPLO registry in real time, and shall block account creation if the permit is not found, is expired, or belongs to a non-food-service establishment outside Valenzuela City.

REQ-8: Upon account creation, the system shall automatically deliver login credentials and onboarding instructions to the establishment's registered contact (via email or SMS), and shall record the creating officer, timestamp, and associated permit number in the system log.

REQ-9: The BPLO officer shall have authority to activate, suspend, or revoke any establishment account at any time through the BPLO administrative dashboard, without accessing flagged ID records, beneficiary personal information, or verification transaction records.

REQ-10: The system shall send automated notifications to establishment staff upon any account status change (activation, suspension, revocation) initiated by a BPLO officer, including the stated reason for any adverse action.

REQ-11: The system shall prevent BPLO administrators from viewing flagged credential data, beneficiary personal information, or verification transaction records at all times, maintaining strict data compartmentalization between account management and verification intelligence functions.

REQ-12: The BPLO Portal shall display an operational dashboard showing total active establishment accounts, recently provisioned accounts, pending administrative actions,

and technical performance metrics to support BPLO oversight responsibilities without exposing any verification or beneficiary data.

REQ-13: The system shall allow an establishment staff to reset their Client Portal password through a self-service mechanism by submitting their registered email address or mobile number, receiving a one-time reset credential, and setting a new password within a defined expiry window, with the event logged in the audit trail.

REQ-14: The BPLO officer shall have authority to update an establishment's registered contact details and trigger a credential reset through the BPLO Portal when self-service password reset cannot be completed by the establishment staff.

REQ-15: The system shall monitor the business permit expiry dates of all active establishment accounts and display a flagged alert on the BPLO Portal dashboard for any establishment whose permit is approaching expiry within 30 days or has already expired, without automatically suspending the account, so that the BPLO officer may take appropriate manual action.

3. Capturing and Preprocessing ID Images

3.1 Description and Priority

This feature allows establishment staff to capture a Senior Citizen or PWD ID image using the Client Portal mobile application camera. The system performs real-time image quality assessment and applies preprocessing steps: deskewing, contrast enhancement, and noise reduction to maximize OCR accuracy before extraction.

Priority: High

3.2 Stimulus / Response

3.2.1 Positive Flow

3.2.1.1 An establishment staff opens the Client Portal and points the device camera at a Senior Citizen or PWD identification card.

Response: The system shall display a real-time camera viewfinder with on-screen guidance indicating optimal ID positioning, lighting conditions, and distance, and provide instant visual feedback confirming image quality is sufficient before capturing.

3.2.1.2 The establishment staff captures an image that passes the image quality assessment threshold.

Response: The system shall automatically apply preprocessing algorithms including deskewing, contrast enhancement, and noise reduction to the captured image before passing it to the OCR extraction engine.

3.2.2 Negative Flow

3.2.2.1 The establishment staff captures an image that is blurry, poorly lit, or significantly skewed beyond correctable limits.

Response: The system shall display an image quality warning, reject the captured image, and prompt the staff to recapture the ID under better conditions before proceeding to OCR extraction.

3.2.2.2 The establishment staff attempts to capture a physically damaged or heavily worn ID with illegible sections.

Response: The system shall notify the staff that image quality is insufficient for automated OCR processing and offer the manual data entry fallback option to continue the verification process.

3.3 Functional Requirements

REQ-16: The system shall provide an establishment staff with a real-time camera viewfinder overlay displaying guidance on ID positioning, distance, and lighting to facilitate optimal image capture before OCR processing within the Client Portal mobile application.

REQ-17: The system shall perform automated image quality assessment on each captured frame and reject images that fall below the defined quality threshold, prompting the establishment staff to recapture the ID before extraction proceeds.

REQ-18: The system shall apply an automated preprocessing pipeline including deskewing, contrast enhancement, and noise reduction to all accepted captured ID images before passing them to the OCR engine to maximize extraction accuracy against non-standardized Philippine LGU credential formats.

REQ-19: The system shall allow an establishment staff to switch to manual data entry as a fallback mechanism when OCR confidence scoring falls below the acceptable processing threshold, enabling continuation of the verification workflow without automated extraction.

REQ-20: The Client Portal mobile application shall support Android 8.0 and later and iOS 12.0 and later to ensure compatibility with devices used across food service establishment environments in Valenzuela City.

4. Extracting and Validating OCR Data Fields

4.1 Description and Priority

This feature covers the automatic extraction of credential data from preprocessed ID images using OCR engine, applying confidence scoring and field-level pattern validation to verify that extracted data conforms to expected formats before presenting results for staff confirmation.

Priority: High

4.2 Stimulus / Response

4.2.1 Positive Flow

4.2.1.1 The preprocessed ID image is passed to the OCR engine by the Client Portal.

Response: The system shall automatically extract the ID number, full name, beneficiary type, validity period, and issuing office

fields, apply confidence scoring to each extracted field, and display the extracted data on the staff's screen for confirmation before database verification.

4.2.1.2 All extracted fields meet or exceed the confidence threshold and pass field-level pattern validation.

Response: The system shall proceed automatically to real-time database verification using the extracted credential data without requiring additional staff input.

4.2.2 Negative Flow

4.2.2.1 The OCR engine extracts data but one or more fields fall below the confidence threshold or fail field-level pattern validation.

Response: The system shall highlight the low-confidence or invalid fields, retain extractable data in the form, and prompt the establishment staff to manually correct or confirm the flagged fields before proceeding to verification.

4.2.2.2 The OCR engine is unable to extract any usable data from the preprocessed image.

Response: The system shall display an extraction failure message and activate the manual data entry fallback, allowing the staff to type in the credential details directly.

4.3 Functional Requirements

REQ-21: The system shall use OCR engine to automatically extract the ID number, full name, beneficiary type, validity period, and issuing office from preprocessed ID images, applying confidence scoring to each extracted field within the data extraction workflow.

REQ-22: The system shall apply field-level pattern validation to all OCR-extracted data, verifying that ID number format, date ranges, and issuing office codes conform to expected Philippine LGU credential structures before proceeding to database verification.

REQ-23: The system shall display OCR-extracted fields to the establishment staff for confirmation only when one or more fields carry a low confidence score or fail field-level pattern validation, highlighting the affected fields for manual review before proceeding to database verification. Fields that pass all confidence and validation checks shall not require additional staff confirmation.

REQ-24: The system shall allow an establishment staff to manually edit any pre-populated field returned by the OCR engine before submitting data for database verification, ensuring staff can correct OCR errors without restarting the capture process.

REQ-25: The system shall proceed to real-time database verification after the staff's confirmation when all extracted fields pass confidence scoring and field-level pattern validation, maintaining the three-action workflow limit of launch, scan, and view result.

5. Verifying Credentials Against Dual Databases

5.1 Description and Priority

This feature performs real-time simultaneous queries against both the OSCA and PDAO databases using the extracted credential data, returning a verification status: VALID, SUSPICIOUS, or INVALID to the establishment staff within 30 seconds without exposing comprehensive beneficiary personal data.

Priority: High

5.2 Stimulus / Response

5.2.1 Positive Flow

5.2.1.1 The establishment staff confirms the OCR-extracted or manually entered credential data and initiates database verification.

Response: The system shall simultaneously query both the OSCA and PDAO databases through secure RESTful API integration and display the verification result: VALID, SUSPICIOUS, or INVALID with a clear visual indicator within 30 seconds from submission.

5.2.1.2 The database query returns a VALID status for the submitted credential.

Response: The system shall display a VALID result with green visual confirmation to the establishment staff and log the transaction in the audit history, without revealing any comprehensive personal data beyond the verification outcome.

5.2.2 Negative Flow

5.2.2.1 The database query returns an INVALID status because the submitted ID number does not match any record in the OSCA or PDAO database.

Response: The system shall display an INVALID result with a red visual indicator, inform the establishment staff of the non-match outcome, and prompt the option to flag the credential as suspicious for LGU investigation.

5.2.2.2 The database query cannot be completed because the network connection is unavailable during the verification attempt.

Response: The system shall display a connectivity error message, check the local cache for a recently verified matching record, and if found, display a cached result with a timestamp of the last online verification.

5.2.2.3 The OSCA and PDAO databases return conflicting results for the same submitted ID number for example, a record is found in one database but not the other.

Response: The system shall return a SUSPICIOUS status to the establishment staff, log the conflict details in the audit trail, and automatically route a discrepancy alert to both the OSCA and PDAO portals for administrative review.

5.3 Functional Requirements

REQ-26: The system shall simultaneously query both the OSCA and PDAO databases through secure RESTful API integration using the submitted ID number and beneficiary type to determine verification status within the dual-database verification engine.

REQ-27: The system shall complete the full verification cycle from credential data submission to result display within 30 seconds under normal network conditions, meeting the real-time point-of-sale performance requirement.

REQ-28: The system shall return a binary verification status of VALID, SUSPICIOUS, or INVALID to the establishment staff without exposing comprehensive beneficiary personal information, implementing a privacy-preserving query design throughout the verification response.

REQ-29: The system shall display the verification result using distinct color-coded visual indicators, green for VALID, amber for SUSPICIOUS, and red for INVALID to enable rapid result interpretation by establishment staff in high-volume point-of-sale environments.

REQ-30: The system shall log each verification transaction including the credential data submitted, verification status returned, timestamp, and establishment identifier in the audit history database immediately upon completion of the verification cycle.

REQ-31: The system shall return a SUSPICIOUS verification status and generate an automatic discrepancy alert routed to both the OSCA and PDAO portals when a

submitted credential produces conflicting results across the two databases, logging the conflict details in the audit trail for administrative resolution.

6. Flagging and Reporting Suspicious Credentials

6.1 Description and Priority

This feature allows establishment staff to flag suspicious or invalid credentials with a single tap after a verification transaction. The system automatically routes the flag report including the captured ID image, extracted data, establishment identifier, and timestamp to the appropriate LGU portal for investigation.

Priority: High

6.2 Stimulus / Response

6.2.1 Positive Flow

6.2.1.1 An establishment staff taps the flag button after receiving a SUSPICIOUS or INVALID verification result.

Response: The system shall display a confirmation prompt and, upon confirmation, transmit the captured ID image, OCR-extracted data, establishment identifier, transaction timestamp, and verification outcome to the OSCA Portal for Senior Citizen IDs or the PDAO Portal for PWD IDs.

6.2.1.2 The flag report is successfully transmitted to the appropriate LGU portal.

Response: The system shall display a confirmation message to the establishment staff, log the flag submission in the establishment's transaction history, and generate a reference number for tracking the report status.

6.2.2 Negative Flow

6.2.2.1 An establishment staff attempts to submit a flag report while the device has no network connection.

Response: The system shall queue the flag report locally on the device with all associated data and automatically transmit it to the appropriate LGU portal upon connectivity restoration, notifying the staff of the pending submission status.

6.2.2.2 An establishment staff accesses the flag submission interface but decides not to proceed and cancels the action.

Response: The system shall discard the flag submission without transmitting any data to LGU portals and return the staff to the verification result screen without logging a flag event.

6.2.2.3 An establishment staff attempts to flag a credential that returned INVALID because no matching record was found in either database, leaving the beneficiary type unconfirmed.

Response: The system shall prompt the establishment staff to manually select whether the credential is a Senior Citizen ID (route

to OSCA) or a PWD ID (route to PDAO) before allowing flag submission.

6.3 Functional Requirements

REQ-32: The system shall allow an establishment staff to flag a suspicious credential with a single tap from the verification result screen, routing the report automatically to the OSCA Portal for Senior Citizen IDs or the PDAO Portal for PWD IDs based on the beneficiary type detected during verification.

REQ-33: The system shall include the captured ID image, OCR-extracted data fields, establishment identifier, transaction timestamp, and verification outcome in every fraud flag transmission sent to the appropriate LGU portal.

REQ-34: The system shall generate a unique reference number for each submitted flag report, display it to the establishment staff upon successful transmission, and associate it with the corresponding transaction entry in the audit history.

REQ-35: The system shall queue fraud flag reports locally when the mobile device is offline and automatically submit them to the appropriate LGU portal upon connectivity restoration, preserving all associated data throughout the queuing period.

REQ-36: The system shall allow an establishment staff to access the establishment's transaction history and compliance report summary from the Client Portal to support internal audit and regulatory compliance monitoring.

REQ-37: The system shall prompt the establishment staff to manually select the beneficiary type: Senior Citizen or PWD before routing a flag report when the verification returned INVALID due to no record found in either database and the credential type could not be confirmed automatically.

REQ-38: The system shall detect when a flag report is submitted for a credential that already has an active open investigation in the OSCA or PDAO portal, and shall log the new submission as a supplementary report linked to the existing investigation record rather than creating a duplicate case.

7. Managing Senior Citizen Records and Investigating Fraud (OSCA)

7.1 Description and Priority

This feature enables OSCA administrators to manage the Senior Citizen ID database, receive real-time notifications of flagged credentials with complete transaction context, review and investigate suspicious records, and mark confirmed fraudulent IDs to prevent future successful verifications across the network.

Priority: High

7.2 Stimulus / Response

7.2.1 Positive Flow

7.2.1.1 A flag report for a Senior Citizen ID is transmitted from the Client Portal to the OSCA Portal.

Response: The system shall immediately notify the OSCA administrator through the portal dashboard, displaying the captured ID image, extracted data fields, flagging establishment details, and transaction timestamp for review.

7.2.1.2 An OSCA administrator reviews a flagged credential and confirms it as fraudulent.

Response: The system shall update the credential status in the OSCA database to FRAUDULENT, propagate the status change in real time to all connected Client Portal instances, and prevent future VALID outcomes for that credential.

7.2.2 Negative Flow

7.2.2.1 An OSCA administrator reviews a flagged credential and determines it is legitimate after investigation.

Response: The system shall dismiss the flag, restore the VALID status in the database, and notify the establishment that submitted the flag of the investigation outcome.

7.2.2.2 An OSCA administrator attempts to access PWD ID records or PDAO Portal data from within the OSCA Portal.

Response: The system shall deny access and display an authorization error, enforcing strict data compartmentalization that limits OSCA administrators to Senior Citizen records only.

7.3 Functional Requirements

REQ-39: The OSCA Portal shall receive real-time notifications of each flag report transmitted from the Client Portal for Senior Citizen IDs, displaying the captured image, extracted data, flagging establishment identifier, and transaction timestamp within the OSCA investigation dashboard.

REQ-40: The OSCA administrator shall investigate flagged credentials and mark confirmed fraudulent IDs as FRAUDULENT within the OSCA Portal, triggering an immediate real-time database status update that propagates across all connected Client Portal instances to prevent future VALID verifications of that credential.

REQ-41: The system shall restrict OSCA Portal access to Senior Citizen ID records only, preventing OSCA administrators from viewing PWD records, PDAO portal data, or BPLO business account information at all times through enforced data compartmentalization.

REQ-42: The OSCA Portal shall enable administrators to add, update, and manage Senior Citizen ID records in the OSCA database, including status changes reflecting investigation outcomes, renewal processing, and record corrections identified through dispute resolutions.

REQ-43: The OSCA Portal shall generate fraud analytics reports displaying temporal patterns, geographic distribution of flagged credentials, establishment-specific flagging trends, and flagged-to-confirmed conversion rates to support evidence-based enforcement strategies.

REQ-44: The system shall return a SUSPICIOUS verification status for any credential that is currently under active investigation in the OSCA or PDAO portal, ensuring that all Client Portal verification attempts during the investigation period reflect the pending review status until a resolution is reached.

8. Managing PWD Records and Investigating Fraud (PDAO)

8.1 Description and Priority

This feature provides the PDAO Portal with parallel functionality to the OSCA Portal specifically for PWD credential management. PDAO administrators manage the PWD ID database, receive flagged credential reports, investigate suspicious records, and mark confirmed fraudulent IDs. PWD data is strictly compartmentalized from Senior Citizen records.

Priority: High

8.2 Stimulus / Response

8.2.1 Positive Flow

8.2.1.1 A flag report for a PWD ID is transmitted from the Client Portal to the PDAO Portal.

Response: The system shall immediately notify the PDAO administrator through the portal dashboard, displaying the captured ID image, extracted data fields, flagging establishment details, and transaction timestamp for investigation.

8.2.1.2 A PDAO administrator reviews a flagged credential and confirms it as fraudulent.

Response: The system shall update the credential status in the PDAO database to FRAUDULENT, synchronize the status change in real time across all connected Client Portal instances, and generate an investigation outcome record for the audit trail.

8.2.2 Negative Flow

8.2.2.1 A PDAO administrator reviews a flagged PWD credential and determines it is legitimate.

Response: The system shall dismiss the flag, restore the VALID status in the PDAO database, and notify the flagging establishment of the resolution outcome.

8.2.2.2 A PDAO administrator attempts to access Senior Citizen records or OSCA Portal data.

Response: The system shall deny access and display an authorization error, enforcing strict data compartmentalization that limits PDAO administrators to PWD records only.

8.3 Functional Requirements

REQ-45: The PDAO Portal shall receive real-time notifications of each flag report transmitted from the Client Portal for PWD IDs, displaying the captured image, extracted

data, flagging establishment identifier, and transaction timestamp within the PDAO investigation dashboard.

REQ-46: The PDAO administrator shall investigate flagged credentials and mark confirmed fraudulent IDs as FRAUDULENT within the PDAO Portal, triggering an immediate real-time status update propagated across all connected Client Portal instances to prevent future VALID verifications of that credential.

REQ-47: The system shall restrict PDAO Portal access to PWD ID records only, preventing PDAO administrators from viewing Senior Citizen records, OSCA portal data, or BPLO business account information through enforced data compartmentalization.

REQ-48: The PDAO Portal shall enable administrators to add, update, and manage PWD ID records in the PDAO database, including status changes from investigation outcomes, renewal processing, and record corrections identified through beneficiary dispute resolutions.

REQ-49: The PDAO Portal shall generate fraud analytics reports displaying temporal patterns, geographic distribution of flagged PWD credentials, establishment-specific trends, and flagged-to-confirmed conversion rates to support evidence-based enforcement and resource allocation.

REQ-50: The system shall return a SUSPICIOUS verification status for any credential that is currently under active investigation in the OSCA or PDAO portal, ensuring that all Client Portal verification attempts during the investigation period reflect the pending review status until a resolution is reached.

9. Synchronizing Fraud Status Across Establishments

9.1 Description and Priority

This feature ensures that when an OSCA or PDAO administrator confirms a flagged credential as fraudulent, the updated FRAUDULENT status is immediately propagated to all connected Client Portal instances across all participating establishments, transforming isolated fraud detection into coordinated network-wide prevention.

Priority: High

9.2 Stimulus / Response

9.2.1 Positive Flow

9.2.1.1 An OSCA or PDAO administrator marks a previously flagged credential as confirmed fraudulent in the portal.

Response: The system shall immediately update the credential status in the master database and propagate the change in real time to all connected Client Portal instances, so that any subsequent verification attempt of that credential returns an INVALID result.

9.2.1.2 An establishment staff attempts to verify a credential that was confirmed fraudulent after the last local cache refresh.

Response: The system shall query the live database, detect the updated FRAUDULENT status, and return an INVALID verification result to the establishment staff.

9.2.2 Negative Flow

9.2.2.1 A Client Portal instance is offline when OSCA or PDAO updates a credential status to FRAUDULENT.

Response: The system shall synchronize the updated status to that Client Portal instance automatically upon connectivity restoration, ensuring the fraudulent credential cannot receive a VALID result once the device reconnects.

9.2.2.2 An OSCA or PDAO administrator reverts a FRAUDULENT credential status to VALID following a successful dispute resolution.

Response: The system shall propagate the reinstated VALID status across all connected Client Portal instances in real time, allowing the credential to once again return a VALID outcome during verification transactions.

9.3 Functional Requirements

REQ-51: The system shall propagate any credential status change confirmed by an OSCA or PDAO administrator in real time to all connected Client Portal instances,

ensuring network-wide synchronization within 10 seconds of the database update under normal network conditions.

REQ-52: The system shall return an INVALID verification result for any credential whose status has been updated to FRAUDULENT in the OSCA or PDAO master database, regardless of which establishment's Client Portal initiates the verification attempt.

REQ-53: The system shall synchronize pending status updates to offline Client Portal instances automatically upon connectivity restoration to ensure fraudulent credential records are applied without requiring manual refresh by establishment staff.

REQ-54: The system shall maintain a complete audit log of all credential status changes, including the administrator who made the change, the timestamp, the previous status, and the new status, for all OSCA and PDAO database modifications within the fraud synchronization workflow.

10. Registering a Beneficiary Account

10.1 Description and Priority

This feature allows Senior Citizens and PWDs to register accounts in the Beneficiary Mobile Application using their ID number and birthdate, verified against the OSCA or PDAO database and confirmed through an OTP sent to their registered mobile number. Registration is optional and does not affect a beneficiary's right to receive discounts.

Note: Beneficiary account deletion (right to erasure under RA 10173 Section 16) is acknowledged as a data subject right but is outside the scope of this capstone implementation. Beneficiaries seeking account removal are directed to their respective OSCA or PDAO office.

Priority: High

10.2 Stimulus / Response

10.2.1 Positive Flow

10.2.1.1 A beneficiary enters their Senior Citizen or PWD ID number and birthdate in the registration screen of the Beneficiary Mobile Application.

Response: The system shall verify the submitted ID number and birthdate against the corresponding OSCA or PDAO database, and if matched, send a one-time PIN to the mobile number registered with that credential for identity confirmation.

10.2.1.2 The beneficiary enters the correct OTP within the allowed time window.

Response: The system shall activate the beneficiary account, link it to the beneficiary's ID record in the OSCA or PDAO database, and navigate the beneficiary to their account dashboard.

10.2.2 Negative Flow

10.2.2.1 The beneficiary enters an ID number and birthdate that do not match any record in the OSCA or PDAO database.

Response: The system shall deny registration and display a message advising the beneficiary to verify their details with their local OSCA or PDAO office.

10.2.2.2 The beneficiary fails to enter the OTP within the allowed time window.

Response: The system shall expire the OTP, display a timeout notification, and offer the beneficiary the option to request a new OTP, with a defined maximum number of resend attempts per registration session.

10.3 Functional Requirements

REQ-55: The system shall allow a beneficiary to initiate account registration by submitting their Senior Citizen or PWD ID number and birthdate, which the system verifies against the corresponding OSCA or PDAO database record before proceeding to OTP confirmation.

REQ-56: The system shall send a one-time PIN to the mobile number registered in the beneficiary's OSCA or PDAO database record as the secondary verification factor during account registration, linking the beneficiary account to their registered device.

REQ-57: The system shall activate a beneficiary account only upon successful validation of both the primary authentication factor (ID number and birthdate) and the secondary

factor (OTP) within the registration workflow, ensuring account ownership is confirmed before access is granted.

REQ-58: The system shall treat beneficiary account registration as optional, ensuring that business establishments provide mandated Senior Citizen and PWD discounts regardless of whether a beneficiary has registered in the application, creating no new technological barrier for vulnerable populations.

REQ-59: The system shall direct a beneficiary to their issuing OSCA or PDAO office to request an update to their registered mobile number, as the mobile number used for OTP and notifications is sourced directly from the LGU database record and cannot be modified within the Beneficiary Mobile Application.

11. Sending Real-Time Transaction Notifications

11.1 Description and Priority

This feature delivers immediate SMS and push notifications to beneficiaries upon each verification attempt of their ID at participating establishments, providing the business name, transaction timestamp, and verification outcome including specific denial reasons, enabling beneficiaries to detect unauthorized credential use in near real time.

Priority: High

11.2 Stimulus / Response

11.2.1 Positive Flow

11.2.1.1 An establishment staff completes a verification of a beneficiary's ID through the Client Portal.

Response: The system shall immediately send both an SMS and a push notification to the beneficiary's registered mobile number and linked device, containing the establishment name, transaction date and time, and the verification outcome: Approved or Denied.

11.2.1.2 A verification transaction results in a Denied outcome.

Response: The system shall include the specific denial reason such as ID number mismatch, database record not found, ID expired, or status marked as FRAUDULENT in the notification delivered to the beneficiary.

11.2.2 Negative Flow

11.2.2.1 A verification attempt occurs but the beneficiary's mobile number or device is temporarily unreachable at the time of the transaction.

Response: The system shall log the transaction and queue the notification for delivery once the beneficiary's device is reachable, without delaying or disrupting the establishment's verification workflow.

11.2.2.2 A beneficiary receives a transaction notification for an establishment they confirm they did not visit.

Response: The system shall provide a direct action within the notification interface allowing the beneficiary to submit an unauthorized use alert, which routes immediately to the appropriate OSCA or PDAO portal.

11.3 Functional Requirements

REQ-60: The system shall deliver an SMS and push notification to the beneficiary's registered mobile number and linked device upon each verification transaction at a participating establishment, including the establishment name, transaction timestamp, and verification outcome (Approved or Denied).

REQ-61: The system shall include the specific denial reason in the notification when a verification transaction returns a Denied outcome, providing the beneficiary with actionable information to understand the basis for the denial within the notification message.

REQ-62: The system shall deliver SMS transaction notifications independently of mobile data access, ensuring that beneficiaries without active internet connectivity still receive verification alerts through the SMS channel.

REQ-63: The system shall allow a beneficiary to submit an unauthorized credential use alert directly from within the transaction notification interface, automatically routing the alert to the appropriate OSCA or PDAO portal for immediate investigation.

REQ-64: The system shall maintain a complete transaction notification history accessible within the Beneficiary Mobile Application, displaying all past verification attempts

associated with the beneficiary's ID in chronological order with establishment names, timestamps, and outcomes.

12. Checking ID Verification Status

12.1 Description and Priority

This feature allows authenticated beneficiaries to view their current ID verification status synchronized with OSCA or PDAO master databases, including last successful verification details, upcoming expiration dates, and issuing office contact information, enabling proactive verification of standing before discount transactions.

Priority: High

12.2 Stimulus / Response

12.2.1 Positive Flow

12.2.1.1 An authenticated beneficiary navigates to the ID Status screen in the Beneficiary Mobile Application.

Response: The system shall retrieve and display the beneficiary's current ID status: Active, Under Review, Suspended, or Expired, synchronized with the OSCA or PDAO master database, along with the last successful verification transaction details and upcoming expiration date.

12.2.1.2 A beneficiary checks their status and finds it shows Active with no anomalies.

Response: The system shall display issuing office contact information alongside the status to allow the beneficiary to reach out proactively if they have questions about their record.

12.2.2 Negative Flow

12.2.2.1 A beneficiary checks their ID status and finds it is marked Under Review or Suspended.

Response: The system shall display the status with an explanation of what it means, provide the applicable dispute or appeal pathway if the beneficiary believes the status is incorrect, and display OSCA or PDAO contact details.

12.2.2.2 The Beneficiary Mobile Application cannot retrieve the current ID status due to a network interruption.

Response: The system shall display the last cached status with a timestamp of when it was last retrieved, and a notification that the displayed status may not reflect the latest database record until connectivity is restored.

12.3 Functional Requirements

REQ-65: The system shall display the beneficiary's current ID status: Active, Under Review, Suspended, or Expired, synchronized with the OSCA or PDAO master database

within the Beneficiary Mobile Application upon authenticated access to the ID Status screen.

REQ-66: The system shall display the last successful verification transaction details including the establishment name, date, and outcome alongside the current ID status to provide the beneficiary with recent verification activity context.

REQ-67: The system shall display the beneficiary's ID expiration date and issuing office contact information within the ID Status screen to enable proactive renewal and direct communication with the issuing authority.

REQ-68: The system shall expose only the information printed on the physical credential ID number, name, type, and dates alongside verification status and transaction outcomes within the Beneficiary Mobile Application, implementing minimal data exposure while fulfilling transparency obligations under Republic Act 10173.

REQ-69: The system shall display the last cached ID status with a retrieval timestamp when a beneficiary accesses the ID Status screen without network connectivity, clearly indicating that the displayed information may not reflect the current database record.

13. Submitting and Tracking Dispute Resolutions

13.1 Description and Priority

This feature provides Senior Citizens and PWDs with a structured dispute mechanism to challenge erroneous verification denials. Beneficiaries select a dispute category, provide a written explanation, and optionally upload supporting documentation. The system routes

the dispute to OSCA or PDAO for human-reviewed adjudication with guaranteed acknowledgment and resolution timelines.

Priority: High

13.2 Stimulus / Response

13.2.1 Positive Flow

13.2.1.1 A beneficiary selects a denied transaction from their notification history and taps the Dispute button.

Response: The system shall open the dispute submission form pre-populated with the transaction reference number, establishment details, and denial reason, and present the dispute category selection menu to the beneficiary.

13.2.1.2 The beneficiary completes the dispute form by selecting a category, entering a written explanation, and optionally uploading supporting documentation, then submits the dispute.

Response: The system shall route the dispute to the appropriate OSCA or PDAO investigation queue, generate a dispute reference number, and send an automatic acknowledgment notification to the beneficiary within 24 hours confirming receipt.

13.2.2 Negative Flow

13.2.2.1 The beneficiary attempts to submit a dispute form without providing a written explanation in the required field.

Response: The system shall display a validation error on the explanation field and prevent submission until the mandatory content is provided, with an on-screen message indicating why the form cannot be submitted.

13.2.2.2 The OSCA or PDAO dispute review exceeds the service-level agreement timeframe without resolution.

Response: The system shall escalate the overdue dispute within the LGU portal, send a status update notification to the beneficiary confirming the case is still under active review, and record the escalation event in the audit trail.

13.3 Functional Requirements

REQ-70: The system shall allow a beneficiary to initiate a dispute from a denied transaction record in the notification history, pre-populating the dispute form with the transaction reference number, establishment details, and denial reason to minimize manual input.

REQ-71: The system shall present the beneficiary with three dispute category options — Wrong Denial (I Am Valid), ID Used Without My Authorization, or Database Contains Incorrect Information — as the first step of the dispute submission workflow. Business

discount refusal incidents shall be submitted separately through the fraud reporting feature in Section 14.

REQ-72: The system shall require a beneficiary to provide a written explanation in the dispute form and allow the optional upload of supporting documentation such as physical ID photos or barangay certifications before the dispute submission can proceed.

REQ-73: The system shall route the submitted dispute to the appropriate OSCA or PDAO investigation queue based on the beneficiary's ID type, send an automatic acknowledgment notification within 24 hours of submission, and provide a unique dispute reference number for tracking.

REQ-74: The system shall enable the beneficiary to track the status of a submitted dispute from within the application, displaying status updates as OSCA or PDAO staff review evidence, and delivering a final resolution notification with explanation upon case closure.

REQ-75: The OSCA or PDAO staff shall judge all disputes through human review within the portal, with the system enforcing service-level agreements of 24-hour acknowledgment and 48-to-72-hour resolution for database errors, and automated escalation notifications for cases exceeding defined timelines.

14. Reporting Fraud as a Beneficiary

14.1 Description and Priority

This feature enables beneficiaries to report three types of fraud incidents that businesses and government agencies cannot independently detect: unauthorized credential use by another person, business refusal of a system-approved discount, and database discrepancy alerts for incorrect personal information in their displayed records.

Priority: High

14.2 Stimulus / Response

14.2.1 Positive Flow

14.2.1.1 A beneficiary receives a transaction notification for an establishment they confirm they did not visit and submits an unauthorized use alert.

Response: The system shall route the alert immediately to the appropriate OSCA or PDAO portal with the transaction details, prompt the administrator to investigate, and if confirmed unauthorized, flag the credential for cross-establishment network protection.

14.2.1.2 A beneficiary submits a business refusal report indicating that an establishment denied their discount despite the system returning a VALID result.

Response: The system shall route the report to the BPLO Portal for investigation of a potential DOJ Opinion violation, log the

incident with transaction details, and send an acknowledgment notification to the beneficiary.

14.2.1.3 A beneficiary identifies incorrect personal information in their displayed ID record and submits a database discrepancy alert.

Response: The system shall route the discrepancy report to the appropriate OSCA or PDAO portal for verification and correction, acknowledge receipt to the beneficiary with a reference number, and update the beneficiary's record upon confirmation by the LGU administrator.

14.2.2 Negative Flow

14.2.2.1 A beneficiary submits a fraud report without providing sufficient details to enable investigation.

Response: The system shall display a validation prompt requesting minimum required information before allowing submission, and provide guidance on what details are needed for the report to be actionable by LGU investigators.

14.3 Functional Requirements

REQ-76: The system shall allow a beneficiary to submit an unauthorized credential use alert when their ID is used at an establishment they did not visit, routing the alert

immediately to the appropriate OSCA or PDAO portal with the transaction reference, establishment details, and beneficiary confirmation statement.

REQ-77: The system shall allow a beneficiary to submit a business refusal report when an establishment denies a discount despite the system returning a VALID verification result, routing the report to the BPLO Portal for investigation of potential DOJ Opinion non-compliance.

REQ-78: The system shall allow a beneficiary to submit a database discrepancy alert when they identify incorrect personal information in their displayed record, routing the report to the appropriate OSCA or PDAO portal for verification and correction of the master database entry.

REQ-79: The system shall generate an acknowledgment notification with a reference number for each beneficiary-submitted fraud report, confirming receipt and providing the beneficiary with a means to track the investigation status within the application.

REQ-80: The OSCA or PDAO administrator shall review and act on beneficiary-submitted fraud reports within the LGU portal, with confirmed unauthorized use reports triggering credential flagging in the master database and cross-establishment synchronization.

REQ-81: The BPLO Portal shall receive and display beneficiary-submitted business refusal reports routed from the Beneficiary Mobile Application, showing the transaction reference, establishment identifier, verification outcome, and beneficiary confirmation

statement, enabling the BPLO officer to investigate potential DOJ Opinion non-compliance by the flagged establishment.

15. Operating Offline and Synchronizing Data

15.1 Description and Priority

This feature provides limited offline capability for the Client Portal by caching recently verified credentials locally, and allows the Beneficiary Mobile Application to display previously loaded information when network access is unavailable. Pending actions queue locally and synchronize automatically upon connectivity restoration.

Priority: Medium

15.2 Stimulus / Response

15.2.1 Positive Flow

15.2.1.1 An establishment staff attempts a verification while the device has no network connection.

Response: The system shall check the local cache for a recently verified matching record and, if found, display the cached result with a timestamp indicating when the credential was last verified online and a clear label that this is a cached offline result.

15.2.1.2 The mobile device reconnects to the network after a period of offline operation.

Response: The system shall automatically synchronize all queued fraud flag reports and pending actions with the server, and update the local cache with the latest database verification statuses without requiring manual intervention from the staff.

15.2.2 Negative Flow

15.2.2.1 An establishment staff attempts to verify an ID not present in the local cache while the device is offline.

Response: The system shall display an offline mode warning stating that real-time verification is unavailable, advise the staff to defer the transaction or proceed provisionally in compliance with DOJ provisional approval requirements, and prompt for reconnection.

15.2.2.2 The device fails to synchronize queued reports after connectivity restoration due to a server-side error.

Response: The system shall retain the queued reports locally, retry synchronization automatically at the next available opportunity, and notify the staff of the pending synchronization status through a persistent indicator in the interface.

15.3 Functional Requirements

REQ-82: The system shall cache recently verified credentials locally on the Client Portal mobile device to enable limited offline verification capability, displaying cached results with clear offline mode labeling and the timestamp of the last online verification to the establishment staff.

REQ-83: The system shall queue fraud flag reports and other pending actions locally when the Client Portal device is offline and automatically submit them to the appropriate LGU portal upon connectivity restoration without requiring manual resubmission by staff.

REQ-84: The system shall display a persistent offline mode indicator to the establishment staff and beneficiary whenever real-time database verification is unavailable due to network interruption, distinguishing cached results from live verification outcomes throughout the offline session.

REQ-85: The Beneficiary Mobile Application shall allow beneficiaries to view previously loaded transaction history and ID status information when network access is unavailable, with a clear indication that the displayed data reflects the last synchronized state and may not be current.

REQ-86: The system shall deliver SMS transaction notifications to beneficiaries independently of mobile data connectivity, ensuring verification event notifications reach beneficiaries through the SMS channel even when internet access is unavailable on their device.

16. Logging Audit Trails and Transaction History

16.1 Description and Priority

This feature ensures that all verification transactions, fraud flag submissions, database status changes, administrator actions, and dispute events are permanently logged in a tamper-evident audit trail stored in the system database, supporting compliance reporting, accountability tracking, and fraud pattern analysis.

Priority: High

16.2 Stimulus / Response

16.2.1 Positive Flow

16.2.1.1 A verification transaction is completed through the Client Portal.

Response: The system shall immediately log the transaction record in the audit database including the credential data submitted, verification status returned, establishment identifier, staff account identifier, and timestamp.

16.2.1.2 An OSCA or PDAO administrator marks a credential as fraudulent.

Response: The system shall log the action in the audit trail with the administrator's account identifier, the credential affected, the previous and new status, and the exact timestamp, creating a permanent non-repudiable record of the database modification.

16.2.1.3 An establishment staff requests a compliance report summary for a defined date range from the transaction history screen.

Response: The system shall generate a summary report of all verification transactions within the requested period, including counts of VALID, SUSPICIOUS, and INVALID outcomes, and present it on screen for staff review.

16.2.2 Negative Flow

16.2.2.1 A user attempts to delete or modify a logged audit record.

Response: The system shall deny the modification, maintain the integrity of all audit records, and log the attempted modification as a security event in the audit trail.

16.3 Functional Requirements

REQ-87: The system shall permanently log every verification transaction in the audit database immediately upon completion, including the credential data submitted, verification status, establishment identifier, staff account identifier, and transaction timestamp to support accountability tracking.

REQ-88: The system shall log all database status changes made by OSCA or PDAO administrators in the audit trail, recording the administrator account identifier, the affected credential, the previous status, the new status, and the exact timestamp of each modification.

REQ-89: The system shall log all fraud flag submissions, dispute events, and administrator investigation outcomes in the audit trail with associated reference numbers, timestamps, and user identifiers to maintain complete traceability across the fraud management lifecycle.

REQ-90: The system shall prevent any user from deleting or modifying audit log records, maintaining tamper-evident transaction history that supports regulatory compliance, fraud investigation, and legal proceedings when required.

REQ-91: The system shall allow establishment staff to generate compliance report summaries from their transaction history for defined date ranges, displaying counts and outcomes of verification transactions to support internal audit and regulatory reporting requirements.

REQ-92: The system shall retain all audit log records for a minimum of 5 years from the date of creation, after which records may be archived to long-term storage, supporting regulatory compliance, fraud investigation, and legal evidentiary requirements under applicable Philippine data governance laws.

REQ-93: The system shall restrict audit trail read access to BPLO administrators for business account and system operation logs, OSCA administrators for Senior Citizen verification and fraud investigation logs, and PDAO administrators for PWD verification and fraud investigation logs, in accordance with the data compartmentalization rules enforced across all portals.

17. Securing Data and Ensuring Privacy Compliance

17.1 Description and Priority

This feature covers the security controls, encrypted data transmission, data minimization principles, and Republic Act 10173 (Data Privacy Act of 2012) compliance mechanisms implemented across all iDetect portals, ensuring beneficiary personal information is protected while enabling transparency and access rights for data subjects.

Priority: High

17.2 Stimulus / Response

17.2.1 Positive Flow

17.2.1.1 An establishment staff submits a verification transaction that queries the OSCA or PDAO database.

Response: The system shall transmit all data over encrypted channels using industry-standard protocols, return only the binary verification status to the Client Portal without exposing comprehensive personal data, and store only the minimum necessary transaction data in the audit log.

17.2.1.2 A beneficiary exercises their RA 10173 Section 16 right to access their verification records.

Response: The system shall provide the beneficiary with access to their current ID status, transaction history, and any dispute records

associated with their credential through the authenticated Beneficiary Mobile Application, fulfilling the statutory data access right.

17.2.2 Negative Flow

17.2.2.1 An unauthorized actor attempts to access the OSCA or PDAO database directly through an unauthenticated API request.

Response: The system shall reject the request, log the unauthorized access attempt with the request source and timestamp in the security audit trail, and notify the system administrator of the security event.

17.2.2.2 A user attempts to access another user role's data or portal features beyond their assigned permissions.

Response: The system shall deny access, display an authorization error message, and log the unauthorized access attempt in the audit trail for administrator review.

17.3 Functional Requirements

REQ-94: The system shall transmit all data between the Client Portal, LGU portals, and backend databases over encrypted channels using industry-standard encryption protocols, protecting credential data and verification results from interception during transmission.

REQ-95: The system shall implement role-based access control across all portals ensuring that BPLO administrators access only business account data, OSCA administrators access only Senior Citizen records, PDAO administrators access only PWD records, establishment staff access only verification results, and beneficiaries access only their own records.

REQ-96: The system shall return only binary verification status responses: VALID, SUSPICIOUS, or INVALID to the Client Portal without transmitting comprehensive beneficiary personal data, implementing privacy-preserving query design that minimizes data exposure to business establishments.

REQ-97: The system shall provide beneficiaries with access to their verification status, transaction history, and dispute records through the authenticated Beneficiary Mobile Application, fulfilling the data subject access rights mandated by Republic Act 10173 Section 16 of the Data Privacy Act of 2012.

REQ-98: The system shall log all unauthorized access attempts, authentication failures, and permission violation events in a dedicated security audit trail with the source, timestamp, and type of event, supporting security monitoring and incident response across all portals.

REQ-99: The system shall support generation of a breach incident report by the system administrator in the event of a confirmed data security breach, containing the nature of the breach, affected data categories, and estimated number of affected data subjects, in

accordance with the 72-hour notification obligation to the National Privacy Commission under Republic Act 10173.

REQ-100: The Client Portal mobile application and Beneficiary Mobile Application shall enforce automatic session expiry after a defined period of inactivity, requiring the user to re-authenticate before resuming access, to prevent unauthorized use of unattended authenticated sessions on shared or lost devices.

18. Designing an Accessible and Multilingual Interface

18.1 Description and Priority

This feature ensures all iDetect interfaces employ large touch targets, high-contrast color schemes, bilingual support in English and Filipino, minimal text input requirements, and progressive disclosure of complex features, maintaining WCAG 2.1 AA accessibility standards to accommodate users with varying technical literacy and abilities.

Priority: Medium

18.2 Stimulus / Response

18.2.1 Positive Flow

18.2.1.1 A beneficiary aged 60 or above opens the Beneficiary Mobile Application for the first time.

Response: The system shall display large touch targets, high-contrast text, clear iconography, and simple onboarding

guidance in both English and Filipino, minimizing cognitive load for elderly users with potentially lower digital literacy.

18.2.1.2 An establishment staff with no prior training opens the Client Portal on their first day.

Response: The system shall guide the staff through the three-step verification workflow: launch, scan, view result using inline contextual prompts and clear visual feedback at each step, enabling proficiency without prior formal training.

18.2.2 Negative Flow

18.2.2.1 A beneficiary switches the application language preference from English to Filipino.

Response: The system shall immediately render all interface labels, instructions, notifications, and status messages in Filipino throughout the application without requiring a session restart.

18.2.2.2 A user interacts with a feature beyond the primary verification or status view workflow.

Response: The system shall apply progressive disclosure, surfacing only the information and controls needed for the immediate task and revealing advanced features only upon deliberate navigation.

18.3 Functional Requirements

REQ-101: The Client Portal and Beneficiary Mobile Application shall employ large touch targets meeting WCAG 2.1 AA minimum size guidelines and high-contrast color schemes to accommodate users with varying visual acuity and motor dexterity, including elderly Senior Citizen beneficiaries.

REQ-102: All iDetect portals and mobile applications shall provide full bilingual support in English and Filipino, allowing users to switch language preference at any time with all interface text, notifications, and guidance rendered in the selected language immediately upon change.

REQ-103: The Client Portal verification workflow shall adhere to the three-action limit of launch, scan, and view result by implementing automatic background authentication, immediate camera activation upon launch, and automatic processing without intermediate confirmation dialogs to minimize adoption resistance.

REQ-104: The system shall apply progressive disclosure across all portals and applications, surfacing only the controls and information required for the immediate task during primary workflows and revealing advanced features such as fraud analytics dashboards and bulk management tools only upon deliberate navigation.

REQ-105: All iDetect interfaces shall minimize mandatory text input requirements, pre-populating fields where data is already known from the authenticated session or the current transaction context, to reduce burden on frontline establishment staff and elderly or low-literacy beneficiaries.

Project Description

Project Design

This study employs an Applied and Descriptive Research Design to develop, implement, and evaluate the iDetect system, an OCR-based ID verification application for Business Establishments under the Business Permit and Licensing Office in Valenzuela City.

Applied research is a type of scientific inquiry directed toward solving specific, practical problems by translating theoretical knowledge into actionable solutions within a real-world context, characterized by its intent to produce findings that are immediately useful to practitioners, policymakers, and institutions facing concrete operational challenges (Creswell, 2014). Descriptive research, on the other hand, systematically describes and documents phenomena, conditions, or characteristics as they currently exist without manipulating variables, thereby providing an accurate and comprehensive account of the subject under investigation (Kothari, 2004). The convergence of these two approaches enables the research to simultaneously capture the existing landscape of ID verification practices and develop a concrete, evidence-based solution to the identified problems.

The applied dimension of this study is grounded in the documented inadequacies of current manual ID verification practices in Valenzuela City, including ₱88.2 billion in lost government revenues attributed to fraudulent ID use. In response, iDetect implements automated cross-verification against authoritative databases, eliminating reliance on subjective visual

inspection. Its bidirectional fraud intelligence system enables one-tap flagging with automatic routing to investigation authorities and real-time synchronization across all system users. Through OCR automation, the mobile-first design targets a reduction in verification time to 5-8 seconds, improving operational throughput and reducing cognitive burden on cashiers and frontline personnel.

The descriptive dimension of the study systematically documents the current state of verification workflows, stakeholder experiences, and system performance through multiple data collection methods. Structured interviews conducted with business managers, cashiers, and LGU personnel from OSCA, PDAO, and BPLO describe existing verification practices, pain points, fraud encounter frequencies, and institutional expectations, with data subjected to thematic analysis to identify recurring patterns. Validated survey questionnaires administered to business establishments from District 1 ($n \approx 1$) and District 2 ($n \approx 1$) and beneficiaries such as senior citizens ($n \approx 5$) and PWD ($n \approx 5$) establish pre-implementation baselines and measure post-implementation improvements through descriptive statistics, paired t-tests, ANOVA, and correlation analysis. System performance metrics continuously document verification processing times, OCR accuracy rates, database query response times, fraud flag frequencies, and dispute resolution timelines. Document analysis of LGU verification policies, fraud reports, and regulatory guidelines further contextualizes the institutional frameworks within which iDetect operates.

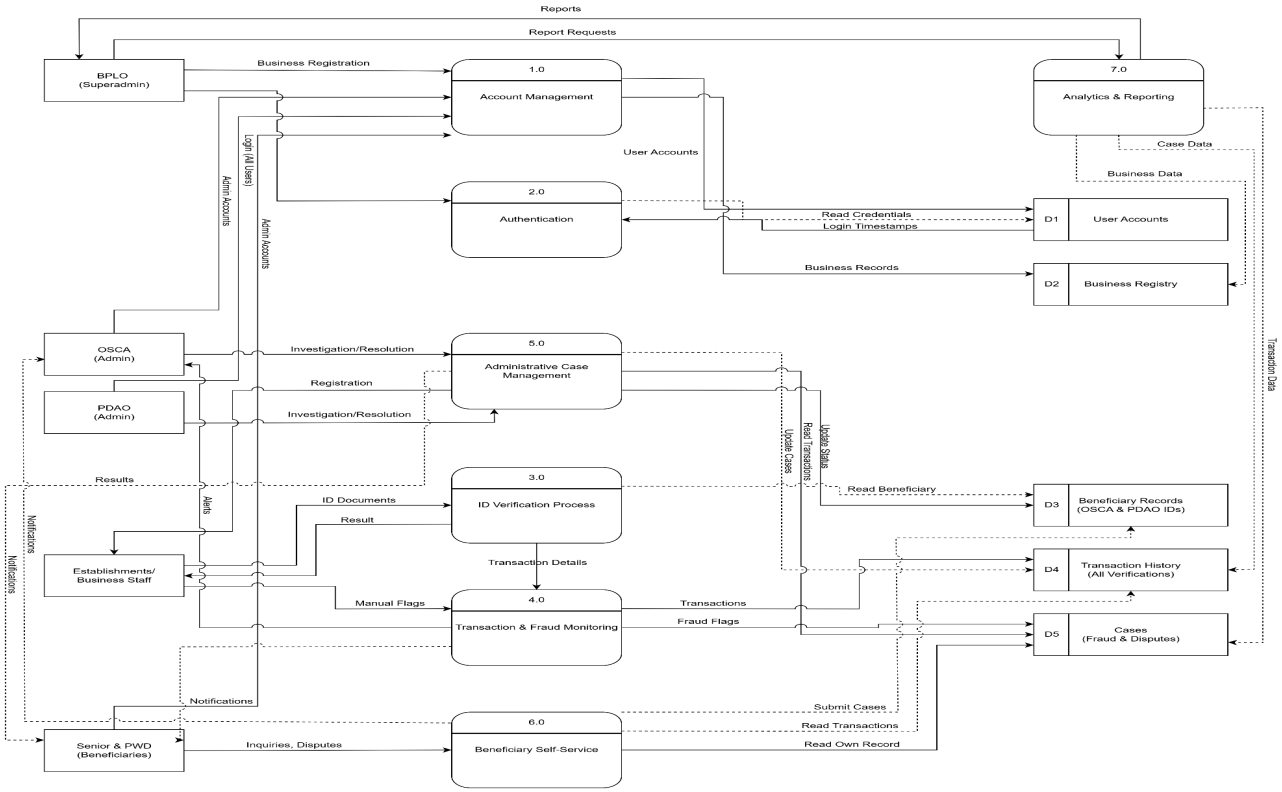
System features emerge directly from both applied and descriptive priorities, with design decisions including offline fallback modes, bilingual interfaces in English and Filipino, minimal text input requirements, and beneficiary accountability mechanisms shaped by direct stakeholder

engagement through interviews, questionnaires, and usability testing. The integration of applied and descriptive approaches produces complementary insights, wherein descriptive data documents what outcomes occurred while the applied framework addresses why such outcomes arise, collectively strengthening the validity and practical relevance of the study's findings through triangulation across multiple data sources and methods.

Proposed System Context Diagram

Figure 13.

Proposed System Level 1 Diagram.



The diagram above presents the context diagram of iDetect, illustrating the system's external actors, core process modules, and data stores.

The external actors on the left represent the four user groups that interact with the system: BPLO (Superadmin), who oversees account and system management; OSCA and PDAO Admins, who handle administrative case management and ID record maintenance; Establishments/Business Staff, who perform ID verification transactions; and Senior & PWD Beneficiaries, who access self-service features for their own credentials.

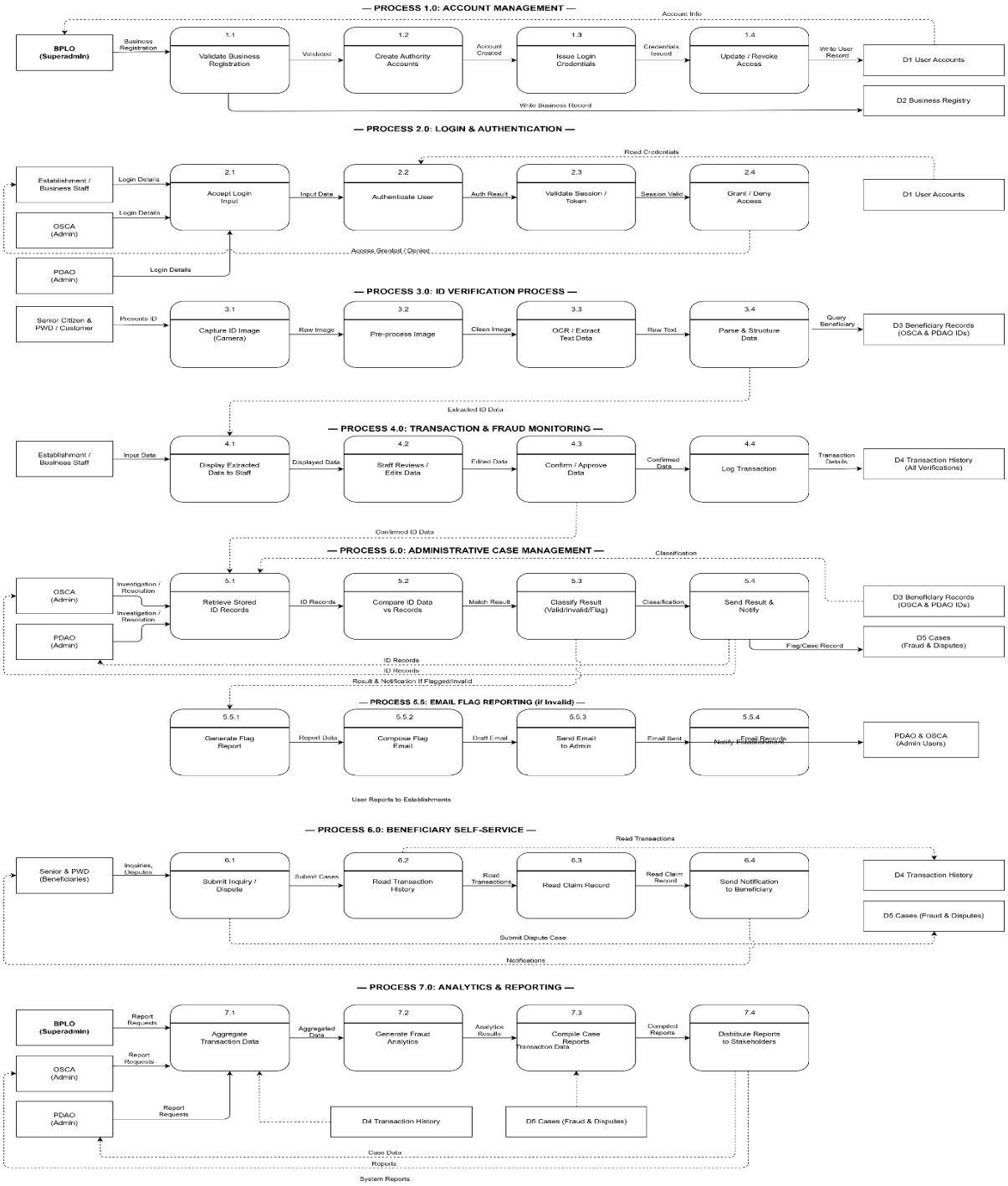
The center of the diagram shows the seven core process modules of iDetect: 1.0 Account Management handles the creation and control of establishment accounts; 2.0 Authentication

manages login and role-based access for all user types; 3.0 ID Verification Process covers the OCR-based scanning and real-time database verification of Senior Citizen and PWD IDs; 4.0 Transaction & Fraud Monitoring tracks verification activity and manages suspicious credential flags; 5.0 Administrative Case Management handles fraud investigations and dispute resolutions by OSCA and PDAO; 6.0 Beneficiary Self-Service allows Senior Citizens and PWDs to check their ID status, receive notifications, and file disputes; and 7.0 Analytics & Reporting generates fraud trends and system performance reports for administrators.

The data stores on the right represent the system's five databases: D1 User Accounts, D2 Business Registry, D3 Beneficiary Records containing OSCA and PDAO ID data, D4 Transaction History logging all verifications, and D5 Cases storing fraud flags and dispute records.

Figure 15.

Proposed System Level 2 Diagram.



The Level 2 DFD expands each process module from Level 1 into its specific sub-processes, showing the detailed flow of data between actors, processes, and data stores.

Process 1.0 – Account Management is handled by the BPLO Superadmin and consists of four sub-processes: validating business registration (1.1), creating authority accounts (1.2), issuing login credentials (1.3), and updating or revoking access (1.4). These processes interact with the D1 User Accounts and D2 Business Registry data stores.

Process 2.0 – Authentication applies to all user types and follows four steps: accepting login input (2.1), authenticating the user (2.2), validating the session token (2.3), and granting or denying access (2.4), all referencing D1 User Accounts.

Process 3.0 – ID Verification Process begins when a Senior Citizen or PWD presents their ID. The system captures the ID image via camera (3.1), preprocesses the image (3.2), extracts text data through OCR (3.3), and parses and structures the extracted data (3.4), querying D3 Beneficiary Records for validation.

Process 4.0 – Transaction & Fraud Monitoring is carried out by establishment staff and involves displaying extracted data (4.1), allowing staff review and edits (4.2), confirming or approving the data (4.3), and logging the transaction (4.4) into D4 Transaction History.

Process 5.0 – Administrative Case Management is handled by OSCA and PDAO admins. It retrieves stored ID records (5.1), compares them against submitted data (5.2), classifies the result as Valid, Invalid, or Flag (5.3), and sends the result and notification (5.4). A

fraud flag sub-flow (5.5) generates a flag report (5.5.1), composes a flag email (5.5.2), sends it to the admin (5.5.3), and notifies the establishment (5.5.4), updating both D3 Beneficiary Records and D5 Cases.

Process 6.0 – Beneficiary Self-Service allows Senior Citizens and PWDs to submit inquiries or disputes (6.1), read their transaction history (6.2), view their claim record (6.3), and receive notifications (6.4), drawing from D4 Transaction History and D5 Cases.

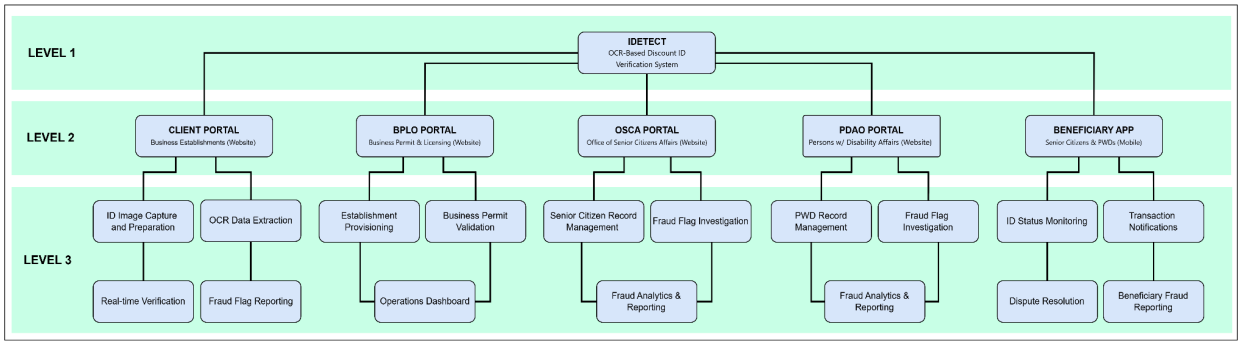
Process 7.0 – Analytics & Reporting is accessible to BPLO, OSCA, and PDAO. It aggregates transaction data (7.1), generates fraud analytics (7.2), compiles case reports (7.3), and distributes reports to stakeholders (7.4), using D4 Transaction History and D5 Cases as its data sources.

Use Case Scenario

Hierarchical Diagram

Figure 16.

Hierarchical Diagram of IDetect



This figure presents the hierarchical decomposition of the iDetect system across three levels, illustrating how the system is organized from its overall purpose down to its specific functional features.

At Level 1, the system is represented as a single entity, iDetect, an OCR-Based Discount ID Verification System developed to address the widespread misuse of fake Senior Citizen and PWD identification cards in food service establishments within Valenzuela City.

Level 2 decomposes the system into five portals serving distinct stakeholder groups: the Client Portal for business establishment staff, the BPLO Portal for the Business Permit and Licensing Office, the OSCA Portal for the Office of Senior Citizens Affairs, the PDAO Portal for the Persons with Disability Affairs Office, and the Beneficiary App for Senior Citizens and PWDs.

Level 3 details the core features of each portal. The Client Portal covers ID image capture, OCR data extraction, real-time verification, fraud flag reporting, and offline sync. The BPLO Portal handles establishment provisioning, permit validation, and an operations dashboard. The OSCA and PDAO Portals each manage their respective ID records, fraud investigations, and analytics. The Beneficiary App provides ID status monitoring, transaction notifications, dispute resolution, and fraud reporting. Together, these portals and features form a unified system that addresses fraud prevention, government coordination, and beneficiary rights protection.

Data Dictionary

Work Breakdown Structure Dictionary

Table 3.

Work Breakdown Structure Dictionary

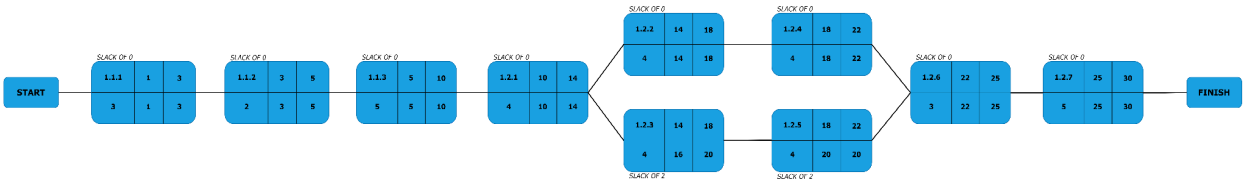
LEVEL	CODE	NAME	DESCRIPTION	PREDECESSORS	DURATION	OWNER
2	1.1	PROJECT INITIATION	DEFINE THE OVERALL SCOPE, STAKEHOLDERS, AND FORMAL AUTHORIZATION DOCUMENT FOR THE IDETECT PROJECT	N/A	N/A	
3	1.1.1	DEFINE SCOPE & OBJECTIVES	IDENTIFY PROJECT BOUNDARIES, GOALS, DELIVERABLES, AND SUCCESS CRITERIA FOR IDETECT	N/A	DAYS 1–3	All Roles
3	1.1.2	IDENTIFY STAKEHOLDERS	LIST ALL INTERNAL AND EXTERNAL STAKEHOLDERS (BPLO, OSCA, PDAO, ESTABLISHMENTS, BENEFICIARIES) AND DEFINE THEIR ROLES	1.1.1	DAYS 3–5	Project Lead Docu Lead
3	1.1.3	DEVELOP PROJECT CHARTER	DRAFT AND OBTAIN SIGN-OFF ON THE FORMAL PROJECT CHARTER AUTHORIZING THE IDETECT CAPSTONE PROJECT	1.1.2	DAYS 5–10	Project Lead Docu Lead
2	1.2	PROJECT PLANNING	PRODUCE ALL PLANNING ARTIFACTS NEEDED TO GUIDE DEVELOPMENT: WBS, WBD, CPM, SEP, AND RISK REGISTER	N/A	N/A	
3	1.2.1	DEVELOP WBS	DECOMPOSE THE ENTIRE IDETECT PROJECT INTO ALL SEVEN PHASES, TASKS, AND SUB-TASKS IN A HIERARCHICAL WORK BREAKDOWN STRUCTURE	1.1.3	DAYS 10–14	Project Lead Docu Lead
3	1.2.2	DEVELOP WBD	WRITE DETAILED DICTIONARY ENTRIES FOR EACH WBS WORK PACKAGE: SCOPE, DELIVERABLES, OWNER, DURATION, DEPENDENCIES, AND ACCEPTANCE CRITERIA	1.2.1	DAYS 14–18	Docu Lead Dev Lead

3	1.2.3	DEVELOP CPM / GANTT CHART	COMPUTE EARLY START, EARLY FINISH, LATE START, LATE FINISH, AND SLACK FOR ALL ACTIVITIES; PRODUCE GANTT CHART (SLACK = 2 DAYS NON-CRITICAL)	1.2.1	DAYS 14–18 (Slack: 2)	Project Lead
3	1.2.4	CREATE SEP	WRITE THE SOFTWARE ENGINEERING PLAN COVERING DEVELOPMENT METHODOLOGY, TOOLS, STANDARDS, AND QUALITY ASSURANCE APPROACH	1.2.2	DAYS 18–22	Dev Lead Docu Lead
3	1.2.5	RISK IDENTIFICATION & PLANNING	IDENTIFY PROJECT RISKS (TECHNICAL, SCHEDULE, RESOURCE), ASSESS PROBABILITY AND IMPACT, AND DEFINE MITIGATION STRATEGIES (SLACK = 2 DAYS NON-CRITICAL)	1.2.2	DAYS 18–22 (Slack: 2)	Project Lead
3	1.2.6	STAKEHOLDER COMMUNICATION PLAN	DOCUMENT COMMUNICATION FREQUENCY, METHODS, AND RESPONSIBILITIES FOR ALL STAKEHOLDERS (WEEKLY REPORTS, STAND-UPS, ADVISER LOG)	1.2.4 1.2.5	DAYS 22–25	Project Lead Docu Lead
3	1.2.7	ADVISER REVIEW & APPROVAL	SUBMIT ALL PLANNING DOCUMENTS TO RESEARCH ADVISER AND TA FOR REVIEW; INCORPORATE FEEDBACK AND OBTAIN FORMAL SIGN-OFF	1.2.6	DAYS 25–30	All Roles

Critical Path Method

Figure 17.

Critical Path Method



The Critical Path Method diagram illustrates the full sequence of iDetect's development activities from START to FINISH, organized around the project's Agile sprint structure. Each node represents a task tied to the system's core development phases, requirements gathering, system design, multi-portal development, and testing with the parallel branching in the middle section reflecting the concurrent development of the five interconnected portals: the Client Portal, BPLO Portal, OSCA Portal, PDAO Portal, and Beneficiary Mobile Application. The critical path traces the zero-float activities that could not be delayed without pushing back the February 2026 submission deadline, particularly the OCR preprocessing pipeline and real-time database API integration, which are the foundational components of iDetect's core function of reducing verification time from 45–60 seconds to under 30 seconds per transaction.

Gantt Chart

Table 5.

Gantt Chart

TASK/ DAY	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1.1.1																														
1.1.2																														
1.1.3																														
1.2.1																														
1.2.2																														
1.2.3																														
1.2.4																														
1.2.5																														
1.2.6																														
1.2.7																														

The Gantt Chart covers a 30-day project window and tracks the scheduled execution of ten tasks from 1.1.1 through 1.2.7, arranged in a staircase pattern that confirms the project's sequential, dependency-driven workflow. The first three tasks spanning days one through nine correspond to the initial phase of background research, problem formulation, and requirements analysis grounded in the documented crisis of 8.5 million fraudulent PWD IDs and ₱88.2 billion in lost government revenues in 2023. The middle tasks from days eight through twenty align with the core system analysis and design deliverables assigned to the Dev Lead and Design Lead in the RACI Matrix, including the Functional Requirements Document, Data Flow Diagrams, ERD, and UI/UX wireframes. The final tasks closing at day thirty correspond to system testing, quality assurance, and final manuscript preparation, with each task's green bar beginning only after its predecessor is substantially complete reflecting the same dependency logic embedded in the CPM diagram.

Stakeholder Analysis

Stakeholder	Power / Interest	Engagement Strategy	Communication Method	Frequency
Research Adviser	High / High	**Manage Closely**: Ensure academic standards and logic traceability (e.g., FR to WBS) are met.	Formal Consultation / Consultation Log	Weekly
Technical Adviser	High / High	**Manage Closely:** Focus on the "Ancestry" of technical artifacts like DFD-to-ERD mapping.	Technical Walkthrough / Email	Bi-weekly
BPLO Officer	High / Low	**Keep Satisfied:** Focus on high-level milestones and how the system enforces business permit compliance.	Executive Summary / Demo	Monthly
OSCA/PDAO Admins	High / High	**Manage Closely:** Involve in Use Case Scenario validation and data compartmentalization rules.	User Interviews / Surveys	As Needed
Establishment Staffs	Low / High	**Keep Informed:** Involve them in UI/UX testing to ensure the 3-action workflow is realistic.	User Testing / Walkthrough	As Needed
Senior Citizen & PWDs	Low / High	**Keep Informed**: Gather feedback on notification clarity and accessibility compliance (WCAG 2.1 AA).	Usability Tests / Surveys	As Needed
Proponents & Students	High / High	**Keep Informed:** Gather feedback on notification clarity and accessibility compliance (WCAG 2.1 AA).	Daily Stand-ups / Discord / Slack	Daily

The Stakeholder Analysis identifies the key groups whose power and interest levels shaped iDetect's engagement and communication strategy throughout development. The Research Adviser and Technical Adviser, both rated High Power and High Interest, are managed closely, the Research Adviser through weekly formal consultations ensuring logic traceability from Functional Requirements to the Work Breakdown Structure, and the Technical Adviser through bi-weekly technical walkthroughs focused on the ancestry of technical artifacts such as the DFD-to-ERD mapping. The BPLO Officer, rated High Power but Low Interest, is kept satisfied through monthly executive summaries centered on high-level milestones and how the system enforces business permit compliance, consistent with the BPLO Portal's defined scope of managing establishment accounts without accessing beneficiary verification data. Establishment staff, rated Low Power and High Interest, are kept informed through user interviews and surveys to validate that the three-action workflow: launch, scan, view accurately reflects real point-of-sale conditions in Valenzuela City food service establishments. The Proponents and Students group, rated High Power and High Interest, are engaged daily through stand-ups and Discord or Slack to maintain accountability for Critical Path assignments and sprint deliverables across the full capstone development cycle.

RACI Matrix

The RACI Matrix below defines the responsibility assignment for each project deliverable across all team roles. Each deliverable is assigned R (Responsible), A (Accountable), C (Consulted), or I (Informed) per role.

LEGEND

R	Responsible
A	Accountable
C	Consulted
I	Informed

Deliverable / Activity	Project Lead	Dev Lead	Design Lead	Docu Lead	TA	Research Adviser
1. Functional Requirements Document (FRD, 105 Requirements)	C	R	C	R	C	C
2. System Analysis (DFD, Use Cases, FRs)	A	R	C	R	C	C
3. System Design (ERD, Schema, UI/UX)	I	R	A/R	I	C	I
4. OCR Engine Integration (Preprocessing Spec, Confidence Scoring)	I	A/R	R	I	C	I
5. Stakeholder Analysis (RACI, Power-Interest Grid)	A/R	I	I	R	C	C
6. Security & Privacy Compliance (RA 10173, Data Privacy Act)	A	R	R	R	C	C
7. UI/UX Prototype (WCAG 2.1 AA, Bilingual EN/FIL)	I	R	A/R	I	C	I

8. System Testing & QA (UAT, Test Cases, Verification Report)	**A**	**R**	**R**	**R**	**I**	**C**
9. Deployment & Integration (API, Portals, Offline Sync)	**I**	**A/R**	**R**	**I**	**C**	**I**
10. Final Manuscript & Defense (Capstone Document, Defense Slides)	**I**	**I**	**I**	**A/R**	**I**	**C**

The RACI Matrix assigns responsibility designations across five major deliverables and six roles: Project Lead, Dev Lead, Design Lead, Documentation Lead, Technical Adviser, and Research Adviser, ensuring every component of iDetect has a clear point of ownership. For Project Management covering the Project Plan, SEP, and WBS, the Project Lead is Accountable and Responsible while the Documentation Lead is co-Responsible for producing the documents. For System Analysis covering the Functional Requirements, Use Cases, and DFD, the Project Lead is Accountable and the Dev Lead is Responsible, reflecting the six core system capabilities detailed in the manuscript.

For System Design covering the ERD, schema, and UI/UX, the Design Lead holds joint Accountable and Responsible status alongside the Dev Lead, consistent with the requirement to achieve WCAG 2.1 AA compliance and a bilingual English and Filipino interface. For Testing and Quality Assurance, the Project Lead is Accountable with the Dev Lead, Design Lead, and Documentation Lead all Responsible, aligned with the ISO 25010 evaluation framework covering eight quality characteristics. For the Final Manuscript and Submission, the

Documentation Lead is solely Accountable and Responsible with the Research Adviser as Consulted. Across all deliverables, both advisers are consistently Consulted or Informed rather than Responsible, reinforcing their supervisory function while the four core team members carry full operational accountability for delivering a system designed to address the ₱88.2 billion annual fraud crisis in Valenzuela City.

Research Locale

This study will be conducted in Valenzuela City, a highly urbanized city in the National Capital Region (NCR) of the Philippines. Valenzuela City serves as an ideal research locale due to its active implementation of anti-fraud measures through the "Bawal ang Fake ID" campaign and the VALID (Valenzuela ID Validation) Portal launched in May 2024, demonstrating local government commitment to addressing fraudulent identification issues.

The research will focus specifically on food service establishments throughout Valenzuela City, including restaurants, fast-food chains, cafes, and dining establishments that regularly process Senior Citizen and PWD discount transactions under Republic Acts 9994 and 10754. These establishments were selected as the primary implementation sites because food service businesses represent the sector most significantly impacted by fraudulent discount claims, with industry data showing discount claims increasing from approximately 5% of transactions in 2019 to over 30% by 2024.

The study will also involve government offices within Valenzuela City, specifically the Office of Senior Citizens Affairs (OSCA), Persons with Disability Affairs Office (PDAO), and Business Permit and Licensing Office (BPLO), which maintain the authoritative databases of legitimate beneficiaries and oversee the regulatory framework for discount program implementation.

Valenzuela City's existing digital infrastructure, including the VALID Portal and newly designed ID cards with holographic seals, provides a conducive environment for implementing and evaluating the iDetect system's integration with local government databases. The city's geographic scope and population size (approximately 714,978 residents as of 2020 census) offer sufficient scale to demonstrate system effectiveness while remaining manageable for a capstone research project.

Sample and Sampling Technique

This study will employ Purposive Sampling, a non-probability sampling technique where respondents are deliberately selected based on specific characteristics, expertise, and relevance to the research objectives. This approach is appropriate for developmental and applied research as it ensures participation of information-rich respondents who possess direct experience with the verification challenges the iDetect system aims to address.

The study will involve approximately 80-100 respondents representing six distinct stakeholder groups: Business Establishment Staff (n≈30-35) including frontline cashiers and managers from food service establishments aged 20-50 years old who process discount

transactions daily; OSCA Personnel (n≈8-10) including administrators and database managers aged 30-60 years old responsible for Senior Citizen ID databases; PDAO Personnel (n≈8-10) including administrators and fraud investigators aged 30-60 years old managing PWD ID databases; BPLO Personnel (n≈5-8) including administrative staff aged 28-55 years old who oversee business establishment accounts; Senior Citizen Beneficiaries (n≈12-15) aged 60 years and above who actively use discount programs; and PWD Beneficiaries (n≈12-15) aged 18 years and above who possess legitimate PWD IDs.

Selection Criteria: Business establishment staff will be selected based on employment at establishments processing significant discount transaction volumes (minimum 10 daily), minimum six months experience in verification roles, and willingness to participate in system testing. OSCA, PDAO, and BPLO personnel will be selected based on official authorization to access databases, involvement in fraud investigation, minimum one year tenure, and technical familiarity with existing verification systems. Beneficiaries will be selected based on possession of legitimate, currently valid IDs from Valenzuela City, active monthly discount usage, and willingness to provide detailed feedback and test the mobile application.

Sampling Procedures: Respondent recruitment will proceed through institutional partnerships. Business establishment staff will be identified through coordination with Valenzuela City BPLO, which maintains records of registered food service establishments. LGU personnel will be recruited through official coordination with respective office heads who will identify appropriate staff meeting selection criteria. Beneficiaries will be recruited through announcements at OSCA and PDAO offices, senior citizen centers, and PWD organization

meetings, with screening to ensure diverse demographic profiles. All respondents will provide informed consent acknowledging voluntary participation, data privacy protections under Republic Act 10173, and the right to withdraw at any time.

The purposive sampling approach ensures engagement of stakeholders most directly impacted by verification system improvements and most capable of providing substantive insights for system development, implementation strategies, and effectiveness evaluation.

CHAPTER 4 ANALYSIS, PRESENTATION, AND INTERPRETATION OF DATA

Demographic Profile of Respondents

Business Establishment Users. The primary target users are frontline cashiers and store managers aged 25–50 years old from retail establishments located in Valenzuela City, including convenience stores, supermarkets, and department stores. This group consists of approximately 50% male and 50% female employees responsible for processing Senior Citizen and PWD discount transactions daily.

LGU Personnel (OSCA and PDAO). Authorized personnel from the Office of Senior Citizens Affairs (OSCA) and Persons with Disability Affairs Office (PDAO) aged 30–60 years old will participate in the study. These administrators and database managers, representing both genders equally, are based in Valenzuela City and manage official ID databases and fraud investigations.

Senior Citizen and PWD Beneficiaries. Senior Citizens aged 60 years and above and Persons with Disabilities aged 18 years and above from Valenzuela City will be included as

respondents. This group, distributed across both genders, represents active users of government-mandated discount programs and will provide feedback on their experience with current verification processes. BPLO (Business Permit and Licensing Office) Personnel Administrative and technical staff from the BPLO aged 28–55 years old will serve as respondents. These personnel, representing both genders, oversee business establishment accounts and monitor system operations within Valenzuela City.

Geographic Location. All respondents will be selected from Valenzuela City, Metro Manila, Philippines, ensuring that feedback reflects the local context and operational environment where iDetect will be implemented. Participants will be recruited from different barangays across the city to represent diverse business establishments and community areas.

Data Gathering Procedures

Interview. The developers will conduct structured interviews with key stakeholders, specifically business establishment staff, managers, and authorized LGU personnel from OSCA and PDAO. These interviews aim to gather in-depth qualitative information regarding their existing ID verification workflows, common challenges encountered, frequency of fraudulent ID incidents, and expectations for an improved verification system. Each interview will follow a pre-defined guide to ensure consistency in the data collected. Responses will be recorded, summarized, and used to finalize the functional and non-functional requirements of the iDetect system.

Questionnaires. The developers will administer structured and validated survey questionnaires to frontline cashiers, business owners, Senior Citizen and PWD beneficiaries, and

LGU personnel. These questionnaires will be designed to gather quantitative data related to user experience, satisfaction with current verification practices, time spent per verification, and perceived usefulness of an automated system. The questionnaire results will be tabulated and analyzed to support the development of user-centered system features, interface design improvements, and system evaluation metrics.

Internet Research. The developers will conduct extensive internet research to collect background information, existing studies, and technological references relevant to the project. This includes reviewing documentation on Optical Character Recognition (OCR), fraud detection systems, database integration practices, and digital verification standards. The developers will also examine online platforms used by various LGUs, such as VALID, OSCA Taguig, and PDAO Taguig systems, to identify their strengths, limitations, and areas where iDetect can provide significant improvements. The information gathered through online research will help establish the theoretical foundation of the study and support the rationale for system development.

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WORK BREAKDOWN STRUCTURE DICTIONARY

LEVEL	COE	NAME	DESCRIPTION	PREDECESSORS	DURATION	OWNER
2	**1.1**	**PROJECT INITIATION**	DEFINE THE OVERALL SCOPE, STAKEHOLDERS, AND FORMAL AUTHORIZATION DOCUMENT FOR THE IDETECT PROJECT	N/A	N/A	
3	1.1.1	DEFINE SCOPE & OBJECTIVES	IDENTIFY PROJECT BOUNDARIES, GOALS, DELIVERABLES, AND SUCCESS CRITERIA FOR IDETECT	N/A	DAYS 1–3	Project Lead
3	1.1.2	IDENTIFY STAKEHOLDERS	LIST ALL INTERNAL AND EXTERNAL STAKEHOLDERS (BPLO, OSCA, PDAO, ESTABLISHMENTS, BENEFICIARIES) AND DEFINE THEIR ROLES	1.1.1	DAYS 3–5	All Roles
3	1.1.3	DEVELOP PROJECT CHARTER (PART 1)	BEGIN DRAFTING THE FORMAL PROJECT CHARTER — COMPLETE INITIAL SECTIONS: PROJECT PURPOSE, SCOPE STATEMENT, AND STAKEHOLDER REGISTER (DAYS 5–7). CONTINUES INTO WEEK 2.	1.1.2	DAYS 5–7 (Part 1 of 2)	Project Lead Docu Lead
2	**1.1**	**PROJECT INITIATION**	DEFINE THE OVERALL SCOPE, STAKEHOLDERS, AND FORMAL AUTHORIZATION DOCUMENT FOR THE IDETECT PROJECT	N/A	N/A	All Roles
3	1.1.1	DEFINE SCOPE & OBJECTIVES	IDENTIFY PROJECT BOUNDARIES, GOALS, DELIVERABLES, AND SUCCESS CRITERIA FOR IDETECT	N/A	DAYS 1–3	Project Lead
3	1.1.2	IDENTIFY STAKEHOLDERS	LIST ALL INTERNAL AND EXTERNAL STAKEHOLDERS (BPLO, OSCA, PDAO, ESTABLISHMENTS, BENEFICIARIES) AND DEFINE THEIR ROLES	1.1.1	DAYS 3–5	Project Lead

2	**1.1**	**PROJECT INITIATION (CONTINUED)**	COMPLETE THE PROJECT CHARTER AND OBTAIN FORMAL SIGN-OFF FROM RESEARCH ADVISER	N/A	N/A	All Roles
3	1.1.3	DEVELOP PROJECT CHARTER (PART 2)	COMPLETE REMAINING CHARTER SECTIONS: OBJECTIVES, MILESTONES, ROLES & RESPONSIBILITIES, RISKS OVERVIEW, AND OBTAIN ADVISER SIGN-OFF.	1.1.3 (Part 1)	DAYS 7–10 (Part 2 of 2)	Project Lead Docu Lead
2	**1.2**	**EARLY PROJECT PLANNING**	BEGIN STRUCTURED DECOMPOSITION OF THE FULL IDETECT PROJECT SCOPE INTO A HIERARCHICAL WORK BREAKDOWN STRUCTURE	N/A	N/A	All Roles
3	1.2.1	DEVELOP WBS	DECOMPOSE THE ENTIRE IDETECT PROJECT INTO ALL SEVEN PHASES, TASKS, AND SUB-TASKS IN A HIERARCHICAL WORK BREAKDOWN STRUCTURE	1.1.3 (Part 2)	DAYS 10–14	Project Lead Docu Lead
2	**1.1**	**PROJECT INITIATION (CONTINUED)**	COMPLETE THE PROJECT CHARTER AND OBTAIN FORMAL SIGN-OFF FROM RESEARCH ADVISER	N/A	N/A	All Roles

GANTT CHART

TASK/ DAY	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1.1.1														
1.1.2														
1.1.3														
1.2.1														